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7

MANAGING PROCESS CONSTRAINTS

A staff member takes bookings for hospital beds. The Radclliffe Infirmary brought about a complete reduction in the number of surgeries that were being routinely cancelled, simply by limiting the number of bookings and appointing a manager to oversee the bookings process.

The Radcliffe Infirmary

The Radcliffe Infirmary, part of the Oxford Radcliffe Hospital Trust, in Oxford, England, is one of the largest teaching hospitals in the United Kingdom. It provides high-quality health care services to the majority of the population in Oxfordshire County, as well as neighboring communities. Several years ago, the hospital was having difficulty keeping up with the demand of this large patient base. Hospital administrators often found their departments to be understaffed and were frequently forced to cancel elective surgeries. The staff that was available was overworked, causing employee sickness rates to rise and accelerating the problem. Hospital administrators knew this dilemma was caused by inefficiencies in hospital's operations; however, no department wanted to admit to causing a bottleneck. To correct the problem, director of operations Jan Elliott decided to initiate a theory of constraints (TOC) workshop. A similar workshop was used at a hospital in Cambridge and generated successful results.

During the workshop, Elliott realized that the problem stemmed from the organization's inconsistent method of booking beds. At the time, standard protocol called for the ward staff to handle the cumbersome task of bed management along with their other responsibilities. It was clear that in order to keep the approach to booking beds consistent, a bed manager needed to be appointed, and new protocol needed to be initiated. Previous protocol allowed up to 11 patients a day to be booked into the unit and to have a bed and surgery time reserved for each. Mondays and Tuesdays were frequently booked to capacity, as most surgeons scheduled surgeries in the beginning of the week

to ensure that their procedures would be performed before the week's end. Because the unit could not work adequately in this capacity, many of these surgeries were performed during off hours or cancelled.

To fix this problem, the new bed management protocol called for the bed manager to book a maximum of six patients a day. Cutting the number almost in half may have seemed counterproductive, but with a smaller number of patients to focus on, staff members were able to carry out their roles more effectively. During a three-month period after implementing this new protocol, not one elective surgery was cancelled. During a three-month period in the previous year, before the new protocol was implemented, nearly 64 surgeries were cancelled.

The new protocol not only improved the hospital's cancellation issues, it also improved the staff's morale. No longer overworked, hospital staff became visibly happier at their jobs.

Although it seemed obvious that initiating the theory of constraints greatly improved, the operations of the Radcliffe Infirmary, conclusive evidence was not formed until nine months later when the position of bed manager became vacant. Without someone to regulate the bed booking for the unit, the situation reverted. Once the position was filled, the improvements returned.

Source: "Our Hospitals," Oxford Radcliffe Hospitals, NHS Trust, at **http://www.oxfordradcliffe.nhs.uk/aboutus/hospitals/hospitals.aspx**, Accessed September 25, 2008; Belinda Phipps, "Hitting the Bottleneck," Health Management, February 1999, at **http://www.goldratt.com/hlthmgmtmn.htm**.

LEARNING GOALS *After reading this chapter, you should be able to:*

1. Explain the theory of constraints.
2. Understand linkage of capacity constraints to financial performance measures.
3. Identify bottlenecks.
4. Apply theory of constraints to product mix decisions.
5. Describe how to manage constraints in an assembly line.

constraint

Any factor that limits the performance of a system and restricts its output. In linear programming, a limitation that restricts the permissible choices for the decision variables.

bottleneck

A capacity constraint resource (CCR) whose available capacity limits the organization's ability to meet the product volume, product mix, or demand fluctuation required by the marketplace.

Suppose one of a firm's processes was recently reengineered, and yet results were disappointing. Costs were still high or customer satisfaction still low. What could be wrong? The answer might be constraints that remain in one or more steps in the firm's processes. A **constraint** is any factor that limits the performance of a system and restricts its output, while *capacity* is the maximum rate of output of a process or a system. When constraints exist at any step, capacity can become imbalanced—too high in some departments and too low in others. As a result, the overall performance of the system suffers.

Constraints can occur up or down the supply chain, with either the firm's suppliers or customers, or within one of the firm's processes like service/product development or order fulfillment. Three kinds of constraints can generally be identified: physical (usually machine, labor, or workstation capacity or material shortages, but could be space or quality), market (demand is less than capacity), or managerial (policy, metrics, or mind-sets that create constraints that impede work flow). A **bottleneck**[1] is a special type of a constraint that relates to the capacity shortage of a process, and is defined as any resource whose available capacity limits the organization's ability to meet the service or product volume, product mix, or fluctuating requirements demanded by the marketplace. A business system or a process would have at least one constraint or a bottleneck; otherwise, its output would be limited only by market demand. The experience of The Radcliffe Infirmary and other firms in the health care, banking, and manufacturing industries demonstrates how important managing constraints can be to an organization's future.

[1]Under certain conditions, a bottleneck is also called a *capacity constrained resource* (CCR). The process with the least capacity is called a bottleneck if its output is less than the market demand, or called a CCR if it is the least capable resource in the system but still has higher capacity than the market demand.

Managing Constraints across the Organization

Firms must manage their constraints and make appropriate capacity choices at the individual-process level, as well as at the organization level. Hence, this process involves inter-functional cooperation. Detailed decisions and choices made within each of these levels affect where resource constraints or bottlenecks show up, both within and across departmental lines. Relieving a bottleneck in one part of an organization might not have the desired effect unless a bottleneck in another part of the organization is also addressed. A bottleneck could be the sales department not getting enough sales or the loan department not processing loans fast enough. The constraint could be a lack of capital or equipment, or it could be planning and scheduling.

Managers throughout the organization must understand how to identify and manage bottlenecks in all types of processes, how to relate the capacity and performance measures of one process to another, and how to use that information to determine the firm's best service or product mix. This chapter explains how managers can best make these decisions.



Traffic bottleneck in Beijing, China.

Lou-Foto/Alamy

The Theory of Constraints

The **theory of constraints (TOC)** is a systematic management approach that focuses on actively managing those constraints that impede a firm's progress toward its goal of maximizing profits and effectively using its resources. The theory was developed nearly three decades ago by Eli Goldratt, a well-known business systems analyst. It outlines a deliberate process for identifying and overcoming constraints. The process focuses not just on the efficiency of individual processes, but also on the bottlenecks that constrain the system as a whole.

TOC methods increase the firm's profits more effectively by focusing on making materials flow rapidly through the entire system. They help firms look at the big picture—how processes can be improved to increase overall work flows, and how inventory and workforce levels can be reduced while still effectively utilizing critical resources. To do this, it is important to understand the relevant performance and capacity measures at the operational level, as well as their relationship to the more broadly understood financial measures at the firm level. These measures and relationships, so critical in successfully applying the principles of the TOC, are defined in Table 7.1.

theory of constraints (TOC)

A systematic management approach that focuses on actively managing those constraints that impede a firm's progress toward its goal.

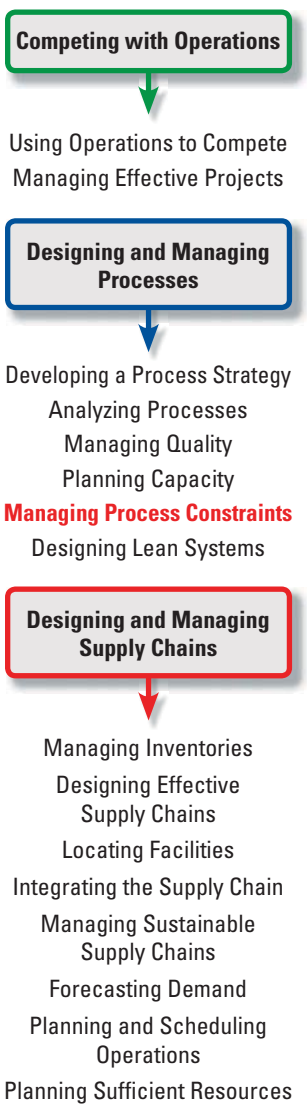


TABLE 7.1 HOW THE FIRM'S OPERATIONAL MEASURES RELATE TO ITS FINANCIAL MEASURES

Operational Measures	TOC View	Relationship to Financial Measures
Inventory (I)	All the money invested in a system in purchasing things that it intends to sell	A decrease in I leads to an increase in net profit, ROI, and cash flow.
Throughput (T)	Rate at which a system generates money through sales	An increase in T leads to an increase in net profit, ROI, and cash flows.
Operating Expense (OE)	All the money a system spends to turn inventory into throughput	A decrease in OE leads to an increase in net profit, ROI, and cash flows.
Utilization (U)	The degree to which equipment, space, or workforce is currently being used, and is measured as the ratio of average output rate to maximum capacity, expressed as a percentage	An increase in U at the bottleneck leads to an increase in net profit, ROI, and cash flows.

According to the TOC view, every capital investment in the system, including machines and work-in-process materials, represents inventory because they could all potentially be sold to make money. Producing a product or a service that does not lead to a sale will not increase a firm's throughput, but will increase its inventory and operating expenses. It is always best to manage the system so that utilization at the bottleneck resource is maximized in order to maximize throughput.

Key Principles of the TOC

The chief concept behind the TOC is that the bottlenecks should be scheduled to maximize their throughput of services or products while adhering to promised completion dates. The underlying assumption is that demand is greater or equal to the capacity of the process that produces the service or product, otherwise instead of internal changes, marketing must work towards promoting increasing its demand. For example, manufacturing a garden rake involves attaching a bow to the rake's head. Rake heads must be processed on the blanking press, welded to the bow, cleaned, and attached to the handle to make the rake, which is packaged and finally shipped to Sears, Home Depot, or Walmart, according to a specific delivery schedule. Suppose that the delivery commitments for all styles of rakes for the next month indicate that the welding station is loaded at 105 percent of its capacity, but that the other processes will be used at only 75 percent of their capacities. According to the TOC, the welding station is the bottleneck resource, whereas the blanking, cleaning, handle attaching, packaging, and shipping processes are nonbottleneck resources. Any idle time at the welding station must be eliminated to maximize throughput. Managers should therefore focus on the welding schedule.

Seven key principles of the TOC that revolve around the efficient use and scheduling of bottlenecks and improving flow and throughput are summarized in Table 7.2.

TABLE 7.2 | SEVEN KEY PRINCIPLES OF THE THEORY OF CONSTRAINTS

1. The focus should be on balancing flow, not on balancing capacity.
2. Maximizing the output and efficiency of every resource may not maximize the throughput of the entire system.
3. An hour lost at a bottleneck or a constrained resource is an hour lost for the whole system. In contrast, an hour saved at a nonbottleneck resource is a mirage because it does not make the whole system more productive.
4. Inventory is needed only in front of the bottlenecks in order to prevent them from sitting idle, and in front of assembly and shipping points in order to protect customer schedules. Building inventories elsewhere should be avoided.
5. Work, which can be materials, information to be processed, documents, or customers, should be released into the system only as frequently as the bottlenecks need it. Bottleneck flows should be equal to the market demand. Pacing everything to the slowest resource minimizes inventory and operating expenses.
6. Activating a nonbottleneck resource (using it for improved efficiency that does not increase throughput) is not the same as utilizing a bottleneck resource (that does lead to increased throughput). Activation of nonbottleneck resources cannot increase throughput, nor promote better performance on financial measures outlined in Table 7.1.
7. Every capital investment must be viewed from the perspective of its global impact on overall throughput (T), inventory (I), and operating expense (OE).

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Bal Seal Engineering is a designer and manufacturer of custom seals and canted-coil™ springs for aerospace, automotive, transportation, medical and other industries. By applying many modern management principles including the theory of constraints (TOC), the company has been able to grow and improve customer satisfaction.

Practical application of the TOC involves the implementation of the following steps.

1. *Identify the System Bottleneck(s).* For the rake example, the bottleneck is the welding station because it is restricting the firm's ability to meet the shipping schedule and, hence, total value-added funds. Other ways of identifying the bottleneck will be looked at in more detail a little later in this chapter.
2. *Exploit the Bottleneck(s).* Create schedules that maximize the throughput of the bottleneck(s). For the rake example, schedule the welding station to maximize its utilization while meeting the shipping commitments to the extent possible. Also make sure that only good quality parts are passed on to the bottleneck.
3. *Subordinate All Other Decisions to Step 2.* Nonbottleneck resources should be scheduled to support the schedule of the bottleneck and not produce more than the bottleneck can handle. That is, the blanking press should not produce more than the welding station can handle, and the activities of the cleaning and subsequent operations should be based on the output rate of the welding station.

4. *Elevate the Bottleneck(s).* After the scheduling improvements in steps 1–3 have been exhausted and the bottleneck is still a constraint to throughput, management should consider increasing the capacity of the bottleneck. For example, if the welding station is still a constraint after exhausting schedule improvements, consider increasing its capacity by adding another shift or another welding machine. Other mechanisms are also available for increasing bottleneck capacity, and we address them a little later.
5. *Do Not Let Inertia Set In.* Actions taken in steps 3 and 4 will improve the welder throughput and may alter the loads on other processes. Consequently, the system constraint(s) may shift. Then, the practical application of steps 1–4 must be repeated in order to identify and manage the new set of constraints.

Because of its potential for improving performance dramatically, many manufacturers have applied the principles of the theory of constraints. All manufacturers implementing TOC principles can also dramatically change the mind-set of employees and managers. Instead of focusing solely on their own functions, they can see the "big picture" and where other improvements in the system might lie.

Identification and Management of Bottlenecks

Bottlenecks can both be internal or external to the firm, and typically represent a process, a step, or a workstation with the lowest capacity. **Throughput time** is the total elapsed time from the start to the finish of a job or a customer being processed at one or more workcenters. Where a bottleneck lies in a given service or manufacturing process can be identified in two ways. A workstation in a process is a bottleneck if (1) it has the highest total time per unit processed, or (2) it has the highest average utilization and total workload

throughput time
Total elapsed time from the start to the finish of a job or a customer being processed at one or more workcenters.

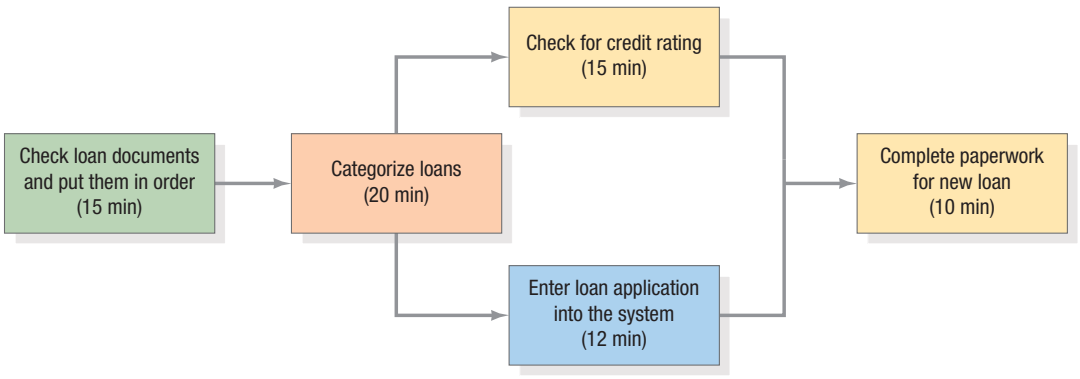
Managing Bottlenecks in Service Processes

Example 7.1 illustrates how a bottleneck step or activity can be identified for a loan approval process at a bank.

EXAMPLE 7.1 **Identifying the Bottleneck in a Service Process**

Managers at the First Community Bank are attempting to shorten the time it takes customers with approved loan applications to get their paperwork processed. The flowchart for this process, consisting of several different activities, each performed by a different bank employee, is shown in Figure 7.1. Approved loan applications first arrive at activity or step 1, where they are checked for completeness and put in order. At step 2, the loans are categorized into different classes according to the loan amount and whether they are being requested for personal or commercial reasons. While credit checking commences at step 3, loan application data are entered in parallel into the information system for record-keeping purposes at step 4. Finally, all paperwork for setting up the new loan is finished at step 5. The time taken in minutes is given in parentheses.

Which single step is the bottleneck, assuming that market demand for loan applications exceeds the capacity of the process? The management is also interested in knowing the maximum number of approved loans this system can process in a 5-hour work day.



▲ **FIGURE 7.1**
Processing Credit Loan Applications at First Community Bank

SOLUTION

We define the bottleneck as step 2, which has the highest time per loan processed. The throughput time to complete an approved loan application is 15 + 20 + max (15, 12) + 10 = **60** minutes. Although we assume no waiting time in front of any step, in practice such a smooth process flow is not always the case. So the actual time taken for completing an approved loan will be longer than 60 minutes due to nonuniform arrival of applications, variations in actual processing times, and the related factors.

The capacity for loan completions is derived by translating the "minutes per customer" at the bottleneck step to "customer per hour." At First Community Bank, it is 3 customers per hour because the bottleneck step 2 can process only 1 customer every 20 minutes (60/3).

DECISION POINT

Step 2 is the bottleneck constraint. The bank will be able to complete a maximum of only 3 loan accounts per hour, or 15 new loan accounts in a 5-hour day. Management can increase the flow of loan applications by increasing the capacity of step 2 up to the point where another step becomes the bottleneck.



Todd Bigelow/Aurora Photos/Alamy

Due to constrained resources like doctors, nurses, and equipment, patients wait for medical care in a crowded waiting room at South Central Family Health Center in Los Angeles, California.

A front-office process with high customer contact and divergence does not enjoy the simple line flows shown in Example 7.1. Its operations may serve many different customer types, and the demands on any one operation could vary considerably from one day to the next. However, bottlenecks can still be identified by computing the average utilization of each operation. However, the variability in workload also creates *floating bottlenecks*. One week the mix of work may make operation 1 a bottleneck, and the next week it may make operation 3 the bottleneck. This type of variability increases the complexity of day-to-day scheduling. In this situation, management prefers lower utilization rates, which allow greater slack to absorb unexpected surges in demand.

TOC principles outlined here are fairly broad-based and widely applicable. They can be useful for evaluating individual processes as well as large systems for both manufacturers as well as service providers. Service organizations, such as Delta Airlines, United Airlines, and major hospitals across the United States, including the U.S. Air Force health care system, use the TOC to their advantage.

Managing Bottlenecks in Manufacturing Processes

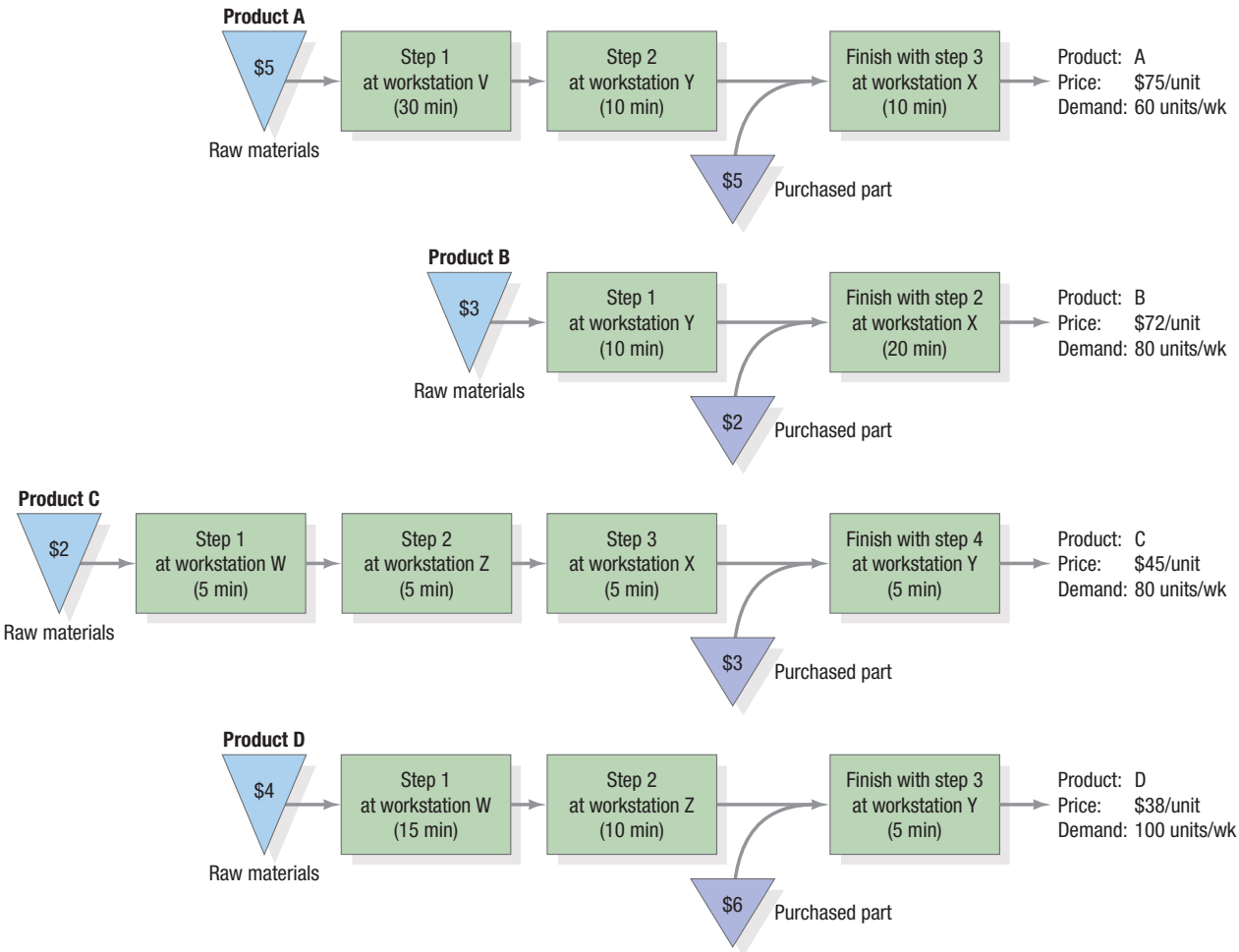
Bottlenecks can exist in all types of manufacturing processes, including the job process, batch process, line process, and continuous process. Since these processes differ in their design, strategic intent, and allocation of resources (see Chapter 3, "Developing a Process Strategy," for additional details), identification and management of bottlenecks will also differ accordingly with process type. We first discuss in this section issues surrounding management of bottlenecks in job and batch processes, while relegating constraint management in line processes for a later section.

Identifying Bottlenecks Manufacturing processes often pose some complexities when identifying bottlenecks. If multiple services or products are involved, extra setup time at a workstation is usually needed to change over from one service or product to the next, which in turn increases the overload at the workstation being changed over. *Setup times* and their associated costs affect the size of the lots traveling through the job or batch processes. Management tries to reduce setup times because they represent unproductive time for workers or machines and thereby allow for smaller, more economic, batches. Nonetheless, whether setup times are significant or not, one way to identify a bottleneck operation is by its utilization. Example 7.2 illustrates how a bottleneck can be identified in a manufacturing setting where setups are negligible.

EXAMPLE 7.2 Identifying the Bottleneck in a Batch Process

Diablo Electronics manufactures four unique products (A, B, C, and D) that are fabricated and assembled in five different workstations (V, W, X, Y, and Z) using a small batch process. Each workstation is staffed by a worker who is dedicated to work a single shift per day at an assigned workstation. Batch setup times have been reduced to such an extent that they can be considered negligible. A flowchart denotes the path each product follows through the manufacturing process as shown in Figure 7.2, where each product's price, demand per week, and processing times per unit are indicated as well. Inverted triangles represent purchased parts and raw materials consumed per unit at different workstations. Diablo can make and sell up to the limit of its demand per week, and no penalties are incurred for not being able to meet all the demand.

Which of the five workstations (V, W, X, Y, or Z) has the highest utilization, and thus serves as the bottleneck for Diablo Electronics?



▲ **FIGURE 7.2**
Flowchart for Products A, B, C, and D

SOLUTION

Because the denominator in the utilization ratio is the same for every workstation, with one worker per machine at each step in the process, we can simply identify the bottleneck by computing aggregate workloads at each workstation.

The firm wants to satisfy as much of the product demand in a week as it can. Each week consists of 2,400 minutes of available production time. Multiplying the processing time at each station for a given product with the number of units demanded per week yields the workload represented by that product. These loads are summed across all products going through a workstation to arrive at the total load for the workstation, which is then compared with the others and the existing capacity of 2,400 minutes.

Workstation	Load from Product A	Load from Product B	Load from Product C	Load from Product D	Total Load (min)
V	60 × 30 = 1800	0	0	0	1,800
W	0	0	80 × 5 = 400	100 × 15 = 1,500	1,900
X	60 × 10 = 600	80 × 20 = 1,600	80 × 5 = 400	0	2,600
Y	60 × 10 = 600	80 × 10 = 800	80 × 5 = 400	100 × 5 = 500	2,300
Z	0	0	80 × 5 = 400	100 × 10 = 1,000	1,400

DECISION POINT

Workstation X is the bottleneck for Diablo Electronics because the aggregate workload at X is larger than the aggregate workloads of workstations V, W, Y, and Z and the maximum available capacity of 2,400 minutes per week.

Identifying the bottlenecks becomes considerably harder when setup times are lengthy and the degree of divergence in the process is greater than that shown in Example 7.2. When the setup time is large, the operation with the highest total time per unit processed would typically tend to be the bottleneck. Variability in the workloads will again likely create floating bottlenecks, especially if most processes involve multiple operations, and often their capacities are not identical. In practice, these bottlenecks can also be determined by asking workers and supervisors in the plant where the bottlenecks might lie and looking for piled up material in front of different workstations.

Relieving Bottlenecks The key to preserving bottleneck capacity is to carefully monitor short-term schedules and keep bottleneck resource as busy as is practical. Managers should minimize idle time at the bottlenecks, caused by delays elsewhere in the system and make sure that the bottleneck has all the resources it needs to stay busy. When a changeover or setup is made at a bottleneck, the number of units or customers processed before the next changeover should be large compared to the number processed at less critical operations. Maximizing the number of units processed per setup means fewer setups per year and, thus, less total time lost to setups. The number of setups also depends on the required product variety; more variety necessitates more frequent changeovers.

The long-term capacity of bottleneck operations can be expanded in various ways. Investments can be made in new equipment and in brick-and-mortar facility expansions. The bottleneck's capacity also can be expanded by operating it more hours per week, such as by hiring more employees and going from a one-shift operation to multiple shifts, or by hiring more employees and operating the plant 6 or 7 days per week versus 5 days per week. Managers also might relieve the bottleneck by redesigning the process, either through *process reengineering* or *process improvement*, or by purchasing additional machines or machines that can handle more capacity.

Product Mix Decisions Managers might be tempted to produce the products with the highest contribution margins or unit sales. *Contribution margin* is the amount each product contributes to profits and overhead; no fixed costs are considered when making the product mix decision. We call this approach the *traditional method*. The problem with this approach is that the firm's actual throughput and overall profitability depend more upon the contribution margin generated at the bottleneck than by the contribution margin of each individual product produced. We call this latter approach the *bottleneck method*. Example 7.3 illustrates both of these methods.

EXAMPLE 7.3 Determining the Product Mix Using Contribution Margin

The senior management at Diablo Electronics (see Exercise 7.2) wants to improve profitability by accepting the right set of orders, and so collected some additional financial data. Variable overhead costs are $8,500 per week. Each worker is paid $18 per hour and is paid for an entire week, regardless of how much the worker is used. Consequently, labor costs are fixed expenses. The plant operates one 8-hour shift per day, or 40 hours each week. Currently, decisions are made using the traditional method, which is to accept as much of the highest contribution margin product as possible (up to the limit of its demand), followed by the next highest contribution margin product, and so on until no more capacity is available. Pedro Rodriguez, the newly hired production supervisor, is knowledgeable about the theory of constraints and bottleneck-based scheduling.

He believes that profitability can indeed be improved if bottleneck resources were exploited to determine the product mix. What is the change in profits if, instead of the traditional method used by Diablo Electronics, the bottleneck method advocated by Pedro is used to select the product mix?

SOLUTION

Decision Rule 1: Traditional Method
Select the best product mix according to the highest overall contribution margin of each product.

Step 1: Calculate the contribution margin per unit of each product as shown here.

	A	B	C	D
Price	$75.00	$72.00	$45.00	$38.00
Raw material and purchased parts	–10.00	–5.00	–5.00	–10.00
= **Contribution margin**	$65.00	$67.00	$40.00	$28.00

When ordered from highest to lowest, the contribution margin per unit sequence of these products is B, A, C, D.

Step 2: Allocate resources V, W, X, Y, and Z to the products in the order decided in step 1. Satisfy each demand until the bottleneck resource (workstation X) is encountered. Subtract minutes away from 2,400 minutes available for each week at each stage.

Work Center	Minutes at the Start	Minutes Left After Making 80 B	Minutes Left After Making 60 A	Can Only Make 40 C	Can Still Make 100 D
V	2,400	2,400	600	600	600
W	2,400	2,400	2,400	2,200	700
X	2,400	800	200	0	0
Y	2,400	1,600	1,000	800	300
Z	2,400	2,400	2,400	2,200	1,200

The best product mix according to this traditional approach is then 60 A, 80 B, 40 C, and 100 D.

Step 3: Compute profitability for the selected product mix.

Profits		
Revenue	$(60 \times \$75) + (80 \times \$72) + (40 \times \$45) + (100 \times \$38)$	= $15,860
Materials	$(60 \times \$10) + (80 \times \$5) + (40 \times \$5) + (100 \times \$10)$	= –$2,200
Labor	(5 workers) × (8 hours/day) × (5 days/week) × ($18/hour)	= –$3,600
Overhead		= –$8,500
Profit		= $1,560

Manufacturing the product mix of 60 A, 80 B, 40 C, and 100 D will yield a profit of $1,560 per week.

Decision Rule 2: Bottleneck Method
Select the best product mix according to the dollar contribution margin per minute of processing time at the bottleneck workstation X. This method would take advantage of the principles outlined in the theory of constraints and get the most dollar benefit from the bottleneck.

Step 1: Calculate the contribution margin/minute of processing time at bottleneck workstation X:

	Product A	Product B	Product C	Product D
Contribution margin	$65.00	$67.00	$40.00	$28.00
Time at bottleneck	10 minutes	20 minutes	5 minutes	0 minutes
Contribution margin per minute	$6.50	$3.35	$8.00	Not defined

When ordered from highest to lowest contribution margin/minute at the bottleneck, the manufacturing sequence of these products is D, C, A, B, which is reverse of the earlier order. Product D is scheduled first because it does not consume any resources at the bottleneck.

Step 2: Allocate resources V, W, X, Y, and Z to the products in the order decided in step 1. Satisfy each demand until the bottleneck resource (workstation X) is encountered. Subtract minutes away from 2,400 minutes available for each week at each stage.

Work Center	Minutes at the Start	Minutes Left After Making 100 D	Minutes Left After Making 80 C	Minutes Left After Making 60 A	Can Only Make 70 B
V	2,400	2,400	2,400	600	600
W	2,400	900	500	500	500
X	2,400	2,400	2,000	1,400	0
Y	2,400	1,900	1,500	900	200
Z	2,400	1,400	1,000	1,000	1,000

The best product mix according to this bottleneck-based approach is then 60 A, 70 B, 80 C, and 100 D.

Step 3: Compute profitability for the selected product mix.

	Profits	
Revenue	$(60 \times \$75) + (70 \times \$72) + (80 \times \$45) + (100 \times \$38)$	$= \$16,940$
Materials	$(60 \times \$10) + (70 \times \$5) + (80 \times \$5) + (100 \times \$10)$	$= -\$2,350$
Labor	(5 workers) × (8 hours/day) × (5 days/week) × ($18/hour)	$= -\$3,600$
Overhead		$= -\$8,500$
Profit		$= \$2,490$

Manufacturing the product mix of 60 A, 70 B, 80 C, and 100 D will yield a profit of $2,490 per week.

DECISION POINT

By focusing on the bottleneck resources in accepting customer orders and determining the product mix, the sequence in which products are selected for production is reversed from **B, A, C, D** to **D, C, A, B**. Consequently, the product mix is changed from 60 A, 80 B, 40 C, and 100 D to 60 A, 70 B, 80 C, and 100 D. The increase in profits by using the bottleneck method is $930, ($2,490 − $1,560), or almost 60 percent over the traditional approach.

Linear programming (see Supplement D) could also be used to find the best product mix in Example 7.3. It must be noted, however, that the problem in Example 7.3 did not involve significant setup times. Otherwise, they must be taken into consideration for not only identifying the bottleneck, but also in determining the product mix. The experiential learning exercise of Min-Yo Garment Company at the end of this chapter provides an interesting illustration of how the product mix can be determined when setup times are significant. In this way, the principles behind the theory of constraints can be exploited for making better decisions about a firm's most profitable product mix.

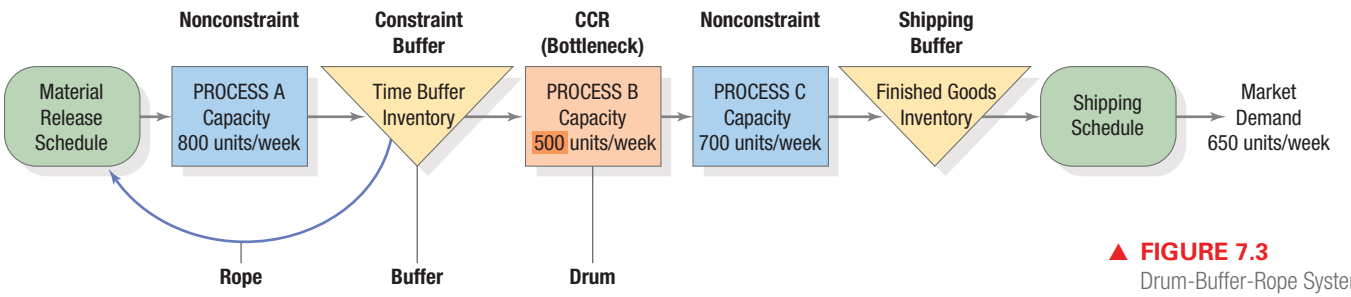
drum-buffer-rope (DBR)

A planning and control system that regulates the flow of work-in-process materials at the bottleneck or the capacity constrained resource (CCR) in a productive system.

Drum-Buffer-Rope Systems **Drum-Buffer-Rope (DBR)** is a planning and control system based on the theory of constraints that is often used in manufacturing firms to plan and schedule production. It works by regulating the flow of work-in-process materials at the bottleneck or the capacity constrained resource (CCR). The bottleneck schedule is the *drum* because it sets the beat or the production rate for the entire plant and is linked to the market demand. The *buffer* is a time buffer that plans early flows to the bottleneck and thus protects it from disruption. It also ensures that the bottleneck is never starved for work. A finished-goods inventory buffer can also be placed in front of the shipping point in order to protect customer shipping schedules. Finally, the *rope* represents the tying of material release to the drum beat, which is the rate at which the bottleneck controls the throughput of the entire plant. It is thus a communication device to ensure

that raw material is not introduced into the system at a rate faster than what the bottleneck can handle. Completing the loop, *buffer management* constantly monitors the execution of incoming bottleneck work. Working together, the drum, the buffer, and the rope can help managers create a production schedule that reduces lead times and inventories while simultaneously increasing throughput and on-time delivery.

To better understand the drum-buffer-rope system, consider the schematic layout shown in Figure 7.3. Process B, with a capacity of only 500 units per week, is the bottleneck because the upstream Process A and downstream Process C have capacities of 800 units per week and 700 units per week, respectively, and the market demand is 650 units per week, on average. In this case, because the capacity at process B is less than the market demand, it is the bottleneck. A constraint time buffer, which can be in the form of materials arriving earlier than needed, is placed right in front of the bottleneck (Process B). A shipping buffer, in the form of finished goods inventory, can also be placed prior to the shipping schedule in order to protect customer orders that are firm. Finally, a rope ties the material release schedule to match the schedule, or drum beat, at the bottleneck. The material flow is pulled forward by the drum beat prior to the bottleneck, while it is pushed downstream toward the customer subsequent to the bottleneck.



▲ **FIGURE 7.3**
Drum-Buffer-Rope System

DBR specifically strives to improve throughput by better utilizing the bottleneck resource and protecting it from disruption through the time buffer and protective buffer capacity elsewhere. So while the process batch in the DBR is any size that minimizes setups and improves utilization at the bottleneck, at nonconstrained resources the process batches are equal to what is needed for production at that time. The material can consequently be released in small batches known as transfer batches at the release point, which then combine at the constraint buffer to make a full process batch at the bottleneck. Transfer batches can be as small as one unit each, to allow a downstream workstation to start work on a batch before it is completely finished at the prior process. Using transfer batches typically facilitates a reduction in overall lead time.

DBR can be an effective system to use when the product the firm produces is relatively simple and the production process has more line flows. Planning is greatly simplified in this case and primarily revolves around scheduling the constrained resource and triggering other points to meet that bottleneck's schedule. Effectively implementing a DBR system requires an understanding of the TOC principles. However, such a system can be utilized in many different kinds of manufacturing and service organizations, either by itself or in conjunction with other planning and control systems. Managerial Practice 7.2 illustrates how the use of a DBR system improved the performance of the Marine Corps Maintenance Center in Albany, Georgia.

MANAGERIAL PRACTICE 7.1 The Drum-Buffer-Rope System at a U.S. Marine Corps Maintenance Center

The U.S. Marine Corps Maintenance Center in Albany, Georgia, overhauls and repairs vehicles used by the corps, such as fuel tankers, trucks, earthmoving equipment, amphibious vehicles, and light armored vehicles. The overhaul process starts with the disassembly of each vehicle to determine the amount and nature of work that needs to be performed. The type and duration of the repair work can vary tremendously for even the same type of vehicle. Faced with such uncertainty, the Center was struggling until four years ago to complete its equipment repairs on time, and it had an increasing backlog to boot. For instance, the Center was able to repair only about 5 MK-48s (heavy-duty haulers) per month, when twice as many MK-48s—10 per month—typically needed repair. Different units of the corps were threatening to divert their orders to the private-sector repair companies.

TOC principles were used to identify the bottlenecks on the shop floor. After the Center's operations were studied in depth, however, contrary to everyone's expectations, it was discovered that more than enough capacity was available to repair and overhaul 10 MK-48s per month. The problem was not capacity; it was the Center's scheduling system. Products were being pushed onto the shop floor without regard for the status of the resources on the floor. Thus, what the Center had was a policy constraint related to the scheduling process, not an actual physical resource constraint.

In order to improve the Center's performance, its managers implemented a simplified form of a drum-buffer-rope system as shown in Figure 7.3. Since the Marine Corps Maintenance Center was not constrained by any internal resource, the drum in such a simplified system was based on firm orders. As orders came in, a quick check was done to measure the total load the Center's least-capable resource was handling. If the resource was not too heavily loaded, the order was accepted and released onto the shop floor for processing. The rope tied the shipping schedule directly to the material release schedule instead of the bottleneck schedule, and the only buffer maintained was the shipping buffer. Such a simplified DBR system did not require any specialized software. It focused simply on the market demand for repairs.

The Center's results following the change were impressive. Repair cycle times were reduced from an average of 167 days to 58 days, work-in-process levels were reduced from 550 percent of demand to 140 percent, and the cost to repair products went down by 25 to 30 percent due to an increased throughput. The Center's ability to repair MK-48s became much more flexible, too. In fact, it can now repair as many as 23 MK-48s per month. The Center is on schedule for 99 percent of the production lines where the TOC principles have been implemented, and the repair costs have decreased by 25 percent. Carrying out these simple improvements made the Albany Maintenance Center a world-class overhaul and repair operation.



Marti Gatlin/U.S. Marine Corps

Repairs to assault vehicles can vary tremendously at the U.S. Marine Corps Maintenance Center in Albany, Georgia. The center struggled to keep up with its repairs until managers implemented the simplified form of a drum-buffer-rope system. The result? Repair times fell from 167 days to just 58 days, on average.

Source: Mandyam Srinivasan, Darren Jones, and Alex Miller, "Applying Theory of Constraints Principles and Lean Thinking at the Marine Corps Maintenance Center," *Defense Acquisition Review Journal,* August–November 2004; M. Srinivasan, Darren Jones, and Alex Miller, "Corps Capabilities," APICS Magazine (March 2005), pp. 46–50.

Managing Constraints in a Line Process

As noted in Chapter 3, "Developing a Process Strategy," products created by a line process include the assembly of computers, automobiles, appliances, and toys. Such assembly lines can exist in providing services as well. For instance, putting together a standardized hamburger with a fixed sequence of steps is akin to operating an assembly line. While the product mix or demand volumes do not change as rapidly for line processes as for job or batch processes, the load can shift between work centers in a line as the end product being assembled is changed or the total output rate of the line is altered. Constraints arising out of such actions can be managed by balancing the workload between different stations in a line, which we explain next in greater detail.

Line Balancing

line balancing
The assignment of work to stations in a line process so as to achieve the desired output rate with the smallest number of workstations.

Line balancing is the assignment of work to stations in a line process so as to achieve the desired output rate with the smallest number of workstations. Normally, one worker is assigned to a station. Thus, the line that produces at the desired pace with the fewest workers is the most efficient one. Achieving this goal is much like the theory of constraints, because both approaches are concerned about bottlenecks. Line balancing differs in how it addresses bottlenecks. Rather than (1) taking on new customer orders to best use bottleneck capacity or (2) scheduling so that bottleneck resources are conserved, line balancing takes a third approach. It (3) creates workstations with workloads as evenly balanced as possible. It seeks to create workstations so that the capacity utilization for the bottleneck is not much higher than for the other workstations in the line. Another difference is that line balancing applies only to line processes that do assembly work, or to work that can be bundled in many ways to create the jobs for each workstation in the line. The latter situation can be found both in manufacturing and service settings.

Line balancing must be performed when a line is set up initially, when a line is rebalanced to change its hourly output rate, or when a product or process changes. The goal is to obtain workstations with well-balanced workloads (e.g., every station takes roughly 3 minutes per customer in a cafeteria line with different food stations).

work elements
The smallest units of work that can be performed independently.

immediate predecessors
Work elements that must be done before the next element can begin.

The analyst begins by separating the work into **work elements**, which are the smallest units of work that can be performed independently. The analyst then obtains the time standard for each element and identifies the work elements, called **immediate predecessors**, which must be done before the next element can begin.

Precedence Diagram Most lines must satisfy some technological precedence requirements; that is, certain work elements must be done before the next can begin. However, most lines also allow for some latitude and more than one sequence of operations. To help you better visualize immediate predecessors, let us run through the construction of a **precedence diagram**.[2] We denote the work elements by circles, with the time required to perform the work shown below each circle. Arrows lead from immediate predecessors to the next work element. Example 7.4 illustrates a manufacturing process, but a back office line-flow process in a service setting can be approached similarly.

precedence diagram
A diagram that allows one to visualize immediate predecessors better; work elements are denoted by circles, with the time required to perform the work shown below each circle.

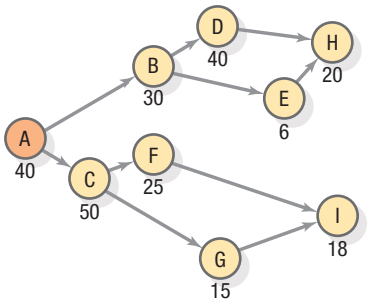
EXAMPLE 7.4 Constructing a Precedence Diagram

Green Grass, Inc., a manufacturer of lawn and garden equipment, is designing an assembly line to produce a new fertilizer spreader, the Big Broadcaster. Using the following information on the production process, construct a precedence diagram for the Big Broadcaster.

Work Element	Description	Time (sec)	Immediate Predecessor(s)
A	Bolt leg frame to hopper	40	None
B	Insert impeller shaft	30	A
C	Attach axle	50	A
D	Attach agitator	40	B
E	Attach drive wheel	6	B
F	Attach free wheel	25	C
G	Mount lower post	15	C
H	Attach controls	20	D, E
I	Mount nameplate	18	F, G
		Total 244	

SOLUTION

Figure 7.4 shows the complete diagram. We begin with work element A, which has no immediate predecessors. Next, we add elements B and C, for which element A is the only immediate predecessor. After entering time standards and arrows showing precedence, we add elements D and E, and so on. The diagram simplifies interpretation. Work element F, for example, can be done anywhere on the line after element C is completed. However, element I must await completion of elements F and G.



◀ **FIGURE 7.4**
Precedence Diagram for Assembling the Big Broadcaster

DECISION POINT

Management now has enough information to develop a line-flow layout that clusters work elements to form workstations, with a goal being to balance the workloads and, in the process, minimize the number of workstations required.

Desired Output Rate The goal of line balancing is to match the output rate to the staffing or production plan. For example, if the plan calls for 4,800 units or customers per week and the line operates 80 hours per week, the desired output rate ideally would be 60 units or customers (4,800/80) per hour. Matching output to the plan ensures on-time delivery and prevents buildup of unwanted inventory or customer delays. However, managers should avoid rebalancing a line too frequently because each time a line is rebalanced many workers' jobs on the line must be redesigned, temporarily hurting productivity and sometimes even requiring a new detailed layout for some stations.

[2]Precedence relationships and precedence diagrams are also important in the entirely different context of project management covered in Chapter 2.

Cycle Time After determining the desired output rate for a line, the analyst can calculate the line's cycle time. A line's **cycle time** is the maximum time allowed for work on a unit at each station.[3] If the time required for work elements at a station exceeds the line's cycle time, the station will be a bottleneck, preventing the line from reaching its desired output rate. The target cycle time is the reciprocal of the desired hourly output rate:

cycle time
The maximum time allowed for work on a unit at each station.

$$c = \frac{1}{r}$$

where

$c =$ cycle time in hours per unit

$r =$ desired output rate in units per hour

For example, if the line's desired output rate is 60 units per hour, the cycle time is $c = 1/60$ hour per unit, or 1 minute.

Theoretical Minimum To achieve the desired output rate, managers use line balancing to assign every work element to a station, making sure to satisfy all precedence requirements and to minimize the number of stations, n, formed. If each station is operated by a different worker, minimizing n also maximizes worker productivity. Perfect balance is achieved when the sum of the work-element times at each station equals the cycle time, c, and no station has any idle time. For example, if the sum of each station's work-element times is 1 minute, which is also the cycle time, the line achieves perfect balance. Although perfect balance usually is unachievable in practice, owing to the unevenness of work-element times and the inflexibility of precedence requirements, it sets a benchmark, or goal, for the smallest number of stations possible. The **theoretical minimum (TM)** for the number of stations is

theoretical minimum (TM)
A benchmark or goal for the smallest number of stations possible, where the total time required to assemble each unit (the sum of all work-element standard times) is divided by the cycle time.

$$\text{TM} = \frac{\Sigma t}{c}$$

where

$\Sigma t =$ total time required to assemble each unit (the sum of all work-element standard times)

$c =$ cycle time

For example, if the sum of the work-element times is 15 minutes and the cycle time is 1 minute, TM = 15/1, or 15 stations. Any fractional values obtained for TM are rounded up because fractional stations are impossible.

Idle Time, Efficiency, and Balance Delay Minimizing n automatically ensures (1) minimal idle time, (2) maximal efficiency, and (3) minimal balance delay. Idle time is the total unproductive time for all stations in the assembly of each unit:

$$\text{Idle time} = nc - \Sigma t$$

where

$n =$ number of stations

$c =$ cycle time

$\Sigma t =$ total standard time required to assemble each unit

Efficiency is the ratio of productive time to total time, expressed as a percent:

$$\text{Efficiency (\%)} = \frac{\Sigma t}{nc}(100)$$

Balance delay is the amount by which efficiency falls short of 100 percent:

balance delay
The amount by which efficiency falls short of 100 percent.

$$\text{Balance delay (\%)} = 100 - \text{Efficiency}$$

As long as c is fixed, we can optimize all three goals by minimizing n.

[3]Except in the context of line balancing, *cycle time* has a different meaning. It is the elapsed time between starting and completing a job. Some researchers and practitioners prefer the term *lead time*.



Nataliya Hora/Shutterstock.com

Skoda Automotive assembly line in Mlada Boleslav. Since 1991, Skoda has been a part of Volkswagen for over 20 years.

EXAMPLE 7.5 Calculating the Cycle Time, Theoretical Minimum, and Efficiency

Green Grass's plant manager just received marketing's latest forecasts of Big Broadcaster sales for the next year. She wants its production line to be designed to make 2,400 spreaders per week for at least the next 3 months. The plant will operate 40 hours per week.

a. What should be the line's cycle time?

b. What is the smallest number of workstations that she could hope for in designing the line for this cycle time?

c. Suppose that she finds a solution that requires only five stations. What would be the line's efficiency?

MyOMLab

Tutor 7.1 in MyOMLab provides another example to calculate these line-balancing measures.

SOLUTION

a. First, convert the desired output rate (2,400 units per week) to an hourly rate by dividing the weekly output rate by 40 hours per week to get $r = 60$ units per hour. Then, the cycle time is

$$c = 1/r = 1/60 \text{ (hour/unit)} = 1 \text{ minute/unit} = 60 \text{ seconds/unit}$$

b. Now, calculate the theoretical minimum for the number of stations by dividing the total time, Σt, by the cycle time, $c = 60$ seconds. Assuming perfect balance, we have

$$\text{TM} = \frac{\Sigma t}{c} = \frac{244 \text{ seconds}}{60 \text{ seconds}} = 4.067 \text{ or } 5 \text{ stations}$$

c. Now, calculate the efficiency of a five-station solution, assuming for now that one can be found:

$$\text{Efficiency (\%)} = \frac{\Sigma t}{nc}(100) = \frac{244}{5(60)}(100) = 81.3\%$$

DECISION POINT

If the manager finds a solution with five stations that satisfies all precedence constraints, then that is the optimal solution; it has the minimum number of stations possible. However, the efficiency (sometimes called the *theoretical maximum efficiency*) will be only 81.3 percent. Perhaps the line should be operated less than 40 hours per week (thereby adjusting the cycle time) and the employees transferred to other kinds of work when the line does not operate.

Finding a Solution Often, many assembly-line solutions are possible, even for such simple problems as Green Grass's. The goal is to cluster the work elements into workstations so that (1) the number of workstations required is minimized, and (2) the precedence and cycle-time requirements are not violated. The idea is to assign work elements to workstations subject to the precedence requirements so that the work content for the station is equal (or nearly so, but less than) the cycle time for the line. In this way, the number of workstations will be minimized.

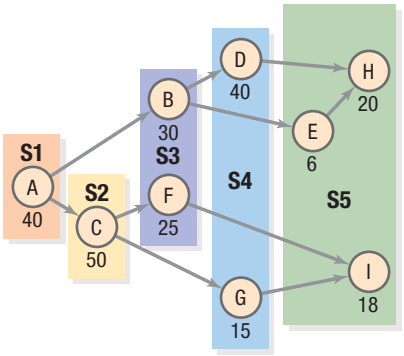
Here we use the trial-and-error method to find a solution, although commercial software packages are also available. Most of these packages use different decision rules in picking which work element to assign next to a workstation being created. The ones used by POM for Windows are described in Table 7.3. The solutions can be examined for improvement, because there is no guarantee that they are optimal or even feasible. Some work elements cannot be assigned to the same station, some changes can be made to reduce the number of stations, or some shifts can provide better balance between stations.

TABLE 7.3 HEURISTIC DECISION RULES IN ASSIGNING THE NEXT WORK ELEMENT TO A WORKSTATION BEING CREATED

Create one station at a time. For the station now being created, identify the unassigned work elements that qualify for assignment: They are candidates if

1. **All of their predecessors have been assigned to this station or stations already created.**
2. **Adding them to the workstation being created will not create a workload that exceeds the cycle time.**

Decision Rule	Logic
Longest work element	Picking the candidate with the longest time to complete is an effort to fit in the most difficult elements first, leaving the ones with short times to "fill out" the station.
Shortest work element	This rule is the opposite of the longest work element rule because it gives preference in workstation assignments to those work elements that are quicker. It can be tried because no single rule guarantees the best solution. It might provide another solution for the planner to consider.
Most followers	When picking the next work element to assign to a station being created, choose the element that has the most *followers* (due to precedence requirements). In Figure 7.4, item C has three followers (F, G, and I) whereas item D has only one follower (H). This rule seeks to maintain flexibility so that good choices remain for creating the last few workstations at the end of the line.
Fewest followers	Picking the candidate with the fewest followers is the opposite of the most followers rule.



▲ **FIGURE 7.5**
Big Broadcaster Precedence Diagram Solution

Figure 7.5 shows a solution that creates just five workstations. We know that five is the minimum possible, because five is the theoretical minimum found in Example 5. All of the precedence and cycle-time requirements are also satisfied. Consequently, the solution is optimal for this problem. Each worker at each station must perform the work elements in the proper sequence. For example, workstation S5 consists of one worker who will perform work elements E, H, and I on each unit that comes along the assembly line. The processing time per unit is 44 seconds (6 + 20 + 18) which does not exceed the cycle time of 60 seconds (see Example 5). Furthermore, the immediate predecessors of these three work elements are assigned to this workstation or upstream workstations, so their precedence requirements are satisfied. The worker at workstation S5 can do element I at any time but will not start element H until element E is finished.

Managerial Considerations

In addition to balancing a line for a given cycle time, managers have four other considerations: (1) pacing, (2) behavioral factors, (3) number of models produced, and (4) different cycle times.

Pacing The movement of product from one station to the next as soon as the cycle time has elapsed is called **pacing**. Pacing manufacturing processes allows materials handling to be automated and requires less inventory storage area. However, it is less flexible in handling unexpected delays that require either slowing down the entire line or pulling the unfinished work off the line to be completed later.

pacing
The movement of product from one station to the next as soon as the cycle time has elapsed.

Behavioral Factors The most controversial aspect of line-flow layouts is behavioral response. Studies show that installing production lines increases absenteeism, turnover, and grievances.

Paced production and high specialization (say, cycle times of less than 2 minutes) lower job satisfaction. Workers generally favor inventory buffers as a means of avoiding mechanical pacing. One study even showed that productivity increased on unpaced lines.

Number of Models Produced A line that produces several items belonging to the same family is called a **mixed-model line**. In contrast, a single-model line produces one model with no variations. Mixed-model production enables a plant to achieve both high-volume production *and* product variety. However, it complicates scheduling and increases the need for good communication about the specific parts to be produced at each station.

mixed-model line
A production line that produces several items belonging to the same family.

Cycle Times A line's cycle time depends on the desired output rate (or sometimes on the maximum number of workstations allowed). In turn, the maximum line efficiency varies considerably with the cycle time selected. Thus, exploring a range of cycle times makes sense. A manager might go with a particularly efficient solution even if it does not match the desired output rate. The manager can compensate for the mismatch by varying the number of hours the line operates through overtime, extending shifts, or adding shifts. Multiple lines might even be the answer.

LEARNING GOALS IN REVIEW

1. **Explain the theory of constraints.** Constraints or bottlenecks can exist in the form of internal resources or market demand in both manufacturing and service organizations, and in turn play an important role in determining system performance. See the section on "The Theory of Constraints (TOC)," pp. 265–267. Review opening vignette on The Radcliffe Infirmary for an application of TOC.
2. **Understand linkage of capacity constraints to financial performance measures**. Review and understand Table 7.1, p. 265.
3. **Identify bottlenecks.** The TOC provides guidelines on how to identify and manage constraints. The section "Identification and Management of Bottlenecks," pp. 267–274, shows you how to identify bottlenecks in both service as well as manufacturing firms.
4. **Apply theory of constraints to product mix decisions**. Review Example 7.3, pp. 270–273, to understand how using a bottleneck based method for allocating resources and determining the product mix leads to greater profits.
5. **Describe how to manage constraints in an assembly line**. Assembly line balancing, as a special form of a constraint in managing a line process within both manufacturing and services, can also be an effective mechanism for matching output to a plan and running such processes more efficiently. The section "Managing Constraints in a Line Process," pp. 274–279, shows you how to balance assembly lines and create work stations. Review Solved Problem 1, pp. 280–281, for an application of line balancing principles.

MyOMLab helps you develop analytical skills and assesses your progress with multiple problems on processing time for average customer, bottleneck activity, maximum customers served at bottleneck per hour, average capacity of system, theoretical maximum of stations in an assembly line, longest work element decision rule, line's efficiency, and cycle time.

MyOMLab Resources	Titles	Link to the Book
Video	*Constraint Management at Southwest Airlines* *1st Bank Villa Italia: Waiting Lines*	Managing Constraints Across the Organization; The Theory of Constraints Identification and Management of Bottlenecks
OM Explorer Solvers	Min-Yo Garment Company Spreadsheet	Estimate Capacity Requirements; Example 6.1 (pp. 229–230); Managing Bottlenecks in Manufacturing Processes; Example 7.2 (pp. 269–270)
OM Explorer Tutors	7.1 Line Balancing	Line Balancing; Example 7.5 (p. 277); Solved Problem 1 (pp. 280–281)
POM for Windows	Line Balancing	Line Balancing; Example 7.5 (p. 277); Solved Problem 1 (pp. 280–281)
SimQuick Simulation Exercises	Simulating process of making jewelry boxes and making choice on investing in new machine	The Theory of Constraints; Example 7.1 (pp. 267–268)
Internet Exercise	Granite Rock and Chevron	The Theory of Constraints; Identification and Management of Bottlenecks
Key Equations		
Image Library		

Key Equations

1. Cycle time: $c = \frac{1}{r}$
2. Theoretical minimum number of workstations: $\text{TM} = \frac{\Sigma t}{c}$
3. Idle time: $nc - \Sigma t$
4. Efficiency(%): $\frac{\Sigma t}{nc}(100)$
5. Balance delay (%): 100 − Efficiency

Key Terms

balance delay 276
bottleneck 264
constraint 264
cycle time 276
drum-buffer-rope (DBR) 272
immediate predecessors 274
line balancing 274
mixed-model line 279
pacing 278
precedence diagram 275
theoretical minimum (TM) 276
theory of constraints (TOC) 265
throughput time 267
work elements 274

Solved Problem 1

A company is setting up an assembly line to produce 192 units per 8-hour shift. The following table identifies the work elements, times, and immediate predecessors:

Work Element	Time (Sec)	Immediate Predecessor(s)
A	40	None
B	80	A
C	30	D, E, F
D	25	B
E	20	B
F	15	B
G	120	A
H	145	G
I	130	H
J	115	C, I
	Total 720	

a. What are the efficiency and balance delay of the solution found?
b. Use trial and error to work out a solution, and show your solution on a precedence diagram.
c. What is the theoretical minimum number of stations?
d. What is the desired cycle time (in seconds)?

SOLUTION

a. The sum of the work-element times is 720 seconds, so

$$\text{TM} = \frac{\Sigma t}{c} = \frac{720 \text{ seconds/unit}}{150 \text{ seconds/unit-station}} = 4.8 \text{ or } 5 \text{ stations}$$

which may not be achievable.

b. Substituting in the cycle-time formula, we get

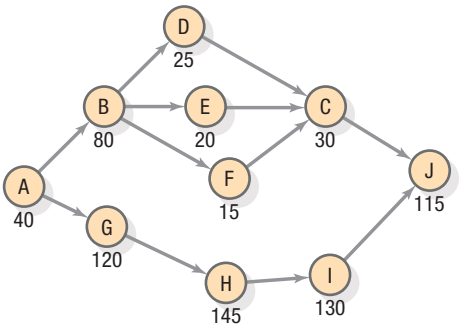
$$c = \frac{1}{r} = \frac{8 \text{ hours}}{192 \text{ units}}(3{,}600 \text{ seconds/hour}) = 150 \text{ seconds/unit}$$

c. The precedence diagram is shown in Figure 7.6. Each row in the following table shows work elements assigned to each of the five workstations in the proposed solution.

d. Calculating the efficiency, we get

$$\text{Efficiency} = \frac{\Sigma t}{nc}(100) = \frac{720 \text{ seconds/unit}}{5[150 \text{ seconds/unit}]} = 96\%$$

Thus, the balance delay is only 4 percent (100–96).

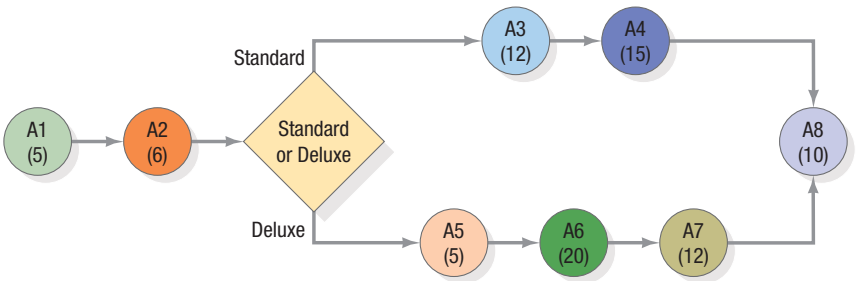


▲ **FIGURE 7.6**
Precedence Diagram

Station	Candidate(s)	Choice	Work-Element Time (Sec)	Cumulative Time (Sec)	Idle Time (c = 150 Sec)
S1	A	A	40	40	110
	B	B	80	120	30
	D, E, F	D	25	145	5
S2	E, F, G	G	120	120	30
	E, F	E	20	140	10
S3	F, H	H	145	145	5
S4	F, I	I	130	130	20
	F	F	15	145	5
S5	C	C	30	30	120
	J	J	115	145	5

Solved Problem 2

Bill's Car Wash offers two types of washes: Standard and Deluxe. The process flow for both types of customers is shown in the following chart. Both wash types are first processed through steps A1 and A2. The Standard wash then goes through steps A3 and A4 while the Deluxe is processed through steps A5, A6, and A7. Both offerings finish at the drying station (A8). The numbers in parentheses indicate the minutes it takes for that activity to process a customer.



a. If 60 percent of the customers are Standard and 40 percent are Deluxe, what is the average capacity of the car wash in customers per hour?

b. What is the capacity (measured as customers served per hour) of Bill's Car Wash to process Standard and Deluxe customers? Assume that no customers are waiting at step A1, A2, or A8.

c. Which step is the bottleneck for the Standard car wash process? For the Deluxe car wash process?

d. Where would you expect Standard wash customers to experience waiting lines, assuming that new customers are always entering the shop and that no Deluxe customers are in the shop? Where would the Deluxe customers have to wait, assuming no Standard customers?

SOLUTION

a. Step A4 is the bottleneck for the Standard car wash process, and Step A6 is the bottleneck for the Deluxe car wash process, because these steps take the longest time in the flow.

b. Standard wash customers would wait before steps A1, A2, A3, and A4 because the activities that immediately precede them have a higher rate of output (i.e., smaller processing times). Deluxe wash customers would experience a wait in front of steps A1, A2, and A6 for the same reasons. A1 is included for both types of washes because the arrival rate of customers could always exceed the capacity of A1.

c. The average capacity of the car wash is $(0.60 \times 4) + (0.40 \times 3) = 3.6$ customers per hour.

d. The capacity for Standard washes is 4 customers per hour because the bottleneck step A4 can process 1 customer every 15 minutes (60/15). The capacity for Deluxe car washes is 3 customers per hour (60/20). These capacities are derived by translating the "minutes per customer" of each bottleneck activity to "customers per hour."

Discussion Questions

1. Take a process that you encounter on a daily basis, such as the lunch cafeteria or the journey from your home to school/work, and identify the bottlenecks that limit the throughput of this process.
2. Using the same process as in question 1, identify conditions that would lead to the bottlenecks changing or shifting away from the existing bottleneck.
3. How could the efficiency of the redesigned process be improved further?

Problems

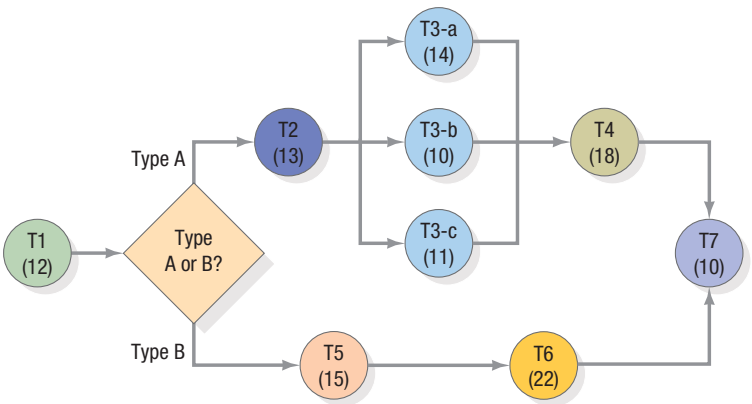
The OM Explorer and POM for Windows software is available to all students using the 10th edition of this textbook. Go to **www.pearsonglobaleditions.com/krajewski** to download these computer packages. If you purchased MyOMLab, you also have access to Active Models software and significant help in doing the following problems. Check with your instructor on how best to use these resources. In many cases, the instructor wants you to understand how to do the calculations by hand. At the least, the software provides a check on your calculations. When calculations are particularly complex and the goal is interpreting the results in making decision, the software entirely replaces the manual calculations.

1. Figure 7.7 details the process flow for two types of customers who enter Barbara's Boutique shop for customized dress alterations. After step T1, Type A customers proceed to step T2 and then to any of the three workstations at T3, followed by steps T4 and T7. After step T1, Type B customers proceed to step T5 and then steps T6 and T7. The numbers in parentheses are the minutes it takes to process a customer.

   a. What is the capacity of Barbara's shop in terms of the numbers of Type A customers who can be served in an hour? Assume no customers are waiting at steps T1 or T7.

   b. If 30 percent of the customers are Type A customers and 70 percent are Type B customers, what is the average capacity of Barbara's shop in customers per hour?

   c. Assuming that the arrival rate is greater than five customers per hour, when would you expect Type A customers to experience waiting lines, assuming no Type B customers in the shop? Where would the Type B customers have to wait, assuming no Type A customers?



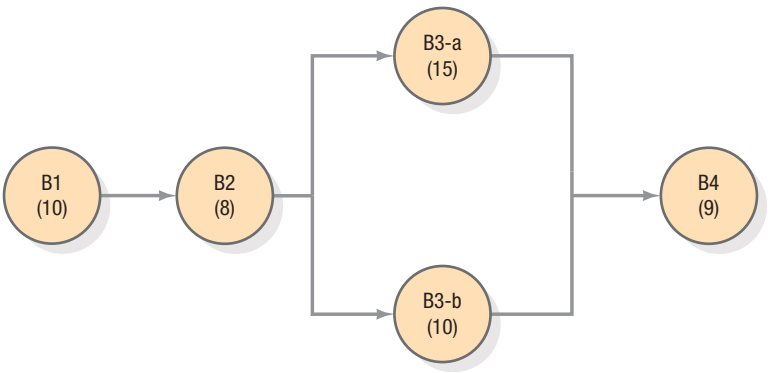
◀ **FIGURE 7.7**
Process Flow for Barbara's Boutique Customers

2. Bill's Barbershop has two barbers available to cut customers' hair. Both barbers provide roughly the same experience and skill, but one is just a little bit slower than the other. The process flow in Figure 7.8 shows that all customers go through steps B1 and B2 and then can be served at either of the two barbers at step B3. The process ends for all customers at step B4. The numbers in parentheses indicate the minutes it takes that activity to process a customer.

a. How long does it take the average customer to complete this process?

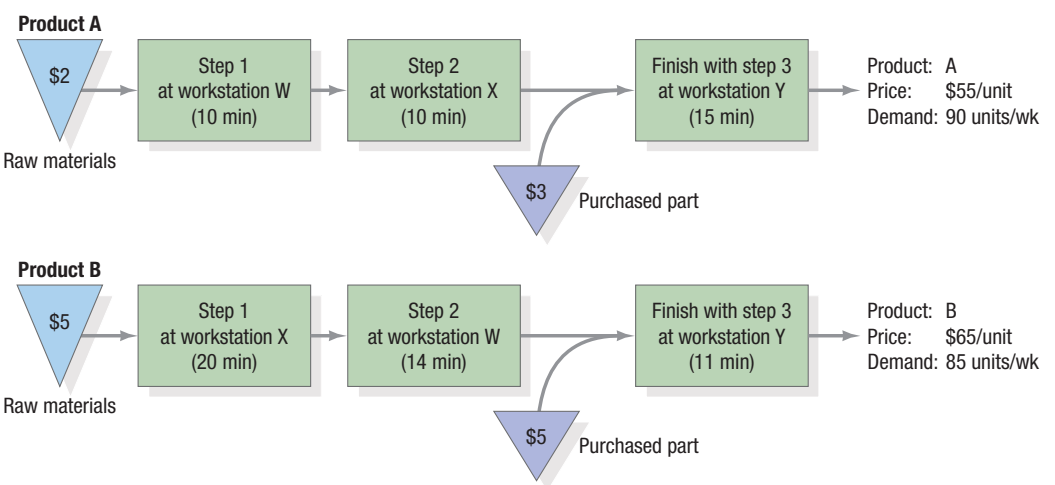
b. What single activity is the bottleneck for the entire process?

c. How many customers can this process serve in an hour?



▲ **FIGURE 7.8**
Process Flow for Bill's Barbershop

3. Canine Kernels Company (CKC) manufactures two different types of dog chew toys (A and B, sold in 1,000-count boxes) that are manufactured and assembled on three different workstations (W, X, and Y) using a small-batch process (see Figure 7.9). Batch setup times are negligible. The flowchart denotes the path each product follows through the manufacturing process, and each product's price, demand per week, and processing times per unit are indicated as well. Purchased parts and raw materials consumed during production are represented by inverted triangles. CKC can make and sell up to the limit of its demand per week; no penalties are incurred for not being able to meet all the demand. Each workstation is staffed by a worker who is dedicated to work on that workstation alone and is paid $6 per hour. Total labor costs per week are fixed. Variable overhead costs are $3,500/week. The plant operates one 8-hour shift per day, or 40 hours/week. Which of the three workstations, W, X, or Y, has the highest aggregate workload, and thus serves as the bottleneck for CKC?



▲ **FIGURE 7.9**
Flowchart for Canine Kernels Company (CKC)

4. Johnson Cogs wants to set up a line to serve 60 customers per hour. The work elements and their precedence relationships are shown in the following table.

a. What is the theoretical minimum number of stations?

b. How many stations are required using the longest work element decision rule?

c. Suppose that a solution requiring five stations is obtained. What is its efficiency?

Work Element	Time (Sec)	Immediate Predecessor(s)
A	40	None
B	30	A
C	50	A
D	40	B
E	6	B
F	25	C
G	15	C
H	20	D, E
I	18	F, G
J	30	H, I
	Total 274	

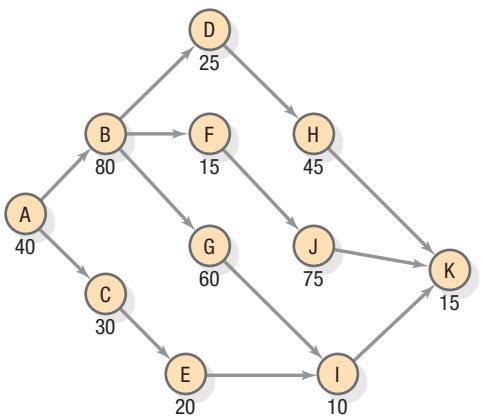
5. Use the longest work element rule to balance the assembly line described in the following table and Figure 7.10 so that it will produce 40 units per hour.

a. What is the cycle time?

b. What is the theoretical minimum number of workstations?

c. Which work elements are assigned to each workstation?

d. What are the resulting efficiency and balance delay percentages?



▲ **FIGURE 7.10**
Precedence Diagram

e. Use the shortest work element rule to balance the assembly line. Do you note any changes in solution?

Work Element	Time (Sec)	Immediate Predecessor(s)
A	40	None
B	80	A
C	30	A
D	25	B
E	20	C
F	15	B
G	60	B
H	45	D
I	10	E, G
J	75	F
K	15	H, I, J
	Total 415	

6. The senior management at Canine Kernels Company (CKC) is concerned with the existing capacity limitation, so they want to accept the mix of orders that maximizes the company's profits. Traditionally, CKC has utilized a method whereby decisions are made to produce as much of the product with the highest contribution margin as possible (up to the limit of its demand), followed by the next highest contribution margin product, and so on until no more capacity is available. Because capacity is limited, choosing the proper product mix is crucial. Troy Hendrix, the newly hired production supervisor, is an avid follower of the theory of constraints philosophy and the bottleneck method for scheduling. He believes that profitability can indeed be approved if bottleneck resources are exploited to determine the product mix.

a. What is the profit if the traditional contribution margin method is used for determining CKC's product mix?

b. What is the profit if the bottleneck method advocated by Troy is used for selecting the product mix?

c. Calculate the profit gain, both in absolute dollars as well as in terms of percentage gains, by using TOC principles for determining product mix.

7. The *trim line* at PW is a small subassembly line that, along with other such lines, feeds into the final chassis line. The entire assembly line, which consists of more than 900 workstations, is to make PW's new E cars. The trim line itself involves only 13 work elements and must handle 20 cars per hour. Work-element data are as follows:

Work Element	Time (Sec)	Immediate Predecessor(s)
A	1.8	None
B	0.4	None
C	1.6	None
D	1.5	A
E	0.7	A
F	0.5	E
G	0.8	B
H	1.4	C
I	1.4	D
J	1.4	F, G
K	0.5	H
L	1.0	J
M	0.8	I, K, L

a. Use the most followers work element decision rule to balance the line and calculate the efficiency of your solution.

b. What cycle time (in minutes) results in the desired output rate?

c. What is the theoretical minimum number of stations?

d. Use the longest work element decision rule to balance the line and calculate the efficiency of your solution.

e. Draw a precedence diagram.

8. In order to meet holiday demand, Penny's Pie Shop requires a production line that is capable of producing 50 pecan pies per week, while operating only 40 hours per week. There are only 4 steps required to produce a single pecan pie with respective processing times of 5 min, 5 min, 45 min, and 15 min.

a. What should be the line's cycle time?

b. What is the smallest number of workstations Penny could hope for in designing the line considering this cycle time?

c. Suppose that Penny finds a solution that requires only four stations. What would be the efficiency of this line?

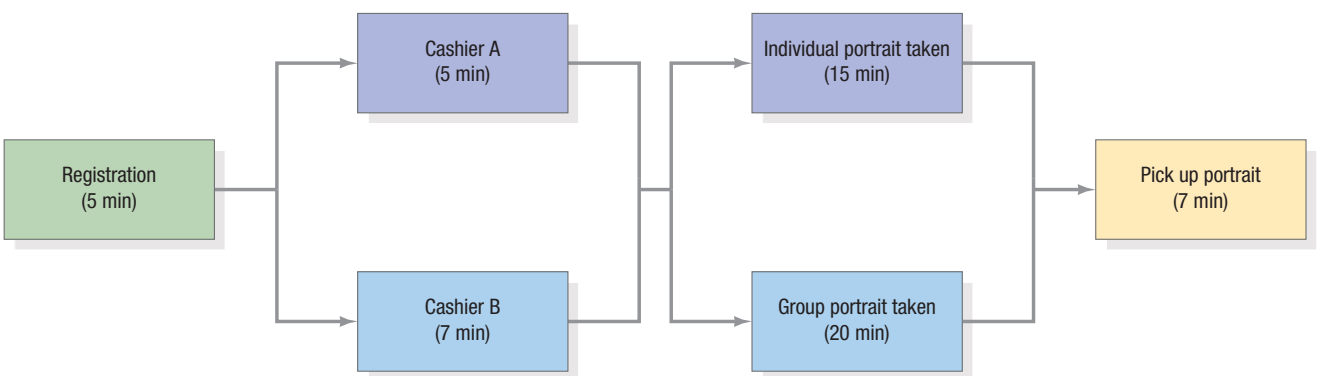
Advanced Problems

9. Melissa's Photo Studio offers both individual and group portrait options. The process flow diagram in Figure 7.11 shows that all customers must first register and then pay at one of two cashiers. Then, depending on whether they want a single or group portrait they go to different rooms. Finally, everyone picks up their own finished portrait.

a. How long does it take to complete the entire process for a group portrait?

b. What single activity is the bottleneck for the entire process, assuming the process receives equal amounts of both groups and individuals?

c. What is the capacity of the bottleneck for both groups and individuals?

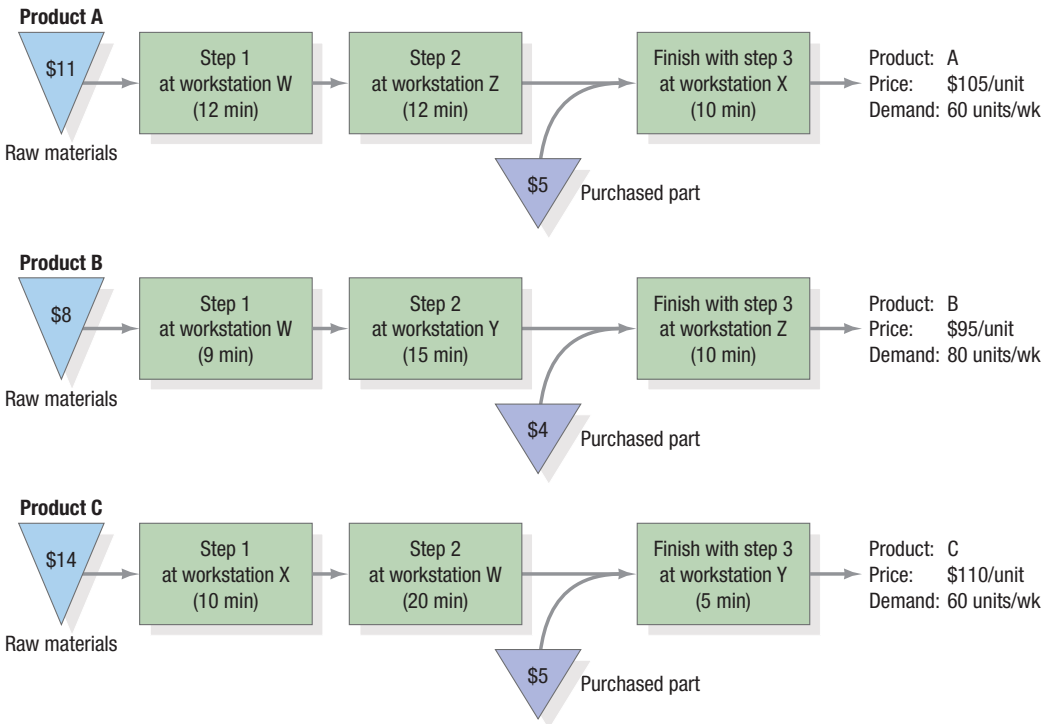


▲ **FIGURE 7.11**
Melissa's Photo Studio

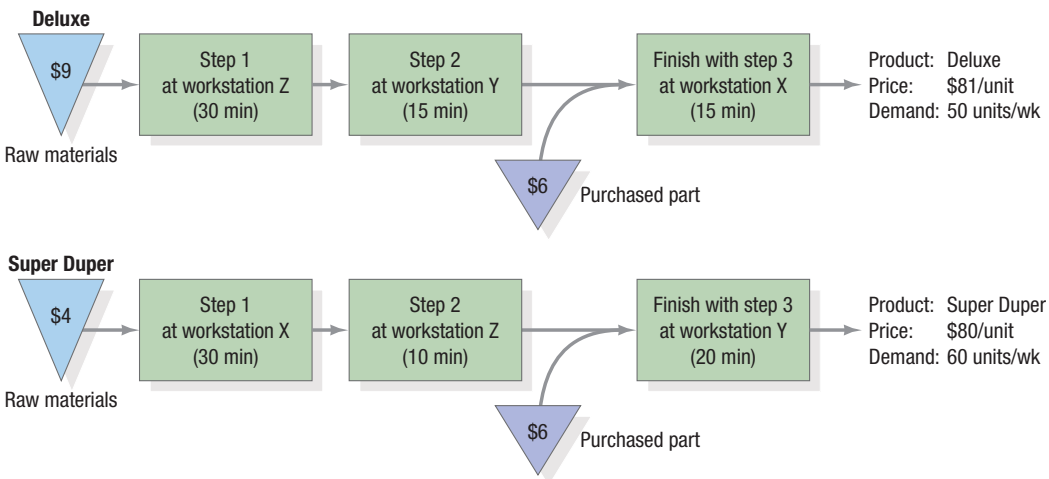
10. Yost-Perry Industries' (YPI) senior management team wants to improve the profitability of the firm by accepting the right set of orders. Currently, decisions are made using the traditional method, which is to accept as much of the highest contribution margin product as possible (up to the limit of its demand), followed by the next highest contribution margin product, and so on until all available capacity is utilized. Because the firm cannot satisfy all the demand, the product mix must be chosen carefully. Jay Perry, the newly promoted production supervisor, is knowledgeable about the theory of constraints and the bottleneck-based method for scheduling. He believes that profitability can indeed be improved if bottleneck resources are exploited to determine the product mix. What is the change in profits if, instead of the traditional method that YPI has used thus far, the bottleneck method advocated by Jay is used for selecting the product mix?

11. Yost-Perry Industries (YPI) manufactures a mix of affordable guitars (A, B, C) that are fabricated and assembled at four different processing stations (W, X, Y, Z). The operation is a batch process with small setup times that can be considered negligible. The product information (price, weekly demand, and processing times) and process sequences are shown in Figure 7.12. Raw materials and purchased parts (shown as a per-unit consumption rate) are represented by inverted triangles. YPI is able to make and sell up to the limit of its demand per week with no penalties incurred for not meeting the full demand. Each workstation is staffed by one highly skilled worker who is dedicated to work on that workstation alone and is paid $15 per hour. The plant operates one 8-hour shift per day and operates on a 5-day work week (i.e., 40 hours of production per person per week). Overhead costs are $9,000/week. Which of the four workstations, W, X, Y, or Z, has the highest aggregate workload, and thus serves as the bottleneck for YPI?

▶ **FIGURE 7.12**
Flowchart for Yost-Perry Industries (YPI)



12. A.J.'s Wildlife Emporium manufactures two unique birdfeeders (Deluxe and Super Duper) that are manufactured and assembled in up to three different workstations (X, Y, Z) using a small batch process. Each of the products is produced according to the flowchart in Figure 7.13. Additionally, the flowchart indicates each product's price, weekly demand, and processing times per unit. Batch setup times are negligible. A.J. can make and sell up to the limit of its weekly demand and there are no penalties for not being able to meet all of the demand. Each



▲ **FIGURE 7.13**
A.J.'s Wildlife Emporium Flowchart

workstation is staffed by a worker who is dedicated to work on that workstation alone and is paid $16 per hour. The plant operates 40 hours per week, with no overtime. Overhead costs are $2,000 per week. Based on the information provided, as well as the information contained in the flowchart, answer the following questions.

a. Using the traditional method, which bases decisions solely on a product's contribution to profits and overhead, what is the optimal product mix and what is the overall profitability?

b. Using the bottleneck-based method, what is the optimal product mix and what is the overall profitability?

13. Cooper River Glass Works (CRGW) produces four different models of desk lamps as shown in Figure 7.14. The operations manager knows that total monthly demand exceeds the capacity available for production. Thus, she is interested in determining the product mix which will maximize profits. Each model's price, routing, processing times, and material cost is provided in Figure 7.14. Demand next month is estimated to be 200 units of model Alpha, 250 units of model Bravo, 150 units of model Charlie, and 225 units of model Delta. CRGW operates only one 8-hour shift per day and is scheduled to work 20 days next month (no overtime). Further, each station requires a 10 percent capacity cushion.

a. Using the bottleneck-based method, what is the optimal product mix and what is the overall profitability?

b. Using the traditional method, which bases decisions solely on a product's contribution to profits and overhead, what is the optimal product mix and what is the overall profitability?

c. Which station is the bottleneck?

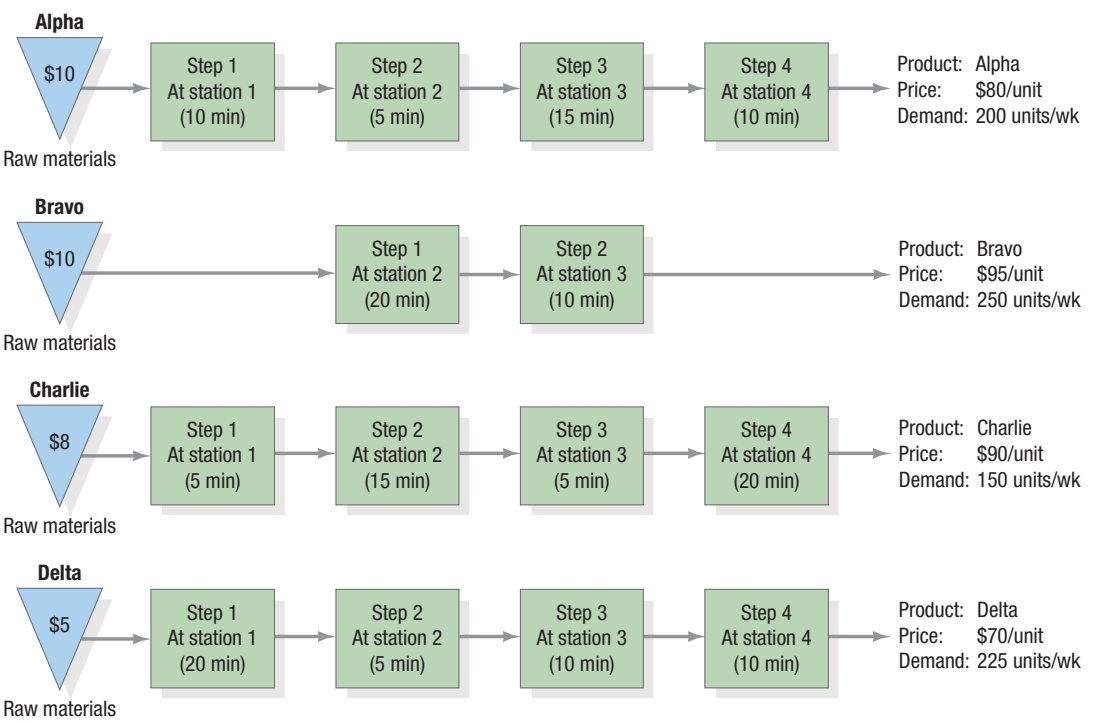
14. The senior management at Davis Watercraft would like to determine if it is possible to improve firm profitability by changing their existing product mix. Currently, the product mix is determined by giving resource priority to the highest contribution margin watercraft. Davis Watercraft always has a contingent of 10 workers on hand; each worker is paid $25 per hour. Overhead costs are $35,000 per week. The plant operates 18 hours per day and 6 days per week. Labor is considered a fixed expense because workers are paid for their time regardless of their utilization. The production manager has determined that workstation 1 is the bottleneck. Detailed production information is provided below.

	Model		
	A	**B**	**C**
Price	$450	$400	$500
Material Cost	$50	$40	$110
Weekly Demand	100	75	40
Processing Time Station 1	60 min	0 min	30 min
Processing Time Station 2	0 min	0 min	60 min
Processing Time Station 3	10 min	60 min	0 min
Processing Time Station 4	20 min	30 min	40 min

a. Using the bottleneck-based method, what is the product mix that yields the highest total profit? What is the resulting profit?

b. Using the traditional method, which bases decisions solely on a product's contribution to profits and overhead, what is the product mix that yields the highest total profit? What is the resulting profit?

▶ FIGURE 7.14
Cooper River Glass Works Flowchart



15. A paced assembly line has been devised to manufacture calculators, as the following data show:

Station	Work Element Assigned	Work Element Time (min)
S1	A	2.7
S2	D, E	0.6, 0.9
S3	C	3.0
S4	B, F, G	0.7, 0.7, 0.9
S5	H, I, J	0.7, 0.3, 1.2
S6	K	2.4

a. What cycle time corresponds to this maximum output rate?

b. What is the maximum hourly output rate from this line? (*Hint:* The line can go only as fast as its slowest workstation.)

c. If a worker is at each station and the line operates at this maximum output rate, how much idle time is lost during each 10-hour shift?

d. What is the line's efficiency?

16. Jane produces custom greeting cards using six distinct work elements. She would like to produce 10 cards in each 8-hour card-making session. Figure 7.15 details each work element and its associated durations in minutes as well as their precedence relationships.

a. What cycle time is required to satisfy the required output rate?

b. What is the theoretical minimum number of workstations required?

c. If Jane identifies a five-station solution, what is the associated efficiency and balance delay?

d. If the cycle time increased by 100 percent, would the theoretical minimum number of workstations also increase by 100 percent?

17. Greg Davis, a business major at the University of South Carolina (USC), has opened Six Points Saco (SPS), a specialty subs–taco restaurant, at the rim of the USC campus. SPS has grown in popularity over the one year that it has been in operation, and Greg is trying to perfect the business model before making it into a franchise. He wants to maximize the productivity of his staff, as well as serve customers well in a timely fashion. One area of concern is the drive-thru operation during the 11:30 A.M. to 12:30 P.M. lunch hour.

The process of fulfilling an order involves fulfilling the tasks listed below.

Greg is interested in getting a better understanding of the staffing patterns that will be needed in order to operate his restaurant. After taking a course in operations management at the university, he knows that fulfilling a customer order at SPS is very similar to operating an assembly line. He has also used the POM for Windows software before, and wants to apply it for examining different demand scenarios for serving his customers.

a. Assuming that no task is assigned to more than one employee, what is the "maximum output capacity" from this assembly line? How many employees will be needed to actually accomplish this maximum output capacity?

b. If Greg wants to process 45 customers per hour, how many employees will he need during the peak period?

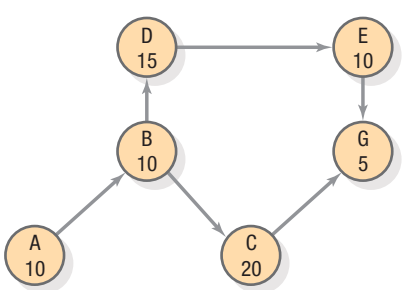
c. With the number of employees determined in part b, what is the maximum number of customers who could be served every hour (i.e., what is the maximum output capacity)?

d. If all the seven tasks are handled by one employee, how many customers could be served per hour?

e. Beyond the output accomplished in part d, if Greg decides to add one additional worker to help out with a bottleneck task, where should he add that worker? With that addition, would he be able to process more customers per hour? If so, what is the new maximum output capacity for the drive-thru?

Task	Time (Seconds)	Immediate Predecessors
A. Take an order at the booth. Most orders are for a taco and a sub.	25	
B. Collect money at the window.	20	A
C. Gather drinks.	35	B
D. Assemble taco order.	32	B
E. Assemble sub order.	30	B
F. Put drinks, taco, and sub in a bag.	25	C, D, E
G. Give the bag to the customer.	10	F

FIGURE 7.15 ▶
Precedence Diagram for Custom Greeting Cards



18. Refer back to problem 7. Suppose that in addition to the usual precedence constraints, there are two zoning constraints within the trim line. First, work elements K and L should be assigned to the same station; both use a common component, and assigning them to the same station conserves storage space. Second, work elements H and J cannot be performed at the same station.

a. What is the efficiency of your solution?

b. Using trial and error, balance the line as best you can.

EXPERIENTIAL LEARNING Min-Yo Garment Company

The Min-Yo Garment Company is a small firm in Taiwan that produces sportswear for sale in the wholesale and retail markets. Min-Yo's garments are unique because they offer fine embroidery and fabrics with a variety of striped and solid patterns. Over the 20 years of its existence, the Min-Yo Garment Company has become known as a quality producer of sports shirts with dependable deliveries. However, during that same period, the nature of the apparel industry has undergone change. In the past, firms could be successful producing standardized shirts in high volumes with few pattern or color choices and long production lead times. Currently, with the advent of regionalized merchandising and intense competition at the retail level, buyers of the shirts are looking for shorter lead times and much more variety in patterns and colors. Consequently, many more business opportunities are available today than ever before to a respected company such as Min-Yo.

Even though the opportunity for business success seemed bright, the management meeting last week was gloomy. Min-Yo Lee, president and owner of Min-Yo Garment, expressed concerns over the performance of the company: "We are facing strong competition for our products. Large apparel firms are driving prices down on high-volume licensed brands. Each day more firms enter the customized shirt business. Our profits are lower than expected, and delivery performance is deteriorating. We must reexamine our capabilities and decide what we can do best."

Products

Min-Yo has divided its product line into three categories: licensed brands, subcontracted brands, and special garments.

Licensed Brands

Licensed brands are brands that are owned by one company but, through a licensing agreement, are produced by another firm that also markets the brand in a specific geographic region. The licenser may have licensees all over the world. The licensee pays the licenser a fee for the privilege of marketing the brand in its region, and the licenser agrees to provide some advertising for the product, typically through media outlets that have international exposure. A key aspect of the licensing agreement is that the licensee must agree to provide sufficient quantities of product at the retail level. Running out of stock hurts the image of the brand name.

Currently, only one licensed brand is manufactured by Min-Yo. The brand, called the Muscle Shirt, is owned by a large "virtual corporation" in Italy that has no manufacturing facilities of its own. Min-Yo has been licensed to manufacture Muscle Shirts and sell them to large retail chains in Taiwan. The retail chains require prompt shipments at the end of each week. Because of competitive pressures from other licensed brands, low prices are important. Min-Yo sells each Muscle Shirt to retail chains for $6.

The demand for Muscle Shirts averages 900 shirts per week. The following demand for Muscle Shirts has been forecasted for the next 12 weeks.

Min-Yo's forecasts of Muscle Shirts are typically accurate to within ±200 shirts per week. If demand exceeds supply in any week, the excess demand is lost. No backorders are taken, and Min-Yo incurs no cost penalty for lost sales.

Subcontracted Brands

Manufacturers in the apparel industry often face uncertain demand. To maintain level production at their plants, many manufacturers seek subcontractors to produce their brands. Min-Yo is often considered a subcontractor because of its reputation in the industry. Although price is a consideration, the owners of subcontracted brands emphasize dependable delivery and the ability of the subcontractor to adjust order quantities on short notice.

Week	Demand	Week	Demand
1*	700	7	1,100
2	800	8	1,100
3	900	9	900
4	900	10	900
5	1,000	11	800
6	1,100	12	700

*In other words, the company expects to sell 700 Muscle Shirts at the end of week 1.

Currently, Min-Yo manufactures only one subcontracted brand, called the Thunder Shirt because of its bright colors. Thunder Shirts are manufactured to order for a company in Singapore. Min-Yo's price to this company is $7 per shirt. When orders are placed, usually twice a month, the customer specifies the delivery of certain quantities in each of the next 2 weeks. The last order the customer placed is overdue, forcing Min-Yo to pay a penalty charge. To avoid another penalty, 200 shirts must be shipped in week 1. The Singapore company is expected to specify the quantities it requires for weeks 2 and 3 at the beginning of week 1. The delivery schedule containing the orders for weeks 4 and 5 is expected to arrive at the beginning of week 3, and so on. The customer has estimated its average weekly needs for the year to be 200 shirts per week, although its estimates are frequently inaccurate.

Because of the importance of this large customer to Min-Yo and the lengthy negotiations of the sales department to get the business, management always tries to satisfy its needs. Management believes that if Min-Yo Garment ever refuses to accept an order from this customer, Min-Yo will lose the Thunder Shirt business. Under the terms of the sales contract, Min-Yo agreed to pay this customer $1 for every shirt not shipped on time for each week the shipment of the shirt is delinquent. Delinquent shipments must be made up.

Special Garments

Special garments are made only to customer order because of their low volume and specialized nature. Customers come to Min-Yo Garment to manufacture shirts for special promotions or special company occasions. Min-Yo's special garments are known as Dragon Shirts because of the elaborate embroidery and oriental flair of the designs. Because each shirt is made to a particular customer's specifications and requires a separate setup, special garments cannot be produced in advance of a firm customer order.

Although price is not a major concern for the customers of special garments, Min-Yo sells Dragon Shirts for $8 a shirt to ward off other companies seeking to enter the custom shirt market. Its customers come to Min-Yo because the company can produce almost any design with high quality and deliver an entire order on time. When placing an order for a Dragon Shirt, a customer specifies the design of the shirt (or chooses from Min-Yo's catalog), supplies specific designs for logos, and specifies the quantity of the order and the delivery date. In the past, management checked to see whether such an order would fit into the schedule, and then either accepted or rejected it on that basis. If Min-Yo accepts an order for delivery at the *end* of a certain week and fails to meet this commitment, it pays a penalty of $2 per shirt for each week delivery is delayed. This penalty is incurred weekly until the delinquent order is delivered. The company tried to forecast demand for specific designs of Dragon Shirts but has given up. Last week, Min-Yo had four Dragon Shirt

opportunities of 50, 75, 200, and 60 units but chose not to accept any of the orders. Dragon Shirt orders in the past ranged from 50 units to 300 units with varying lead times.

Figure 7.16, Min-Yo's current open-order file, shows that in some prior week Min-Yo accepted an order of 400 Thunder Shirts for delivery last week. The open-order file is important because it contains the commitment management made to customers. Commitments are for a certain quantity and a date of delivery. As customer orders are accepted, management enters the quantity in the green cell representing the week that they are due. Because Dragon Shirts are unique unto themselves, they each have their own order number for future use. No Dragon Shirt orders appear in the open-order file because Min-Yo has not committed to any in the past several weeks.

Manufacturing

Process

The Min-Yo Garment Company has the latest process technology in the industry—a machine, called a garment maker, that is run by one operator on each of three shifts. This single machine process can make every garment Min-Yo produces; however, the changeover times consume a substantial amount of capacity. Company policy is to run the machine three shifts a day, five days a week. If business is insufficient to keep the machine busy, the workers are idle because Min-Yo is committed to never fire or lay off a worker. By the same token, the firm has a policy of never working on weekends. Thus, the capacity of the process is 5 days × 24 hours = 120 hours per week. The hourly wage is $10 per hour, so the firm is committed to a fixed labor cost of $10 × 120 = $1,200 per week. Once the machine has been set up to make a particular type of garment, it can produce that garment at the rate of 10 garments per hour, regardless of type. The cost of the material in each garment, regardless of type, is $4. Raw materials are never a problem and can be obtained overnight.

Scheduling the Garment Maker

Scheduling at Min-Yo is done once each week, after production for the week has been completed and shipped, after new orders from customers have arrived, and before production for the next week has started. Scheduling results in two documents.

The first is a production schedule, shown in Figure 7.17. The schedule shows what management wants the garment maker process to produce in a given week. Two spreadsheet entries are required for each product that is to be produced in a given week. They are in the green shaded cells. The first is the production quantity. In Figure 7.17, the schedule shows that Min-Yo produced quantities of 800 units for Muscle and 200 units for Thunder last week. The second input is a "1" if the machine is to be set up for a given product or a "blank" if no changeover is required. Figure 7.17 shows that last week changeovers were required for the Muscle and Thunder production runs. The changeover information is important because, at the end of a week, the garment maker process will be set up for the last product produced. If the same product is to be produced first the following week, no new changeover will be required. Management must keep track of the sequence of production each week to take advantage of this savings. The only exception to this rule is Dragon Shirts, which are unique orders that always require a changeover. In week 0, Min-Yo did not produce any Dragon Shirts; however, it did produce 800 Muscle Shirts, followed by 200 Thunder Shirts. Finally, the spreadsheet calculates the hours required for the proposed

MIN-YO GARMENT COMPANY

Open Order File (Record of commitments)

	Week Order is Due									
Product	1	2	3	4	5	6	7	8	9	10
Thunder Orders	400									
Dragon Order 1										
Dragon Order 2										
Dragon Order 3										
Dragon Order 4										
Dragon Order 5										
Dragon Order 6										
Dragon Order 7										
Dragon Order 8										
Dragon Order 9										
Dragon Order 10										
Dragon Order 11										
Dragon Order 12										
Dragon Order 13										
Dragon Order 14										
Dragon Order 15										

Intro | Open Order File | Week 1 | Week 2 | Week 3 | Week 4 | Week 5 | Week 6 | Week 7 | Week 8 | Week 9 | Week 10 | Si

▲ **FIGURE 7.16**

Min-Yo's Open Order File

Note: All orders are to be delivered at the end of the week indicated, after production for the week has been completed and before next week's production is started.

MIN-YO GARMENT COMPANY

PRODUCTION SCHEDULE

The two inputs to the Production Schedule table are:
1. The quantity you decide to produce this time period
2. Whether there is a setup/changover required (1 or 0)

PRODUCT		
	Changeover	Quantity
Muscle	1	800
Hours		88
Thunder	1	200
Hours		30

				Changeover	Quantity		Changeover	Quantity
Dragon Order 1			Dragon Order 11			Dragon Order 21		
Dragon Order 2			Dragon Order 12			Dragon Order 22		
Dragon Order 3			Dragon Order 13			Dragon Order 23		
Dragon Order 4			Dragon Order 14			Dragon Order 24		
Dragon Order 5			Dragon Order 15			Dragon Order 25		
Dragon Order 6			Dragon Order 16			Dragon Order 26		
Dragon Order 7			Dragon Order 17			Dragon Order 27		
Dragon Order 8			Dragon Order 18			Dragon Order 28		
Dragon Order 9			Dragon Order 19			Dragon Order 29		
Dragon Order 10			Dragon Order 20			Dragon Order 30		
Total Dragon Hours		0						
Total Dragon Production		0						

Total Hours scheduled		118

Is production within capacity? Yes

▲ **FIGURE 7.17**
Min-Yo's Production Schedule

schedule. Changeover times for Muscle, Thunder, and Dragon Shirts are 8, 10, and 25 hours respectively. Because the garment maker process produces 10 garments per hour regardless of type, the production hours required for Muscle Shirts is 8 + 800/10 = 88 hours, and the production hours for Thunder Shirts is 10 + 200/10 = 30 hours, as shown in Figure 7.17. The total time spent on the garment maker process on all products in a week cannot exceed 120 hours. The spreadsheet will not allow you to proceed if this constraint is violated.

The second document is a weekly profit and loss (P&L) statement that factors in sales and production costs, including penalty charges and inventory carrying costs, as shown in Figure 7.18. The inventory carrying cost for *any type of product* is $0.10 per shirt per week left in inventory after shipments for the week have been made. The spreadsheet automatically calculates the P&L statement, which links to the open-order file and the production schedule, after the demand for Muscle Shirts is known. Figure 7.18 shows that the actual demand for Muscle Shirts last week was 750 shirts.

P&L STATEMENT

Product	Price	Beg Inv	Production	Available	Demand	Sales	End Inv	Inv/Past due costs
Muscle	$6	550	800	1350	750	4500	600	60
Thunder	$7		200	200	400	1400	-200	200
Dragon Orders	$8		0	0	0	0	0	0
Totals			1000			5900		260

		Current	Cumulative
Sales Total		$5,900	$5,900
Labor	$1,200		
Materials	$4,000		
Inv/Past due	$260		
Total Cost		$5,460	
Profit Contribution		$440	$440

▲ **FIGURE 7.18**
Min-Yo's P&L Schedule

Notes

- The past due quantity of shirts are those shirts not shipped as promised, and appear as a negative number in the "End Inv" column.
- Available = Beginning inventory + Production
- Sales = Demand × Price when demand < available; Available × Price, otherwise
- Inventory cost = \$0.10 times number of shirts in inventory. Past due cost equals past due quantity times the penalty (\$1 for Thunder Shirts; \$2 for Dragon Shirts). These costs are combined in the "Inv/Past Due Costs" column.

The Simulation

At Min-Yo Garment Company, the executive committee meets weekly to discuss the new order possibilities and the load on the garment maker process. The executive committee consists of top management representatives from finance, marketing, and operations. You will be asked to participate on a team and play the role of a member of the executive committee in class. During this exercise, you must decide how far into the future to plan. Some decisions, such as the markets you want to exploit, are long-term in nature. Before class, you may want to think about the markets and their implications for manufacturing. Other decisions are short-term and have an impact on the firm's ability to meet its commitments. In class, the simulation will proceed as follows.

1. Use the Min-Yo Tables spreadsheet in OM Explorer in MyOMLab. It is found in the Solver menu, under Constraint Management. You will start by specifying the production schedule for week 1, based on the forecasts for week 1 in the case narrative for Muscle Shirts and additional information on new and existing orders for the customized shirts from your instructor. *You may assume that your managerial predecessors left the garment machine set up for Thunder Shirts.* The production schedule decision is to be made in collaboration with your executive committee colleagues in class.
2. When all the teams have finalized their production plans for week 1, the instructor will supply the actual demands for Muscle Shirts in week 1. Enter that quantity in the P&L statement in the spreadsheet for week 1.
3. After the P&L statement for week 1 is completed, the instructor will announce the new order requests for Thunder Shirts and Dragon Shirts to be shipped in week 2 and the weeks beyond.
4. You should look at your order requests, accept those that you want, and reject the rest. Add those that you accept for delivery in future periods to your open-order file. Enter the quantity in the cell representing the week the order is due. You are then irrevocably committed to them and their consequences.
5. You should then make out a new production schedule, specifying what you want your garment-maker process to do in the next week (it will be for week 2 at that time).
6. The instructor will impose a time limit for each period of the simulation. When the time limit for one period has been reached, the simulation will proceed to the next week. Each week the spreadsheet will automatically update your production and financial information in the Summary Sheet.

VIDEO CASE Constraint Management at Southwest Airlines

What if you could take a commercial airline flight any time, and anywhere you wanted to go? Just show up at the airport without the need to consider time schedules or layovers. Aside from the potentially cost-prohibitive nature of such travel, there are also constraints in the airline system that preclude this kind of operation. From the lobby check-in process through to boarding at the gate and processing plane turnaround, the process of operating the airline is filled with constraints that must be managed in order for them to be successful and profitable. Flight schedules are tightly orchestrated and controlled, departure and arrival gates at airports are limited, and individual aircraft have seating capacities in each section of the plane, to name a few.

Southwest Airlines is one company that has figured out how to manage its constraints and generate positive customer experiences in the process. No other airline can claim the same level of profitability and customer satisfaction Southwest regularly achieves. What is its secret?

Talk to any loyal Southwest customer and you will hear rave reviews about its low fares, great customer service, and lack of assigned seating that gives customers a chance to choose who they sit next to onboard. From an operations perspective, it is much more than what the customer sees. Behind the scenes, operations managers carefully manage and execute—3,400 times a day in over 60 cities in the United States—a process designed to manage all potential bottleneck areas.

Southwest's famous rapid gate-turnaround of 25 minutes or less demonstrates how attention to the activities that ground operations must complete to clean, fuel, and prepare a plane for flight can become bottlenecks if not properly scheduled. In the terminal at the gate, passenger boarding also can be a bottleneck if the boarding process itself is not carefully managed. Since the individual mix of passengers present a different set of issues with each flight that often are not evident until the passengers actually arrive at the gate, ranging from families with kids and strollers to large quantities of carry-on bags and passengers needing wheelchair assistance, operations managers must be ready for any and all situations to avoid a boarding bottleneck while also assuring a pleasant and stress-free gate experience for all passengers.



Pearson

Passengers boarding a Southwest Airlines flight.

In 2007, as part of the company's continuous improvement activities, Southwest focused its attention on the passenger boarding process to determine whether there was a better way to board. Its existing process consisted of three groups, A, B, C, with no assigned seating. Depending on passenger check-in and arrival time, passengers were given a spot in a group.

Those first to check-in received choice places in the A group. Last to check-in ended up in the C group, and usually had a choice of only middle seats in the back of the plane upon boarding. As passengers arrived at the gate, they queued up in their respective boarding group areas to await the boarding call.

Seven different alternate boarding scenarios were designed and tested. They included

- New family pre-boarding behind the "A" group of first-to-board passengers
- Family pre-boarding before anyone else, but seating choices limited on-board to behind the wing
- Six boarding groups (within A-B-C groups) instead of the original three A-B-C groups
- Assigned boarding gate line positions based on both boarding group and gate arrival time
- Single boarding chute at the gate, but up to nine groups all in one queue
- Boarding with a countdown clock to give customers an incentive to get in line and board quickly; incentives given out if everyone was on time
- Educational boarding video to make the boarding process fun, inform passengers how to board efficiently, and provide the company another way to promote its brand.

QUESTIONS

1. Analyze Southwest's passenger boarding process using the Theory of Constraints.
2. Which boarding scenario among the different ones proposed would you recommend for implementation? Why?
3. How should Southwest evaluate the gate boarding and plane turnaround process?
4. How will Southwest know that the bottleneck had indeed been eliminated after the change in the boarding process?

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16 PLANNING SUFFICIENT RESOURCES

Dow Corning scientists research next generation solar technologies at the company's Solar Application Center

Dow Corning

Dow Corning, created as a joint venture between Corning Incorporated and the Dow Chemical Company in 1943, is a global leader that offers over 7,000 innovative products and services using silicon-based technology in diverse industries such as electronics, aviation and aerospace, textile, automotive, and health care, among others. As an example, Dow Corning's Electrical Insulating Compounds are used for creating moisture proof seals for aircraft, automotive, and marine ignition systems. With 11,500 employees, over 25,000 customers, and 45 manufacturing and warehouse locations worldwide, its annual revenues of $6 billion generated a net income of $866 million in 2010. More than half its sales are outside the United States. In order to integrate different business functions and enhance resource planning across the entire firm and its supply chain, Dow Corning turned to SAP, a leading provider of enterprise resource planning (ERP) software solutions. SAP delivers role based access to crucial data, applications, and analytical tools, and has a suite of applications for Business Processes (like Financial Management, Customer Relationship Management, Human Capital Management, and Supply Chain Management), Business Analytics, and Technology. Dow installed the SAP R/3 and mySAP Supply Chain Management solution, with SAP Advanced Planner and Optimizer (APO) at its core.

Prior to the ERP implementation, limited transparency and redundancy within the existing legacy systems made it difficult to access and analyze data

needed for effective resource planning, and also hampered decision making and responsiveness. A sequenced implementation of the SAP modules facilitated the linking of key processes from order generation to production planning to warehousing to delivery and final billing. The SAP APO solution enabled the SCOR model of plan, source, make, and deliver (see Chapter 12, "Integrating the Supply Chain"), which allowed the linkage of shop floor processes and manufacturing operations with the rest of the business. With a transparent view of orders, materials, equipment, product quality, and cost information, Dow Corning can now coordinate plants and processes with greater ease and better match production to market requirements on a global scale. Employee productivity and satisfaction are also up, due mostly to faster response times and accurate on-time deliveries.

Source: End to End Supply Chain Management at Dow Corning, **http://www.sap.com/usa/solutions/business-suite/erp/operations/pdf/CS_Dow_Corning.pdf**; Dow Corning, Optimizing Operational Performance to Sharpen Competitive Advantage, **www.sap.com**; **http://www.dowcorning.com/content/about/aboutmedia/fastfacts.asp**; May 29, 2011; **http://www12.sap.com/index.epx#/solutions/index.epx**; May 29, 2011.

LEARNING GOALS *After reading this chapter, you should be able to:*

1. Explain how enterprise resource planning (ERP) systems can foster better resource planning.
2. Explain how the concept of dependent demand is fundamental to resource planning.
3. Describe a master production schedule (MPS) and compute available-to-promise quantities.
4. Apply the logic of a material requirements planning (MRP) system to identify production and purchase orders needed for dependent demand items.
5. Apply MRP principles to the provision of services and distribution inventories.

Dow Corning demonstrates that companies can gain a competitive edge by using an effective information system to help with their resource planning. Companies must ensure that all of the resources they need to produce finished services or products are available at the right time. If they are not, a firm risks losing business. For a manufacturer, this task can mean keeping track of thousands of subassemblies, components, and raw materials as well as key equipment capacities. For a service provider, this task can mean keeping track of numerous supplies and carefully scheduling the time and capacity requirements of different employees and types of equipment.

We begin this chapter by describing enterprise resource planning (ERP) systems, which have become a valuable tool for, among other things, resource planning. We then examine a specific approach to resource planning, called material requirements planning (MRP). The concluding section of the chapter illustrates how service providers manage their supplies, human resources, equipment, and financial resources.

resource planning
A process that takes sales and operations plans; processes information in the way of time standards, routings, and other information on how services or products are produced; and then plans the input requirements.

enterprise process
A companywide process that cuts across functional areas, business units, geographical regions, and product lines.

enterprise resource planning (ERP) systems
Large, integrated information systems that support many enterprise processes and data storage needs.

Resource Planning across the Organization

Resource planning lies at the heart of any organization, cutting across all of its different functional areas. It takes sales and operations plans; processes information in the way of time standards, routings, and other information on how services or products are produced; and then plans the input requirements. It also can create reports for managers of the firm's major functional areas, such as human resources, purchasing, sales and marketing, and finance and accounting. In essence, resource planning is a process in and of itself that can be analyzed relative to the firm's competitive priorities.

Enterprise Resource Planning

An **enterprise process** is a companywide process that cuts across functional areas, business units, geographic regions, product lines, suppliers, and customers. **Enterprise resource planning (ERP) systems** are large, integrated information systems that support many enterprise processes and

data storage needs. By integrating the firm's functional areas, ERP systems allow an organization to view its operations as a whole rather than having to try to put together the different information pieces produced by its various functions and divisions. Today, ERP systems are being used by traditional brick-and-mortar organizations such as manufacturers, restaurants, hospitals, and hotels, as well as by Internet companies that rely extensively on Web connectivity to link their customers and suppliers.

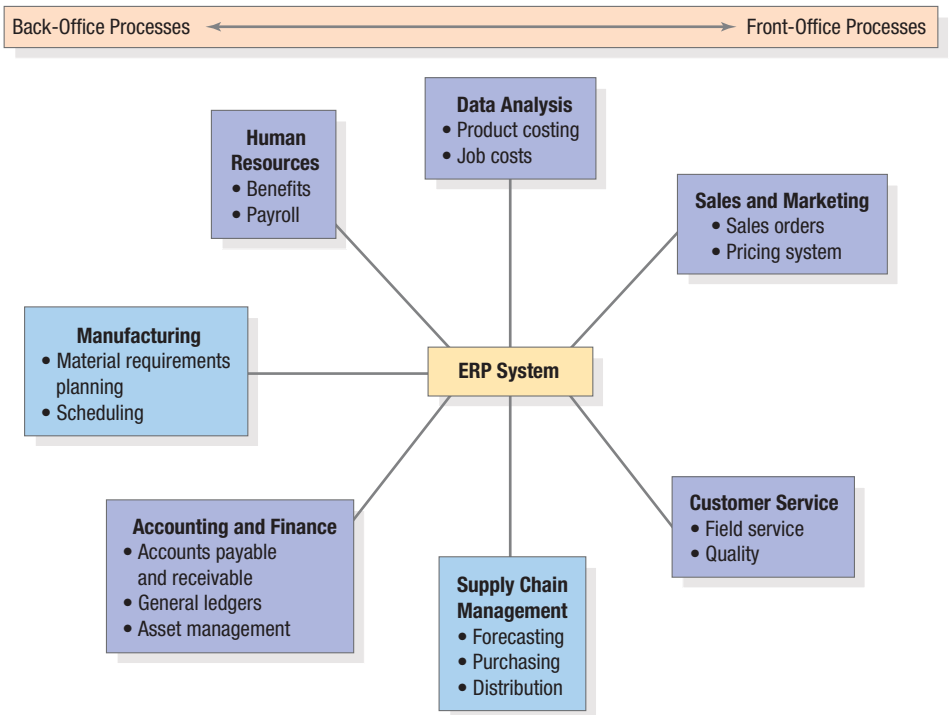
How ERP Systems Are Designed

ERP revolves around a single comprehensive database that can be made available across the entire organization (or enterprise). Passwords are generally issued to allow certain personnel to access certain areas of the system. Having a single database for all of the firm's information makes it much easier for managers to monitor all of the company's products at all locations and at all times. The database collects data and feeds them into the various modular applications (or suites) of the software system. As new information is entered as a *transaction* in one application, related information is automatically updated in the other applications, including the firm's financial and accounting databases, its human resource and payroll databases, sales, supplier and customer databases, and so forth. In this way, the ERP system streamlines the data flows throughout the organization and supply chain and provides employees with direct access to a wealth of real-time operating information scattered across different functions in the organization. Figure 16.1 shows some of the typical applications with a few subprocesses nested within each one. Some of the applications are for back-office operations such as manufacturing and payroll, while others are for front-office operations such as customer service.

Amazon.com is one company that uses an ERP system. The supply chain application of Amazon's system is particularly important because it allows Amazon.com to link customer orders to warehouse shipments and, ultimately, to supplier replenishment orders. Other applications are more important in other businesses. For example, universities put particular emphasis on the human resources and accounting and finance applications, and manufacturers have an interest in almost every application suite. Not all applications in Figure 16.1 need to be integrated into an ERP system, but those left out will not share their information with the ERP system. Sometimes, however, ERP systems are designed to interface with a firm's existing, older information systems (called "legacy systems").

Designing an ERP system requires that a company carefully analyze its major processes so that appropriate decisions about the coordination of legacy systems and new software can be made. Sometimes, a company's processes that involve redundancies and convoluted information flows must be completely reengineered before the firm can enjoy the benefits of an integrated

Competing with Operations
Using Operations to Compete
Managing Effective Projects
Designing and Managing Processes
Developing a Process Strategy
Analyzing Processes
Managing Quality
Planning Capacity
Managing Process Constraints
Designing Lean Systems
Designing and Managing Supply Chains
Managing Inventories
Designing Effective Supply Chains
Locating Facilities
Integrating the Supply Chain
Managing Sustainable Supply Chains
Forecasting Demand
Planning and Scheduling Operations
Planning Sufficient Resources



◀ **FIGURE 16.1**
ERP Application Modules
Source: Based on *Enterprise Resource Planning (ERP)* by Scalle and Cotteleer, Harvard Business School Press. Boston, MA, 1999, No. 9-699-020.



Stefan Kiefer/Imagebroker/Alamy

Headquarters of the software company SAP AG in Walldorf, Baden-Wuerttemberg, Germany, Europe

information system. However, a recent study showed that companies reap the greatest rewards when they keep their ERP implementations simple, work with a small number of software vendors, and use standardized systems rather than customizing them extensively. Firms can otherwise end up spending excessive amounts of money on ERP systems that are complex to use and costly to manage. UK confectionary giant Cadbury had to take a 12-million pound hit to their profits due to a build-up of excess inventory of chocolate bars caused by information technology problems arising from the roll out of a new SAP-based ERP system.

Most ERP systems today use a graphical user interface, although the older, keyboard-driven, text-based systems are still popular because of their dependability and technical simplicity. Users navigate through various screens and menus. Training, such as during ERP implementation, focuses on these screens and how users can utilize them to get their jobs done. The biggest suppliers of these off-the-shelf commercial ERP packages are SAP AG, a German company that was also used by Dow Corning in the opening vignette, followed by Oracle Corporation.

Material Requirements Planning

material requirements planning (MRP)

A computerized information system developed specifically to help manufacturers manage dependent demand inventory and schedule replenishment orders.

MRP explosion

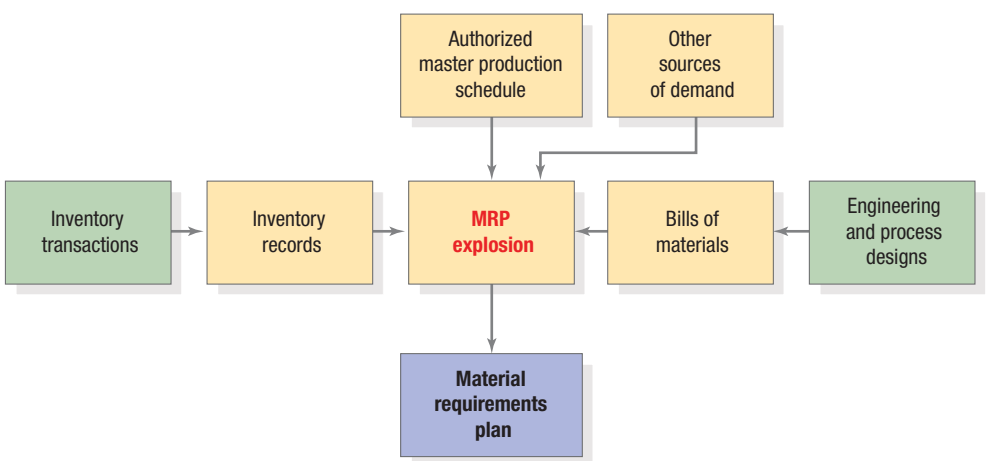
A process that converts the requirements of various final products into a material requirements plan that specifies the replenishment schedules of all the subassemblies, components, and raw materials needed to produce final products.

The Manufacturing and Supply Chain Management modules in Figure 16.1 deal with resource planning. Understanding resource planning begins with the concept of *dependent demand*, which sets it apart from the techniques covered in Chapter 9, "Managing Inventories." **Material requirements planning (MRP)** is a computerized information system developed specifically to help manufacturers manage dependent demand inventory and schedule replenishment orders. The key inputs of an MRP system are a bill of materials database, a master production schedule, and an inventory record database, as shown in Figure 16.2. Using this information, the MRP system identifies the actions planners must take to stay on schedule, such as releasing new production orders, adjusting order quantities, and expediting late orders.

An MRP system translates the master production schedule and other sources of demand, such as independent demand for replacement parts and maintenance items, into the requirements for all subassemblies, components, and raw materials needed to produce the required parent items. This process is called an **MRP explosion** because it converts the requirements of various final products into a material requirements plan that specifies the replenishment schedules of all the subassemblies, components, and raw materials needed to produce final products.

We first explore the nature of dependent demand and how it differs from independent demand, followed by a discussion of each of the key inputs to the MRP system shown in Figure 16.2.

FIGURE 16.2 ▶
Material Requirements Plan Inputs



Dependent Demand

For years, many companies tried to manage production and their dependent demand inventories using independent demand systems similar to those discussed in Chapter 9, "Managing Inventories," but the outcome was seldom satisfactory because dependent demand is fundamentally different from independent demand. To illustrate the concept of dependent demand, let us consider a Huffy bicycle produced for retail outlets. Demand for a final product, such as a bicycle, is called *independent demand* because it is influenced only by market conditions. In contrast, the demand for spokes going into the bicycle "depends" on the production planned for its wheels. Huffy must forecast this demand using techniques such as those discussed in Chapter 14, "Forecasting Demand." However, Huffy also keeps many other items in inventory—handlebars, pedals, frames, and wheel rims—used to make completed bicycles. Each of these items has a **dependent demand** because the quantity required varies with the production plans for other items held in the firm's inventory—finished bikes, in this case. For example, the demand for frames, pedals, and wheel rims is *dependent* on the production of completed bicycles. Operations can calculate the demand for dependent demand items once the bicycle production levels are laid out in the sales and operations plan. For example, every bicycle needs two wheel rims, so 1,000 completed bicycles need 1,000(2) = 2,000 rims. Forecasting techniques are not needed for the rims.

dependent demand

The demand for an item that occurs because the quantity required varies with the production plans for other items held in the firm's inventory.

The bicycle, or any other product that is manufactured from one or more components, is called a **parent**. The wheel rim is an example of a **component**—an item that goes through one or more operations to be transformed into or become part of one or more parents. A wheel rim, for example, will have several different parents if the rim is used to make more than one style of bicycle. This parent–component relationship can cause erratic dependent demand patterns for components. Suppose that every time inventory falls to 500 units (a reorder point), an order for 1,000 more bicycles is placed, as shown in Figure 16.3(a). The assembly supervisor then authorizes the withdrawal of 2,000 rims from inventory, along with other components for the finished product. The demand for the rim is shown in Figure 16.3(b). So, even though customer demand for the finished bicycle is continuous and reasonably uniform, the production demand for wheel rims is "lumpy"; that is, it occurs sporadically, usually in relatively large quantities. Thus, the production decisions for the assembly of bicycles, which account for the costs of assembling the bicycles and the projected assembly capacities at the time the decisions are made, determine the demand for wheel rims.

parent

Any product that is manufactured from one or more components.

component

An item that goes through one or more operations to be transformed into or become part of one or more parents.

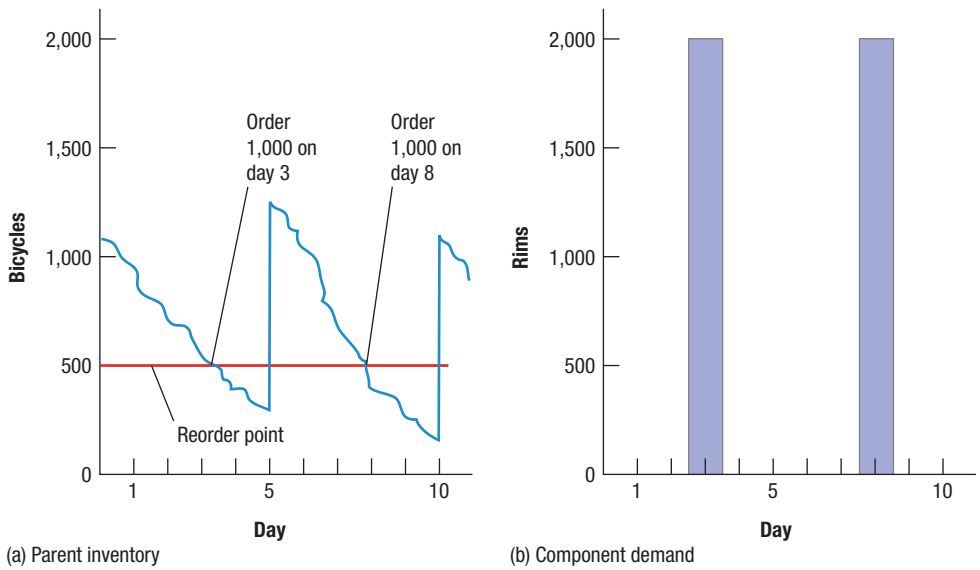
Managing dependent demand inventories is complicated because some components may be subject to both dependent and independent demand. For example, the shop floor needs 2,000 wheel rims for the new bicycles, but the company also sells replacement rims for old bicycles directly to retail outlets. This practice places an independent demand on the inventory of wheel rims. Materials requirements planning can be used in complex situations involving components that may have independent demand as well as dependent demand inventories.

Bill of Materials

The replenishment schedule for a component is determined from the production schedules of its parents. Hence, the system needs accurate information on parent–component relationships. A **bill of materials (BOM)** is a record of all the components of an item, the parent–component

bill of materials (BOM)

A record of all the components of an item, the parent–component relationships, and the usage quantities derived from engineering and process designs.



◀ **FIGURE 16.3**
Lumpy Dependent Demand Resulting from Continuous Independent Demand

relationships, and the usage quantities derived from engineering and process designs. In Figure 16.4, the BOM of a simple ladder-back chair shows that the chair is made from a ladder-back subassembly, a seat subassembly, front legs, and leg supports. In turn, the ladder-back subassembly is made from back legs and back slats, and the seat subassembly is made from a seat frame and a seat cushion. Finally, the seat frame is made from seat-frame boards. For convenience, we refer to these items by the letters shown in Figure 16.4.

usage quantity
The number of units of a component that are needed to make one unit of its immediate parent.

end item
The final product sold to a customer.

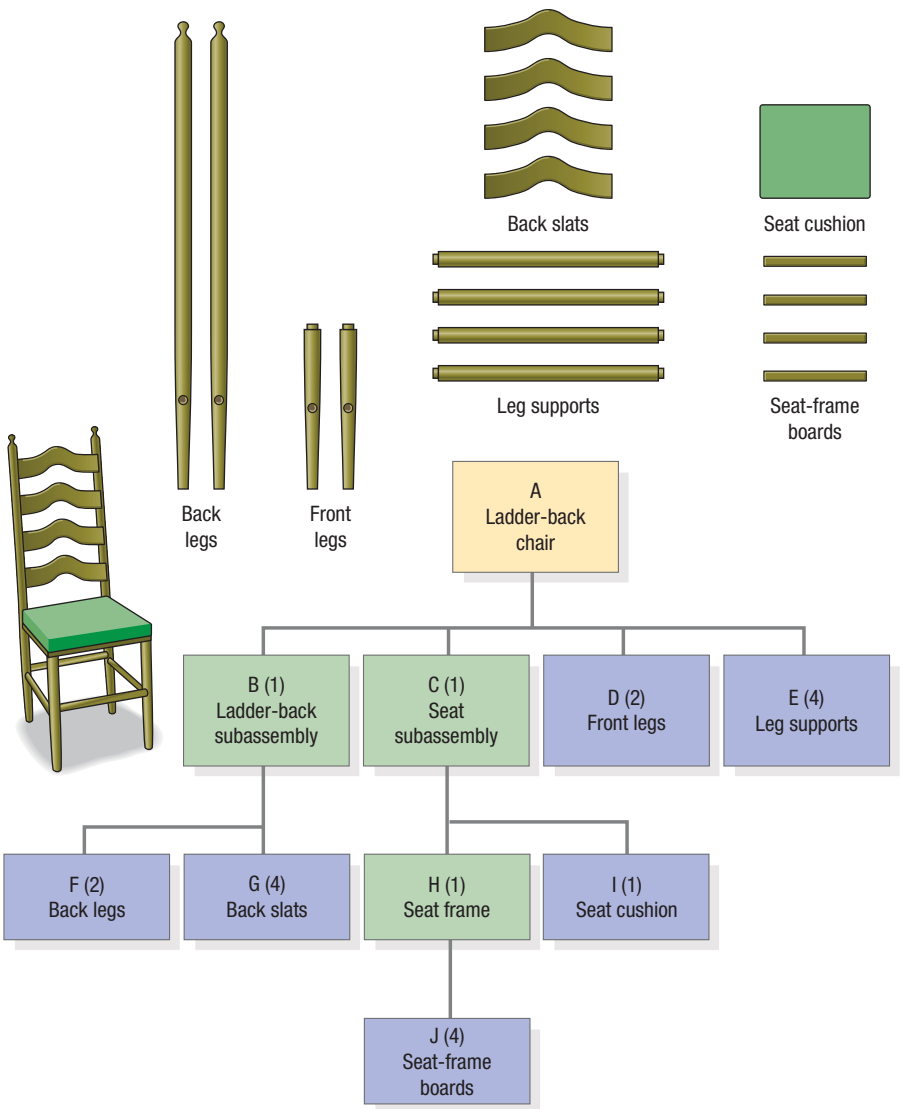
intermediate item
An item that has at least one parent and at least one component.

subassembly
An intermediate item that is *assembled* (as opposed to being transformed by other means) from more than one component.

All items except item A are components because they are needed to make a parent. Items A, B, C, and H are parents because they all have at least one component. The BOM also specifies the **usage quantity**, or the number of units of a component that are needed to make one unit of its immediate parent. Figure 16.4 shows usage quantities for each parent–component relationship in parentheses. Note that one chair (item A) is made from one ladder-back subassembly (item B), one seat subassembly (item C), two front legs (item D), and four leg supports (item E). In addition, item B is made from two back legs (item F) and four back slats (item G). Item C needs one seat frame (item H) and one seat cushion (item I). Finally, item H needs four seat-frame boards (item J).

Four terms frequently used to describe inventory items are *end items*, *intermediate items*, *subassemblies*, and *purchased items*. An **end item** typically is the final product sold to the customer; it is a parent but not a component. Item A in Figure 16.4, the completed ladder-back chair, is an end item. Accounting statements classify inventory of end items as either work-in-process (WIP), if work remains to be done, or finished goods. An **intermediate item**, such as item B, C, or H, has at least one parent and at least one component. Some products have several levels of intermediate items; the parent of one intermediate item can also be an intermediate item. Inventory of intermediate items—whether completed or still on the shop floor—is classified as WIP. A **subassembly** is an intermediate item that is assembled (as opposed to being transformed by other means) from more than

FIGURE 16.4 ▶
BOM for a Ladder-Back Chair



one component. Items B and C are subassemblies. A **purchased item** has no components because it comes from a supplier, but it has one or more parents. Examples are items D, E, F, G, I, and J in Figure 16.4. Inventory of purchased items is treated as raw materials in accounting statements.

purchased item

An item that has one or more parents but no components because it comes from a supplier.

A component may have more than one parent. **Part commonality**, sometimes called *standardization of parts* or *modularity*, is the degree to which a component has more than one immediate parent. As a result of commonality, the same item may appear in several places in the bill of materials for a product, or it may appear in the bills of materials for several different products. For example, the seat subassembly in Figure 16.4 is a component of the ladder-back chair and of a kitchen chair that is part of the same family of products. The usage quantity specified in the bill of materials relates to a specific parent–component relationship. The usage quantity for any component can therefore change, depending on the parent item. Part commonality, or using the same part in many parents, increases its volume and repeatability, which provides several process advantages and helps minimize inventory costs.

part commonality

The degree to which a component has more than one immediate parent.

A careful dissection of the popular iPAD and iPAD 2 products from Apple shows that their bill of materials differ in the enclosure and battery, but are otherwise remarkably similar in their components and design. Same suppliers are used for many components, and costs are very comparable for new revisions of chips found in the previous iPad and iPhones. Standardization of designs and components across different products and generations allows Apple to be very competitive and profitable.

Master Production Scheduling

The second input into a material requirements plan is the **master production schedule (MPS)**, which details how many end items will be produced within specified periods of time. It breaks the sales and operations plan into specific product schedules. Figure 16.5 shows how a sales and operations plan for a family of chairs breaks down into the weekly MPS for each specific chair type (the time period can be hours, days, weeks, or months). The chair example demonstrates the following aspects of master scheduling:

master production schedule (MPS)

A part of the material requirements plan that details how many end items will be produced within specified periods of time.

1. The sums of the quantities in the MPS must equal those in the sales and operations plan. This consistency between the plans is desirable because of the economic analysis done to arrive at the sales and operations plan.
2. The production quantities must be allocated efficiently over time. The specific mix of chair types—the number of each type as a percent of the total family's quantity—is based on historic demand and on marketing and promotional considerations. The planner must select lot sizes for each chair type, taking into consideration economic factors such as production setup costs and inventory carrying costs.

iPad total product costs $270		iPad 2 total product costs $267
	LCD display – 9.7-inch multi-touch screen **$59**	LCD display – 9.7-inch multi-touch screen **$50**
	Camera – None **$0**	Camera – Front & Rear **$4.50**
	Memory – Samsung **$47**	Memory – Samsung/Toshiba **$30**
	Apple Processor – Apple A4 **$17**	Apple Processor – Apple A5 **$25**
	Radio Components Infineon/Broadcom **$26**	Radio Components Infineon/Broadcom or Qualcomm/Broadcom **$25**
	Sensors – STMicroelectronics **$1.00**	Sensors – STMicroelectronics **$2.50**
	Battery **$23**	Battery **$20 – $25**
		

Bill of Materials for Apple's iPad and iPad2.

D. Hurst/Alamy and PSL Images/Alamy

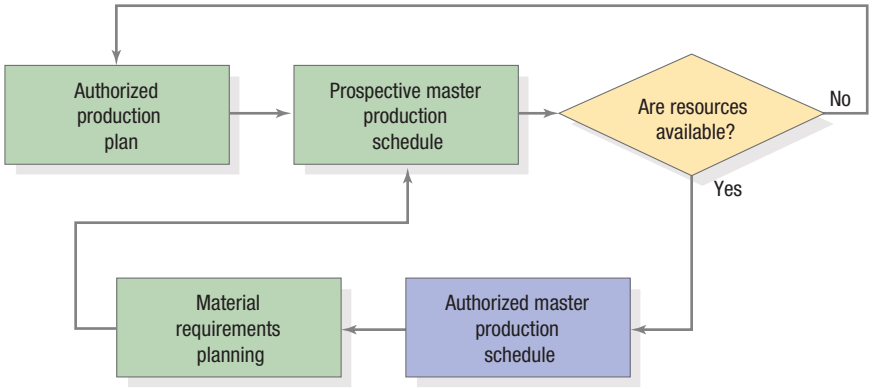
FIGURE 16.5 ▶
MPS for a Family of Chairs

	April				May			
	1	2	3	4	5	6	7	8
Ladder-back chair	150					150		
Kitchen chair				120			120	
Desk chair		200	200		200			200
Sales and operations plan for chair family	670				670			

3. Capacity limitations and bottlenecks, such as machine or labor capacity, storage space, or working capital, may determine the timing and size of MPS quantities. The planner must acknowledge these limitations by recognizing that some chair styles require more resources than others and setting the timing and size of the production quantities accordingly.

Figure 16.6 shows the master production scheduling process. Operations must first create a prospective MPS to test whether it meets the schedule with the resources (e.g., machine capacities, workforce, overtime, and subcontractors) provided for in the sales and operations plan. Operations then revises the MPS until a schedule that satisfies all of the resource limitations is developed or until it is determined that no feasible schedule can be developed. In the latter event, the production plan must be revised to adjust production requirements or increase authorized resources. Once a feasible prospective MPS has been accepted by the firm's managers, operations uses the authorized MPS as input to material requirements planning. Operations can then determine specific schedules for component production and assembly. Actual performance data such as inventory levels and shortages are inputs to preparing the prospective MPS for the next period, and so the master production scheduling process is repeated from one period to the next.

FIGURE 16.6 ▶
Master Production Scheduling Process



Developing a Master Production Schedule The process of developing a master production schedule includes (1) calculating the projected on-hand inventory and (2) determining the timing and size of the production quantities of specific products. We use the manufacturer of the ladder-back chair to illustrate the process. For simplicity, we assume that the firm does not utilize safety stocks for end items, even though many firms do. In addition, we use weeks as our planning periods, even though hours, days, or months could be used.

Step 1. *Calculate Projected On-Hand Inventories.* The first step is to calculate the projected on-hand inventory, which is an estimate of the amount of inventory available each week after demand has been satisfied:

$$\begin{pmatrix}\text{Projected on-hand}\\ \text{inventory at end}\\ \text{of this week}\end{pmatrix} = \begin{pmatrix}\text{On-hand}\\ \text{inventory at}\\ \text{end of last week}\end{pmatrix} + \begin{pmatrix}\text{MPS quantity}\\ \text{due at start}\\ \text{of this week}\end{pmatrix} - \begin{pmatrix}\text{Projected}\\ \text{requirements}\\ \text{this week}\end{pmatrix}$$

In some weeks, no MPS quantity for a product may be needed because sufficient inventory already exists. For the projected requirements for this week, the scheduler uses whichever is

larger—the forecast or the customer orders booked—recognizing that the forecast is subject to error. If actual booked orders exceed the forecast, the projection will be more accurate if the scheduler uses the booked orders because booked orders are a known quantity. Conversely, if the forecast exceeds booked orders for a week, the forecast will provide a better estimate of the requirements needed for that week because some orders are yet to come in.

The manufacturer of the ladder-back chair produces the chair to stock and needs to develop an MPS for it. Marketing has forecasted a demand of 30 chairs for the first week of April, but actual customer orders booked are for 38 chairs. The current on-hand inventory is 55 chairs. No MPS quantity is due in week 1. Figure 16.7 shows an MPS record with these quantities listed. Because actual orders for week 1 are greater than the forecast, the scheduler uses that figure for actual orders to calculate the projected inventory balance at the end of week 1:

Item: Ladder-back chair

Quantity on Hand: 55	April 1	April 2
Forecast	30	30
Customer orders (booked)	38	27
Projected on-hand inventory	17	–13
MPS quantity	0	0
MPS start		

Explanation: Forecast is less than booked orders in week 1; projected on-hand inventory balance = 55 + 0 – 38 = 17.

Explanation: Forecast exceeds booked orders in week 2; projected on-hand inventory balance = 17 + 0 – 30 = –13. The shortage signals a need to schedule an MPS quantity for completion in week 2.

◀ **FIGURE 16.7** Master Production Schedule for Weeks 1 and 2

$$\text{Inventory} = \begin{pmatrix}55 \text{ chairs}\\ \text{currently}\\ \text{in stock}\end{pmatrix} + \begin{pmatrix}\text{MPS quantity}\\ (0 \text{ for week 1}\end{pmatrix} - \begin{pmatrix}38 \text{ chairs already}\\ \text{promised for}\\ \text{delivery in week 1}\end{pmatrix} = 17 \text{ chairs}$$

In week 2, the forecasted quantity exceeds actual orders booked, so the projected on-hand inventory for the end of week 2 is 17 + 0 − 30 = −13. The shortage signals the need for more chairs to be produced and available for week 2.

Step 2. *Determine the Timing and Size of MPS Quantities.* The goal of determining the timing and size of MPS quantities is to maintain a nonnegative projected on-hand inventory balance. As shortages in inventory are detected, MPS quantities should be scheduled to cover them. The first MPS quantity should be scheduled for the week when the projected on-hand inventory reflects a shortage, such as week 2 in Figure 16.7.[1] The scheduler adds the MPS quantity to the projected on-hand inventory and searches for the next period when a shortage occurs. This shortage signals a need for a second MPS quantity, and so on.

Figure 16.8 shows a master production schedule for the ladder-back chair for the next 8 weeks. The order policy requires production lot sizes of 150 units. A shortage of 13 chairs in week 2 will occur unless the scheduler provides for an MPS quantity for that period. Our convention is to show blanks instead of zeroes in all rows, which improves

▼ **FIGURE 16.8** Master Production Schedule for Weeks 1–8

Item: Ladder-back chair **Order Policy:** 150 units **Lead Time:** 1 week

Quantity on Hand: 55	April 1	April 2	April 3	April 4	May 5	May 6	May 7	May 8
Forecast	30	30	30	30	35	35	35	35
Customer orders (booked)	38	27	24	8	0	0	0	0
Projected on-hand inventory	17	137	107	77	42	7	122	87
MPS quantity		150					150	
MPS start	150					150		

Explanation: The time needed to assemble 150 chairs is 1 week. The assembly department must start assembling chairs in week 1 to have them ready by week 2.

Explanation: On-hand inventory balance = 17 + 150 – 30 = 137. The MPS quantity is needed to avoid a shortage of 30 – 17 = 13 chairs in week 2.

[1] In some cases, new orders will be planned before a shortage is encountered. Two such instances occur when safety stocks and anticipation inventories are built up.

readability and is often used in practice. The only exception is in the projected on-hand inventory row, where a number is always shown, even if it is a 0 or negative number.

Once the MPS quantity is scheduled, the updated projected inventory balance for week 2 is

$$\text{Inventory} = \begin{pmatrix}\text{17 chairs in}\\ \text{inventory at the}\\ \text{end of week 1}\end{pmatrix} + \begin{pmatrix}\text{MPS quantity}\\ \text{of 150 chairs}\end{pmatrix} - \begin{pmatrix}\text{Forecast of}\\ \text{30 chairs}\end{pmatrix} = 137 \text{ chairs}$$

The scheduler proceeds column by column through the MPS record until it reaches the end, filling in the MPS quantities as needed to avoid shortages. The 137 units will satisfy forecasted demands until week 7, when the inventory shortage in the absence of an MPS quantity is $7 + 0 - 35 = -28$. This shortage signals the need for another MPS quantity of 150 units. The updated inventory balance is $7 + 150 - 35 = 122$ chairs for week 7.

The last row in Figure 16.8 indicates the periods in which production of the MPS quantities must *begin* so that they will be available when indicated in the MPS quantity row. In the upper-right portion of the MPS record, a lead time of 1 week is indicated for the ladder-back chair; that is, 1 week is needed to assemble 150 ladder-back chairs, assuming that items B, C, D, and E are available. For each MPS quantity, the scheduler works backward through the lead time to determine when the assembly department must start producing chairs. Consequently, a lot of 150 units must be started in week 1 and another in week 6.

available-to-promise (ATP) inventory

The quantity of end items that marketing can promise to deliver on specified dates.

Available-to-Promise Quantities In addition to providing manufacturing with the timing and size of production quantities, the MPS provides marketing with information useful for negotiating delivery dates with customers. The quantity of end items that marketing can promise to deliver on specified dates is called **available-to-promise (ATP) inventory**. It is the difference between the customer orders already booked and the quantity that operations is planning to produce. As new customer orders are accepted, the ATP inventory is reduced to reflect the commitment of the firm to ship those quantities, but the actual inventory stays unchanged until the order is removed from inventory and shipped to the customer. An available-to-promise inventory is associated with each MPS quantity because the MPS quantity specifies the timing and size of new stock that can be earmarked to meet future bookings.

▼ **FIGURE 16.9**
MPS Record with an ATP Row

Item: Ladder-back chair — **Order Policy:** 150 units — **Lead Time:** 1 week

	April				May			
Quantity on Hand: 55	1	2	3	4	5	6	7	8
Forecast	30	30	30	30	35	35	35	35
Customer orders (booked)	38	27	24	8	0	0	0	0
Projected on-hand inventory	17	137	107	77	42	7	122	87
MPS quantity		150					150	
MPS start	150					150		
Available-to-promise (ATP) inventory	17	91					150	

Explanation: The total of customer orders booked until the next MPS receipt is 38 units. The ATP = 55 (on-hand) + 0 (MPS quantity) – 38 = 17.

Explanation: The total of customer orders booked until the next MPS receipt is 27 + 24 + 8 = 59 units. The ATP = 150 (MPS quantity) – 59 = 91 units.

Figure 16.9 shows an MPS record with an additional row for the available-to-promise quantities. The ATP in week 2 is the MPS quantity minus booked customer orders until the next MPS quantity, or $150 - (27 + 24 + 8 + 0 + 0) = 91$ units. The ATP indicates to marketing that, of the 150 units scheduled for completion in week 2, 91 units are uncommitted, and total new orders up to that quantity can be promised for delivery as early as week 2. In week 7, the ATP is 150 units because there are no booked orders in week 7 and beyond.

The procedure for calculating available-to-promise information is slightly different for the first (current) week of the schedule than for other weeks because it accounts for the inventory currently in stock. The ATP inventory for the first week equals *current on-hand inventory* plus the MPS quantity for the first week, minus the cumulative total of booked orders up to (but not including) the week in which the next MPS quantity arrives. So, in Figure 16.9, the ATP for the first week is $55 + 0 - 38 = 17$. This information indicates to the sales department that it can promise as many as 17 units this week, 91 more units sometime in weeks 2

through 6, and 150 more units in week 7 or 8. If customer order requests exceed ATP quantities in those time periods, the MPS must be changed before the customer orders can be booked or the customers must be given a later delivery date—when the next MPS quantity arrives. See Solved Problem 2 at the end of this chapter for an example of decision making using the ATP quantities.

Inventory planners do not create master production plans manually, although they thoroughly understand the logic built into them. Figure 16.10 is typical of the computer support available. It was created with the *Master Production Scheduling* Solver in OM Explorer, and confirms the output shown in Figure 16.9.

Lot Size	150								
Lead Time	1								
Quantity on Hand	55	1	2	3	4	5	6	7	8
Forecast		30	30	30	30	35	35	35	35
Customer Orders (Booked)		38	27	24	8				
Projected On-Hand Inventory		17	137	107	77	42	7	122	87
MPS Quantity			150					150	
MPS Start		150					150		
Available-to-Promise Inv (ATP)		17	91					150	

FIGURE 16.10 ▲
Master Production Scheduling Solver Output Using *OM Explorer*

Freezing the MPS The master production schedule is the basis of all end item, subassembly, component, and materials schedules. For this reason, changes to the MPS can be costly, particularly if they are made to MPS quantities soon to be completed. Increases in an MPS quantity can result in material shortages, delayed shipments to customers, and excessive expediting costs. Decreases in MPS quantities can result in unused materials or components (at least until another need for them arises) and valuable capacity being used to create products not needed. Similar costs occur when forecasted need dates for MPS quantities are changed. For these reasons, many firms, particularly those with a make-to-stock strategy and a focus on low-cost operations, *freeze*, or disallow changes to, the near-term portion of the MPS.

Reconciling the MPS with Sales and Operations Plans Because the master production schedule is based on both forecasts as well as actual orders received, it can differ from the sales and operations plan when summed across different periods in a month. For instance, in Figure 16.5, if the sum total MPS quantities of the three models of chairs in the month of April was 725 instead of 670, then either the management must revise the sales and operations plan upwards by authorizing additional resources to match supply with demand, or reduce the quantities of MPS in the month of April to match the sales and operations plan. Master production schedules drive plant and supplier activity, so they must be synchronized with actual customer demands and the sales and operations plans to ensure that the firm's planning decisions are actually being implemented on an ongoing basis.

MyOMLab

Tutor 16.1 in MyOMLab provides a new example to practice completing a master production schedule.

inventory record

A record that shows an item's lot-size policy, lead time, and various time-phased data.

gross requirements

The total demand derived from *all* parent production plans.

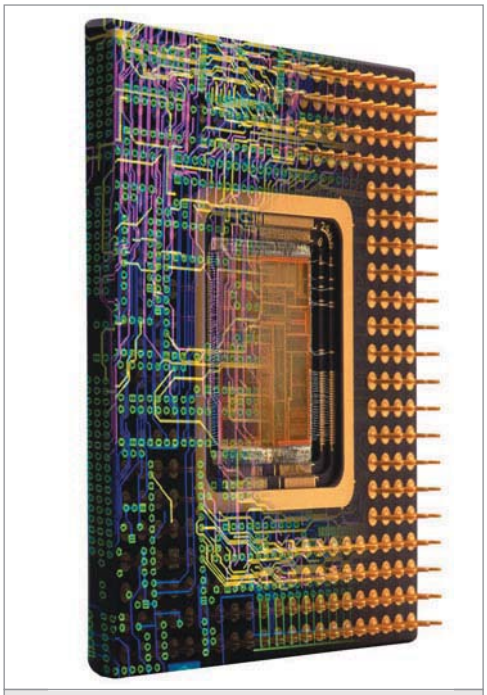
Inventory Record

Inventory records are a third major input to MRP, and inventory transactions are the basic building blocks of up-to-date records (see Figure 16.2). These transactions include releasing new orders, receiving scheduled receipts, adjusting due dates for scheduled receipts, withdrawing inventory, canceling orders, correcting inventory errors, rejecting shipments, and verifying scrap losses and stock returns. Recording the transactions accurately is essential if the firm's on-hand inventory balances are to be correct and its MRP system is to operate effectively.

The **inventory record** divides the future into time periods called *time buckets*. In our discussion, we use weekly time buckets for consistency with our MPS example, although other time periods could as easily be used. The inventory record shows an item's lot-size policy, lead time, and various time-phased data. The purpose of the inventory record is to keep track of inventory levels and component replenishment needs. The time-phased information contained in the inventory record consists of (1) *gross requirements*, (2) *scheduled receipts*, (3) *projected on-hand inventory*, (4) *planned receipts*, and (5) *planned order releases*.

We illustrate the discussion of inventory records with the seat subassembly, item C that was shown in Figure 16.4. Suppose that it is used in two products: a ladder-back chair and a kitchen chair.

Gross Requirements The **gross requirements** are the total demand derived from *all* parent production plans. They also include demand not otherwise accounted for, such as demand for replacement parts for units already sold. Figure 16.11 shows an inventory record for item C, the seat subassembly. Item C is produced in lots of 230 units and has a lead time of 2 weeks. The inventory



WidStock/Alamy

Cypress Semiconductor, a California-based company that manufactures logic devices, USB controllers, general-purpose programmable clocks, memories, and wireless connectivity solutions for consumer and automotive markets, uses commercial software solutions to manage the complexity of its master production scheduling processes.

Item: C
Description: Seat subassembly
Lot Size: 230 units
Lead Time: 2 weeks

		Week							
		1	2	3	4	5	6	7	8
Gross requirements		150	0	0	120	0	150	120	0
Scheduled receipts		230	0	0	0	0	0	0	0
Projected on-hand inventory	37	117	117	117	–3	–3	–153	–273	–273
Planned receipts									
Planned order releases									

Explanation:
Gross requirements are the total demand for the two chairs. Projected on-hand inventory in week 1 is 37 + 230 – 150 = 117 units.

FIGURE 16.11 ▲
MRP Record for the Seat Subassembly

record also shows item C's gross requirements for the next 8 weeks, which come from the master production schedule for the ladder-back and kitchen chairs (see Figure 16.5). The MPS start quantities for each parent are added to arrive at each week's gross requirements. The seat subassembly's gross requirements exhibit lumpy demand: Operations will withdraw seat subassemblies from inventory in only 4 of the 8 weeks.

The MRP system works with release dates to schedule production and delivery for components and subassemblies. Its program logic anticipates the removal of all materials required by a parent's production order from inventory at the beginning of the parent item's lead time—when the scheduler first releases the order to the shop.

Scheduled Receipts Recall that *scheduled receipts* (sometimes called open orders) are orders that have been placed but not yet completed. For a purchased item, the scheduled receipt could be in one of several stages: being processed by a supplier, being transported to the purchaser, or being inspected by the purchaser's receiving department. If the firm is making the item in-house, the order could be on the shop floor being processed, waiting for components, waiting for a machine to become available, or waiting to be moved to its next operation. According to Figure 16.11, one 230-unit order of item C is due in week 1. Given the 2-week lead time, the inventory planner probably released the order 2 weeks ago. Scheduled receipts due in beyond the item's lead time are unusual, caused by events such as a last-minute change in the MPS.

projected on-hand inventory
An estimate of the amount of inventory available each week after gross requirements have been satisfied.

Projected On-Hand Inventory The **projected on-hand inventory** is an estimate of the amount of inventory available each week after gross requirements have been satisfied. The beginning inventory, shown as the first entry (37) in Figure 16.11, indicates the on-hand inventory available at the time the record was computed. As with scheduled receipts, entries are made for each actual withdrawal and receipt to update the MRP database. Then, when the MRP system produces the revised record, the correct inventory will appear.

Other entries in the row show inventory expected in future weeks. Projected on-hand inventory is calculated as

$$\begin{pmatrix}\text{Projected on-hand}\\ \text{inventory balance}\\ \text{at end of week } t\end{pmatrix} = \begin{pmatrix}\text{Inventory on}\\ \text{hand at end of}\\ \text{week } t-1\end{pmatrix} + \begin{pmatrix}\text{Scheduled}\\ \text{or planned}\\ \text{reeipts in}\\ \text{week } t\end{pmatrix} - \begin{pmatrix}\text{Gross}\\ \text{requirements}\\ \text{in week } t\end{pmatrix}$$

planned receipts
Orders that are not yet released to the shop or the supplier.

The projected on-hand calculation includes the consideration of **planned receipts**, which are orders not yet released to the shop or the supplier. Planned receipts should not be confused with scheduled receipts. Planned receipts are still at the planning stage and can still change from one week to the next, whereas scheduled receipts are actual orders that are being acted upon by the shop or supplier. In Figure 16.11, the planned receipts are all zero. The on-hand inventory calculations for each week are as follows:

Week 1: $37 + 230 - 150 = 117$
Weeks 2 and 3: $117 + 0 - 0 = 117$
Week 4: $117 + 0 - 120 = -3$
Week 5: $-3 + 0 - 0 = -3$
Week 6: $-3 + 0 - 150 = -153$
Week 7: $-153 + 0 - 120 = -273$
Week 8: $-273 + 0 - 0 = -273$

In week 4, the balance drops to −3 units, which indicates that a shortage of 3 units will occur unless more seat subassemblies are built. This condition signals the need for a planned receipt to arrive in week 4. In addition, unless more stock is received, the shortage will grow to 273 units in weeks 7 and 8.[2]

Item: C
Description: Seat subassembly
Lot Size: 230 units
Lead Time: 2 weeks

	Week							
	1	2	3	4	5	6	7	8
Gross requirements	150	0	0	120	0	150	120	0
Scheduled receipts	230	0	0	0	0	0	0	0
Projected on-hand inventory (37)	117	117	117	227	227	77	187	187
Planned receipts				230			230	
Planned order releases		230			230			

Explanation:
Without a planned receipt in week 4, a shortage of 3 units will occur: 117 + 0 + 0 – 120 = –3 units. Adding the planned receipt brings the balance to 117 + 0 + 230 – 120 = 227 units. Offsetting for a 2-week lead time puts the corresponding planned order release back to week 2.

Explanation:
The first planned receipt lasts until week 7, when projected inventory would drop to 77 + 0 + 0 – 120 = –43 units. Adding the second planned receipt brings the balance to 77 + 0 + 230 – 120 = 187 units.
The corresponding planned order release is for week 5 (or week 7 minus 2 weeks).

▲ **FIGURE 16.12**
Completed Inventory Record for the Seat Subassembly

Planned Receipts Planning for the receipt of new orders will keep the projected on-hand balance from dropping below zero. The planned receipt row is developed as follows:

1. Weekly on-hand inventory is projected until a shortage appears. Completion of the initial planned receipt is scheduled for the week in which the shortage is projected. The addition of the newly planned receipt should increase the projected on-hand balance so that it equals or exceeds zero. It will exceed zero when the lot size exceeds requirements in the week it is planned to arrive.
2. The projection of on-hand inventory continues until the next shortage occurs. This shortage signals the need for the second planned receipt.

This process is repeated until the end of the planning horizon by proceeding column by column through the MRP record—filling in planned receipts as needed and completing the projected on-hand inventory row. Figure 16.12 shows the planned receipts for the seat subassembly. In week 4, the projected on-hand inventory will drop below zero, so a planned receipt of 230 units is scheduled for week 4. The updated inventory on-hand balance is 117 (inventory at end of week 3) + 230 (planned receipts) − 120 (gross requirements) = 227 units. The projected on-hand inventory remains at 227 for week 5 because no scheduled receipts or gross requirements are anticipated. In week 6, the projected on-hand inventory is 227 (inventory at end of week 5) − 150 (gross requirements) = 77 units. This quantity is greater than zero, so no new planned receipt is needed. In week 7, however, a shortage will occur unless more seat subassemblies are received. With a planned receipt in week 7, the updated inventory balance is 77 (inventory at end of week 6) + 230 (planned receipts) − 120 (gross requirements) = 187 units.

Planned Order Releases A **planned order release** indicates when an order for a specified quantity of an item is to be issued. We must place the planned order release quantity in the proper time bucket. To do so, we must assume that all inventory flows—scheduled receipts, planned receipts, and gross requirements—occur at the same point of time in a time period. Some firms assume that all flows occur at the beginning of a time period; other firms assume that they occur at the end of a time period or at the middle of the time period. Regardless of when the flows are assumed to occur, we find the release date by subtracting the lead time from the receipt date. For example, the release date for the first planned order release in Figure 16.12 is 4 (planned receipt date) −2 (lead time) = 2 (planned order release date). Figure 16.12 shows the planned order releases for the seat subassembly. If all goes according to the plan, we will release an order for 230 seat assemblies next week (in week 2). This order release sets off a series of updates to the inventory record. First, the planned order release for the order is removed. Next, the planned receipt for 230 units in week 4 is also removed. Finally, a new scheduled receipt for 230 units will appear in the scheduled receipt row for week 4.

planned order release
An indication of when an order for a specified quantity of an item is to be issued.

[2]There is an exception to the rule of scheduling a planned receipt whenever the projected inventory otherwise becomes negative. When a scheduled receipt is coming in *after* the inventory becomes negative, the first recourse is to expedite the scheduled receipt (giving it an earlier due date), rather than scheduling a new planned receipt.

Planning Factors

The planning factors in a MRP inventory record play an important role in the overall performance of the MRP system. By manipulating these factors, managers can fine-tune inventory operations. In this section, we discuss planning lead time, lot-sizing rules, and safety stock.

Planning Lead Time Planning lead time is an estimate of the time between placing an order for an item and receiving the item in inventory. Accuracy is important in planning lead time. If an item arrives in inventory sooner than needed, inventory holding costs increase. If an item arrives too late, stockouts, excessive expediting costs, or both may occur.

For purchased items, the planning lead time is the time allowed for receiving a shipment from the supplier after the order has been sent, including the normal time to place the order. Often, the purchasing contract stipulates the delivery date. For items manufactured in-house, a rough-cut estimate of the planning lead time can be obtained by keeping track of the actual lead times for recent orders and computing an average. A more extensive estimating process consists of breaking down each of the following factors:

- Setup time
- Processing time
- Materials handling time between operations
- Waiting time

Each of these times must be estimated for every operation along the item's route. Estimating setup, processing, and materials handling times can be relatively easy, but estimating the waiting time for materials handling equipment or for a workstation to perform a particular operation can be more difficult. In a facility that uses a make-to-order strategy, such as a machine shop, the load on the shop varies considerably over time, causing actual waiting times for a particular order to fluctuate widely. Therefore, being able to accurately estimate the waiting time is especially important when it comes to estimating the planning lead time. However, in a facility that uses a make-to-stock strategy, such as an assembly plant, product routings are more standard and waiting time is more predictable; hence, waiting time generally is a less-troublesome part of planning lead times.

Lot-Sizing Rules A lot-sizing rule determines the timing and size of order quantities. A lot-sizing rule must be assigned to each item before planned receipts and planned order releases can be computed. The choice of lot-sizing rules is important because they determine the number of setups required and the inventory holding costs for each item. We present three lot-sizing rules: (1) fixed order quantity, (2) periodic order quantity, and (3) lot-for-lot.

fixed order quantity (FOQ)

A rule that maintains the same order quantity each time an order is issued.

Fixed Order Quantity The **fixed order quantity (FOQ)** rule maintains the same order quantity each time an order is issued.[3] For example, the lot size might be the size dictated by equipment capacity limits, such as when a full lot must be loaded into a furnace at one time. For purchased items, the FOQ could be determined by the quantity discount level, truckload capacity, or minimum purchase quantity. Alternatively, the lot size could be determined by the economic order quantity (EOQ) formula (see Chapter 9, "Managing Inventories"). Figure 16.12 illustrated the FOQ rule. However, if an item's gross requirement within a week is particularly large, the FOQ might be insufficient to avoid a shortage. In such unusual cases, the inventory planner must increase the lot size beyond the FOQ, typically to a size large enough to avoid a shortage. Another option is to make the order quantity an integer multiple of the FOQ. This option is appropriate when capacity constraints limit production to FOQ sizes (at most).

periodic order quantity (POQ)

A rule that allows a different order quantity for each order issued but issues the order for predetermined time intervals.

Periodic Order Quantity The **periodic order quantity (POQ)** rule allows a different order quantity for each order issued but issues the order for predetermined time intervals, such as every two weeks. The order quantity equals the amount of the item needed during the predetermined time between orders and must be large enough to prevent shortages. Specifically, the POQ is

$$\begin{pmatrix}\text{POQ lot size}\\ \text{to arrive in}\\ \text{week } t\end{pmatrix} = \begin{pmatrix}\text{Total gross requirements}\\ \text{for } P \text{ weeks, including}\\ \text{week } t\end{pmatrix} - \begin{pmatrix}\text{Projected on-hand}\\ \text{inventory balance at}\\ \text{end of week } t-1\end{pmatrix}$$

This amount exactly covers P weeks' worth of gross requirements. That is, the projected on-hand inventory should equal zero at the end of the Pth week.

[3] The *kanban* system essentially uses a FOQ rule, except that the order quantity is very small.

Suppose that we want to switch from the FOQ rule used in Figure 16.12 to the POQ rule. Figure 16.13 was created with the *Single-Item MRP* Solver in OM Explorer. It shows the application of the POQ rule, with $P = 3$ weeks, to the seat subassembly inventory. The first order is required in week 4 because it is the first week that projected inventory balance will fall below zero. The first order using $P = 3$ weeks is

$$\text{(POQ lot size)} = \begin{pmatrix}\text{Gross requirements}\\ \text{for weeks}\\ 4, 5, \text{and } 6\end{pmatrix} - \begin{pmatrix}\text{Inventory at}\\ \text{end of week 3}\end{pmatrix}$$

$$= (120 + 0 + 150) - 117 = 153 \text{ units}$$

The second order must arrive in week 7 with a lot size of $(120 + 0) - 0 = 120$ units. This second order reflects only two weeks' worth of gross requirements—to the end of the planning horizon.

MyOMLab

Tutor 16.2 in MyOMLab provides a new example to practice lot-sizing decisions using FOQ, POQ, and L4L rules.

Periods 8

Item Description: Seat Assembly | Period (P) for POQ: 3 | Lot Size (FOQ): | Lead Time: 2

POQ Rule

		1	2	3	4	5	6	7	8
Gross requirements		150			120		150	120	
Scheduled receipts		230							
Projected on-hand inventory	37	117	117	117	150	150			
Planned receipts					153			120	
Planned order releases			153			120			

◀ **FIGURE 16.13**
The POQ ($P = 3$) Rule for the Seat Subassembly

The POQ rule does not mean that the planner must issue a new order every P weeks. Rather, when an order is planned, its lot size must be enough to cover P successive weeks. One way to select a P value is to divide the average lot size desired, such as the EOQ or some other applicable lot size, by the average weekly demand. That is, express the target lot size as the desired weeks of supply (P) and round to the nearest integer.

Lot for Lot A special case of the POQ rule is the **lot-for-lot (L4L) rule**, under which the lot size ordered covers the gross requirements of a single week. Thus, $P = 1$, and the goal is to minimize inventory levels. This rule ensures that the planned order is just large enough to prevent a shortage in the single week it covers. The L4L lot size is

lot-for-lot (L4L) rule
A rule under which the lot size ordered covers the gross requirements of a single week.

$$\begin{pmatrix}\text{L4L lot size}\\ \text{to arrive in}\\ \text{week } t\end{pmatrix} = \begin{pmatrix}\text{Gross requirements}\\ \text{for week } t\end{pmatrix} - \begin{pmatrix}\text{Projected on-hand}\\ \text{inventory balance at}\\ \text{end of week } t-1\end{pmatrix}$$

The projected on-hand inventory combined with the new order will equal zero at the end of week t. Following the first planned order, an additional planned order will be used to match each subsequent gross requirement.

This time we want to switch from the FOQ rule to the L4L rule. Figure 16.14 shows the application of the L4L rule to the seat subassembly inventory. As before, the first order is needed in week 4:

$$\text{(L4L lot size)} = \begin{pmatrix}\text{Gross requirements}\\ \text{in week 4}\end{pmatrix} - \begin{pmatrix}\text{Inventory balance}\\ \text{at end of week 3}\end{pmatrix}$$

$$= 120 - 117 = 3$$

Periods 8

Item Description: Seat Assembly | Period (P) for POQ: | Lot Size (FOQ): | Lead Time: 2

L4L Rule

		1	2	3	4	5	6	7	8
Gross requirements		150			120		150	120	
Scheduled receipts		230							
Projected on-hand inventory	37	117	117	117					
Planned receipts					3		150	120	
Planned order releases			3		150	120			

◀ **FIGURE 16.14**
The L4L Rule for the Seat Subassembly

The stockroom must receive additional orders in weeks 6 and 7 to satisfy each of the subsequent gross requirements. The planned receipt for week 6 is 150 and for week 7 is 120.

Comparing Lot-Sizing Rules Choosing a lot-sizing rule can have important implications for inventory management. Lot-sizing rules affect inventory costs and setup and ordering costs. The FOQ, POQ, and L4L rules differ from one another in one or both respects. In our example, each rule took effect in week 4, when the first order was placed. Let us compare the projected on-hand inventory averaged over weeks 4 through 8 of the planning horizon. The data are shown in Figures 16.12, 16.13, and 16.14, respectively.

$$\text{FOQ: } \frac{227 + 227 + 77 + 187 + 187}{5} = 181 \text{ units}$$

$$\text{POQ: } \frac{150 + 150 + 0 + 0 + 0}{5} = 60 \text{ units}$$

$$\text{L4L: } \frac{0 + 0 + 0 + 0 + 0}{5} = 0 \text{ units}$$

The performance of the L4L rule with respect to average inventory levels comes at the expense of an additional planned order and its accompanying setup time and cost. We can draw three conclusions from this comparison:

1. The FOQ rule generates a high level of average inventory because it creates inventory *remnants*. A remnant is inventory carried into a week, but it is too small to prevent a shortage. Remnants occur because the FOQ does not match requirements exactly. For example, according to Figure 16.12, the stockroom must receive a planned order in week 7, even though 77 units are on hand at the beginning of that week. The remnant is the 77 units that the stockroom will carry for 3 weeks, beginning with receipt of the first planned order in week 4. Although they increase average inventory levels, inventory remnants introduce stability into the production process by buffering unexpected scrap losses, capacity bottlenecks, inaccurate inventory records, or unstable gross requirements.
2. The POQ rule reduces the amount of average on-hand inventory because it does a better job of matching order quantity to requirements. It adjusts lot sizes as requirements increase or decrease. Figure 16.13 shows that in week 7, when the POQ rule has fully taken effect, the projected on-hand inventory is zero—no remnants.
3. The L4L rule minimizes inventory investment, but it also maximizes the number of orders placed. This rule is most applicable to expensive items or items with small ordering or setup costs. It is the only rule that can be used for a low-volume item made to order. It can also approximate the small-lot inventory levels of a lean system.

By avoiding remnants, both the POQ and the L4L rule may introduce instability by tying the lot-sizing decision so closely to requirements. If any requirement changes, so must the lot size, which can disrupt component schedules. Last-minute increases in parent orders may be hindered by missing components.

Safety Stock An important managerial decision is the quantity of safety stock to carry. It is more complex for dependent demand items than for independent demand items. Safety stock for dependent demand items with lumpy demand (gross requirements) is helpful only when future gross requirements, the timing or size of scheduled receipts, and the amount of scrap that will be produced are uncertain. As these uncertainties are resolved, safety stock should be reduced and ultimately eliminated. The usual policy is to use safety stock for end items and purchased items to protect against fluctuating customer orders and unreliable suppliers of components but to avoid using it as much as possible for intermediate items. Safety stocks can be incorporated in the MRP logic by using the following rule: Schedule a planned receipt whenever the projected on-hand inventory balance drops below the desired safety stock level (rather than zero, as before). The objective is to keep a minimum level of planned inventories equal to the safety stock quantity. Figure 16.15 shows what happens when the safety stock requirement has just been increased from 0 units to 80 units of safety stock for the seat assembly using an FOQ of 230 units. The beginning projected on-hand quantity is still 37 units when the safety stock policy is introduced, and cannot fall below 80 units in any future period thereafter. Compare the

▼ **FIGURE 16.15**
Inventory Record for the Seat Subassembly Showing the Application of a Safety Stock

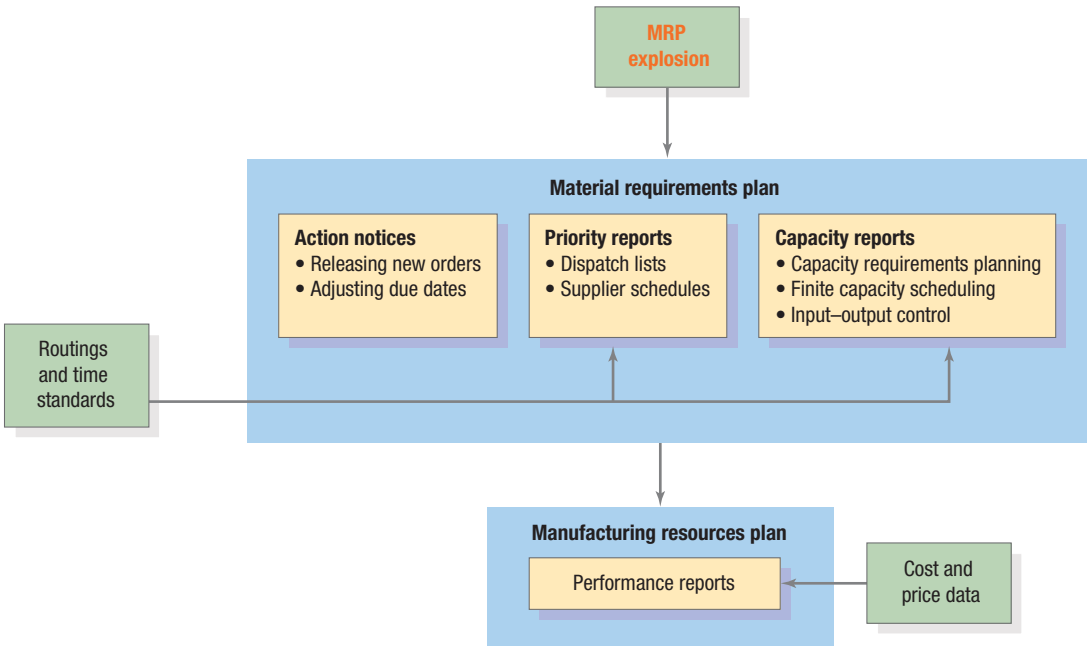
FOQ Rule — **Lot Size:** 230 units; **Lead Time:** 2 weeks; **Safety Stock:** 80 units

	Week							
	1	2	3	4	5	6	7	8
Gross requirements	150	0	0	120	0	150	120	0
Scheduled receipts	230	0	0	0	0	0	0	0
Projected on-hand inventory 37	117	117	117	227	227	307	187	187
Planned receipts	0	0	0	230	0	230	0	0
Planned order releases	0	230	0	230	0	0	0	0

results in Figure 16.15 to Figure 16.12. The net effect is to move the second planned order release from week 5 to week 4 to avoid dropping below 80 units in week 6.

Outputs from MRP

MRP systems provide many reports, schedules, and notices to help planners control dependent demand inventories, as indicated in Figure 16.16. In this section, we discuss the MRP explosion process, notices that alert planners to items needing attention, resource requirement reports, and performance reports.



◀ **FIGURE 16.16**
MRP Outputs

MRP Explosion MRP translates, or *explodes*, the MPS and other sources of demand into the requirements needed for all of the subassemblies, components, and raw materials the firm needs to produce parent items. This process generates the material requirements plan for each component item.

An item's gross requirements are derived from three sources:

1. The MPS for immediate parents that are end items
2. The planned order releases (*not* the gross requirements, scheduled receipts, or planned receipts) for immediate parents below the MPS level
3. Any other requirements not originating in the MPS, such as the demand for replacement parts

Consider the seat subassembly and its inventory record shown in Figure 16.12. The seat subassembly requires a seat cushion and a seat frame, which in turn needs four seat-frame boards. Its BOM is shown in Figure 16.17 (see also Figure 16.4, which shows how the seat subassembly BOM relates to the product as a whole). How many seat cushions should we order from the supplier? How many seat frames should we produce to support the seat subassembly schedule? How many seat-frame boards do we need to make? The answers to these questions depend on the existing inventories of these items and the replenishment orders already in progress. MRP can help answer these questions through the explosion process.

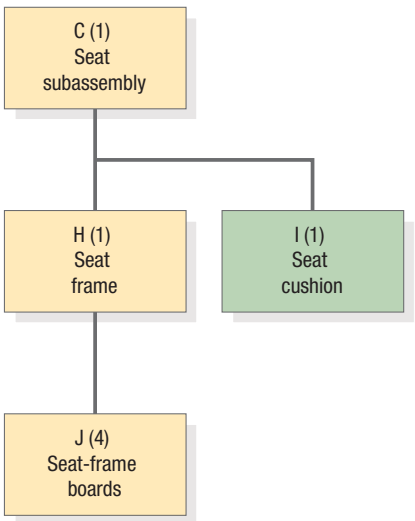
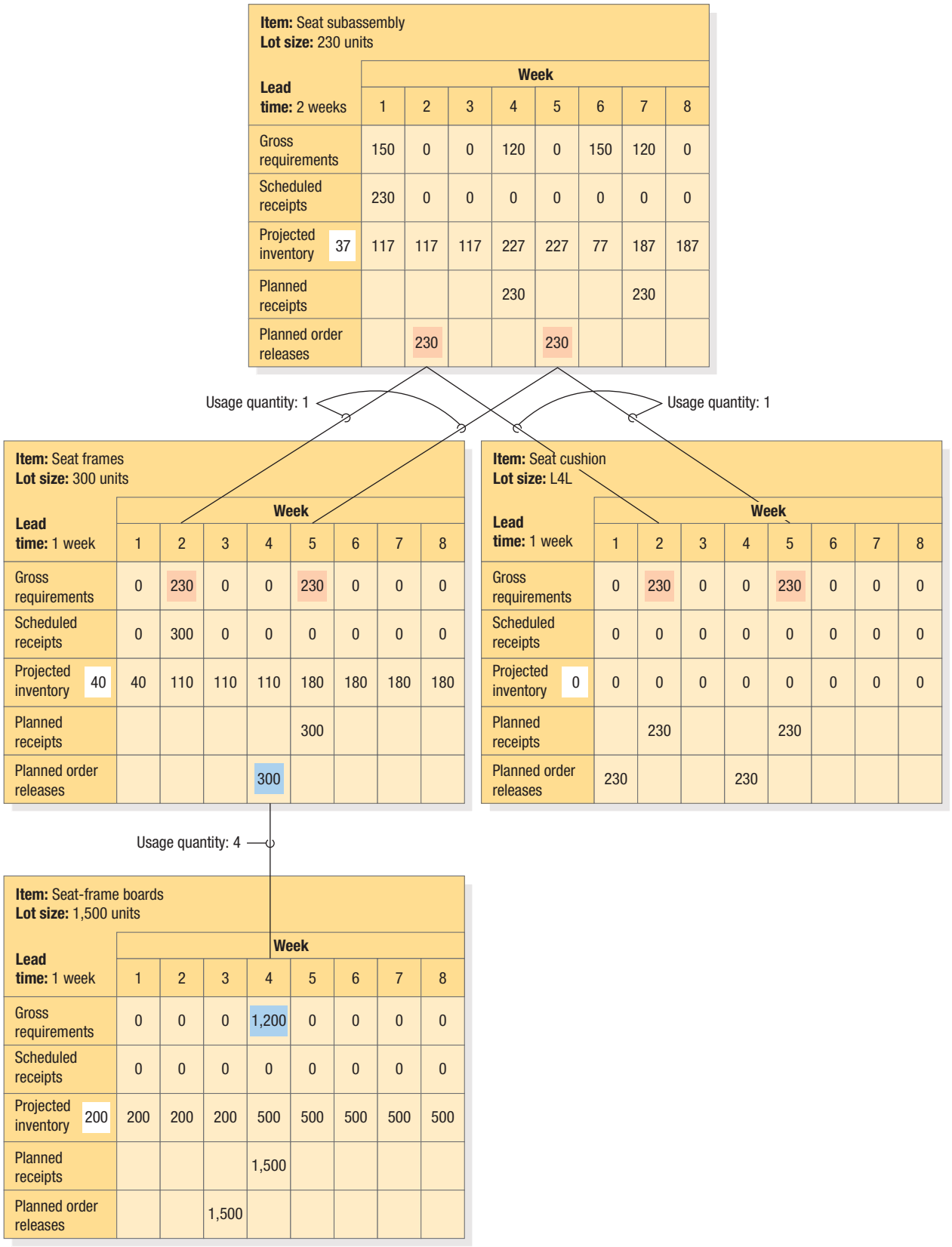


FIGURE 16.17 ▲
BOM for the Seat Subassembly

Figure 16.18 shows the MRP records for the seat subassembly and its components. We already showed how to develop the MRP record for the seat subassembly. We now concentrate on the MRP records of its components. The lot-size rules are an FOQ of 300 units for the seat frame, L4L for the seat cushion, and an FOQ of 1,500 for the seat-frame boards. All three components have a 1-week lead time. The key to the explosion process is to determine the proper timing and size of the gross requirements for each component. After we make those determinations, we can derive the planned order release schedule for each component by using the logic already demonstrated.



Item: Seat subassembly
Lot size: 230 units
Lead time: 2 weeks

	Week							
	1	2	3	4	5	6	7	8
Gross requirements	150	0	0	120	0	150	120	0
Scheduled receipts	230	0	0	0	0	0	0	0
Projected inventory 37	117	117	117	227	227	77	187	187
Planned receipts				230			230	
Planned order releases		230			230			

Usage quantity: 1

Usage quantity: 1

Item: Seat frames
Lot size: 300 units
Lead time: 1 week

	Week							
	1	2	3	4	5	6	7	8
Gross requirements	0	230	0	0	230	0	0	0
Scheduled receipts	0	300	0	0	0	0	0	0
Projected inventory 40	40	110	110	110	180	180	180	180
Planned receipts					300			
Planned order releases				300				

Item: Seat cushion
Lot size: L4L
Lead time: 1 week

	Week							
	1	2	3	4	5	6	7	8
Gross requirements	0	230	0	0	230	0	0	0
Scheduled receipts	0	0	0	0	0	0	0	0
Projected inventory 0	0	0	0	0	0	0	0	0
Planned receipts		230			230			
Planned order releases	230			230				

Usage quantity: 4

Item: Seat-frame boards
Lot size: 1,500 units
Lead time: 1 week

	Week							
	1	2	3	4	5	6	7	8
Gross requirements	0	0	0	1,200	0	0	0	0
Scheduled receipts	0	0	0	0	0	0	0	0
Projected inventory 200	200	200	200	500	500	500	500	500
Planned receipts				1,500				
Planned order releases			1,500					

FIGURE 16.18 ▲
MRP Explosion of Seat Assembly Components

In our example, the components have no independent demand for replacement parts. Consequently, in Figure 16.18 the gross requirements of a component come from the planned order releases of its parents. The seat frame and the seat cushion get their gross requirements from the planned order release schedule of the seat subassembly. Both components have gross requirements of 230 units in weeks 2 and 5, the same weeks in which we will be releasing orders to make more seat subassemblies. In week 2, for example, the materials handler for the assembly department will withdraw 230 seat frames and 230 seat cushions from inventory so that the assembly department can produce the seat subassemblies in time to avoid a stockout in week 4. The materials plans for the seat frame and the seat cushion must allow for that.

Using the gross requirements in weeks 2 and 5, we can develop the MRP records for the seat frame and the seat cushion, as shown in Figure 16.18. For a scheduled receipt of 300 seat frames in week 2, an on-hand quantity of 40 units, and a lead time of 1 week, we need to release an order of 300 seat frames in week 4 to cover the assembly schedule for the seat subassembly. The seat cushion has no scheduled receipts and no inventory on hand; consequently, we must place orders for 230 units in weeks 1 and 4, using the L4L logic with a lead time of 1 week.

After determining the replenishment schedule for the seat frame, we can calculate the gross requirements for the seat-frame boards. We plan to begin producing 300 seat frames in week 4. Each frame requires 4 boards, so we need to have $300(4) = 1{,}200$ boards available in week 4. Consequently, the gross requirement for seat-frame boards is 1,200 in week 4. Given no scheduled receipts, 200 boards in stock, a lead time of 1 week, and an FOQ of 1,500 units, we need a planned order release of 1,500 in week 3.

The questions posed earlier can now be answered. We should plan to release the following orders: 300 seat frames in week 4; 230 seat cushions in each of weeks 1 and 4; and 1,500 seat-frame boards in week 3. If MRP plans are updated weekly, only the planned order for week 1 should be released now. Releasing it creates a scheduled receipt of 230 seat cushions that will appear in the updated inventory record. The other orders remain in the planning stage, and even might be revised by the MRP explosion done next week.

Computer Support In practice a company can have thousands of dependent demand items with an average of six bills of materials levels. Time horizons often stretch out for 30 or more time periods into the future. Doing a MRP explosion by hand (as shown in Figure 16.18) would be impractical. What is needed is massive data processing, the very thing that computers do best, leaving the decision making to the inventory analyst. The *Material Requirements Planning* Solver of OM Explorer in MyOMLab represents a small example of what is done on a much larger scale by commercial packages, can compute requirements for up to two end items. It has the ability to develop inventory records up to 18 items deep with little effort, and can easily re-compute these requirements if there is any change in planning parameters.

MyOMLab

Figure 16.19 shows the output from the *Material Requirements Planning* Solver of OM Explorer. It confirms the same results as Figure 16.18. Based on this output, only one thing would be brought to the attention of the inventory planner—the planned order release of 230 units for seat cushions. Its planned order release is now "mature" for release this week. Unless the planner knows of a problem, the planner would place the order with the supplier. At the same time, the planner would input a transaction that automatically eliminates the planned order release in period 1, removes the planned receipt in period 2, and inserts a scheduled receipt for 230 units in period 2. The planner need not look at the records for seat frames or seat-frame boards, because no action is needed for them.

Other Important Reports Once computed, inventory records for any item appearing in the BOM can be printed in hard copy or displayed on a computer video screen. Inventory planners use a computer-generated memo called an **action notice** to make decisions about releasing new orders and adjusting the due dates of scheduled receipts. These notices are generated every time the system is updated, typically once per week. The action notice alerts planners to only the items that need their attention, such as those items that have a planned order release in the current period or a scheduled receipt that needs its due date revised. Planners can then view the full records for those items and take the necessary actions. An action notice can simply be a list of part numbers for items that need attention; or it can be the full record for such items, with a note at the bottom identifying the action needed.

action notice

A computer-generated memo alerting planners about releasing new orders and adjusting the due dates of scheduled receipts.

By itself, the MRP system does not recognize capacity limitations when computing planned orders; that is, it may call for a planned order release that exceeds the amount that can be physically produced. An essential role of planners is to monitor the capacity requirements of material requirements plans, adjusting a plan when it cannot be met. Particular attention is paid to bottlenecks. The planner can apply theory of constraints (TOC) principles (see Chapter 7, "Managing Process Constraints") to keep bottleneck operations fed by adjusting some lot sizing rules or occasionally overriding planned order releases. To facilitate this process, various types of capacity reports can be provided. For example, **capacity requirements planning (CRP)** reports

capacity requirements planning (CRP)

A technique used for projecting time-phased capacity requirements for work stations; its purpose is to match the materials requirements plan with the capacity of key processes.

FIGURE 16.19 ▶
Material Requirements Planning Solver Output of *OM Explorer* for Seat Assembly Components

Master Production Schedule

		1	2	3	4	5	6	7	8	9	10
Item A MPS Start	Descr: Seat		230			230					
Item B MPS Start	Descr:										

☑ Use second finished item

Material Requirements Planning

Item C — Descr: Seat Frames — Period (P) for POQ — Lot Size (FOQ): 300; Lead Time: 1; Safety Stock
FOQ Rule
Usage Quantity for Item: A 1 B

		1	2	3	4	5	6	7	8	9	10
Gross Requirements			230			230					
Scheduled Receipts			300								
Projected On-Hand Inventory	40	40	110	110	110	180	180	180	180	180	180
Planned Receipts						300					
Planned Order Releases					300						

Item D — Descr: Seat Cushion — Period (P) for POQ: 1 — Lot Size (FOQ); Lead Time: 1; Safety Stock
L4L Rule
Usage Quantity for Item: A 1 B C

		1	2	3	4	5	6	7	8	9	10
Gross Requirements			230			230					
Scheduled Receipts											
Projected On-Hand Inventory	0	0	0	0	0	0	0	0	0	0	0
Planned Receipts			230			230					
Planned Order Releases		230			230						

Item E — Descr: Seat-frame boards — Period (P) for POQ — Lot Size (FOQ): 1500; Lead Time: 1; Safety Stock
FOQ Rule
Usage Quantity for Item: A B C 4 D

		1	2	3	4	5	6	7	8	9	10
Gross Requirements					1200						
Scheduled Receipts											
Projected On-Hand Inventory	200	200	200	200	500	500	500	500	500	500	500
Planned Receipts					1500						
Planned Order Releases				1500							

project time-phased capacity requirements for workstations. They calculate workload according to the work required to complete the scheduled receipts already in the shop and to complete the planned order releases not yet released. Bottlenecks are those workstations at which the projected loads exceed station capacities.

Other types of outputs are also possible, such as priority reports on orders already placed to the shop or with suppliers. Priority reports begin with the due dates assigned to scheduled receipts, which planners keep up to date so that they continue to reflect when receipt is really needed. On a broader scale, the information in an MRP system is useful to functional areas other than operations. MRP evolved into **manufacturing resource planning (MRP II)**, a system that ties the basic MRP system to the company's financial system and to other core and supporting processes. For example, management can project the dollar value of shipments, product costs, overhead allocations, inventories, backlogs, and profits by using the MRP plan along with prices and product and activity costs from the accounting system. Also, information from the MPS, scheduled receipts, and planned orders can be converted into cash flow projections, which are broken down by product families. Similar computations are possible for other performance measures of interest to management. In fact, MRP II ultimately evolved into enterprise resource planning (ERP), which was introduced at the beginning of this chapter. Some firms may, however, forego the cost of vendor-delivered MRP and ERP systems because of the huge budgets and company resources involved in their deployment, and instead create their own MRP system implementations in-house. Managerial Practice 16.1 illustrates how Winnebago created and adapted its own homegrown software for the MRP system to achieve business self sufficiency.

manufacturing resource planning (MRP II)
A system that ties the basic MRP system to the company's financial system and to other core and supporting processes.

MANAGERIAL PRACTICE 16.1 Material Requirements Planning at Winnebago Industries

Winnebago Industries, based in Forest City, Iowa, is a leading manufacturer of motor homes and related products and services in the United States since 1958. In a challenging business environment, the firm improves the quality of its products at lowered production costs by emphasizing employee teamwork and involvement. So it is no surprise that Winnebago has for decades kept away from software packages, preferring instead to create and adapt its own applications. It can thereby achieve a closer fit to the company's business needs, an objective that is at times not met by the ERP vendors described at the beginning of this chapter. It can also be more responsive to users of its systems at lowered costs since it can reuse its code as needed.

An example of Winnebago's homegrown approach is its MRP system, which runs on an IBM zSeries mainframe and is used to plan material needs and schedule production orders. When the company added a new model to its fleet of motor homes, it took only a few hundred hours of development, including a new bill of materials, to make the required changes to the MRP system in order to support production of the new vehicle. Such agility in adapting the MRP system to manufacturing and supplier needs can be especially rewarding in an environment where the change in product variety is constant. In 2007 alone, Winnebago offered 20 different models in 86 different floor plans, each of which can be outfitted with a variety of options including colors, wood stains, and drawer pulls. For the 2011 model year, several new floorplans were added. When you are in the business of building customized homes-on-wheels ranging in prices from $60,000 to $285,000, combining cost consciousness with common sense and employee involvement in developing your own MRP, sales order management, and purchasing systems can prove valuable.



CRAIG BORCK KRT/Newscom

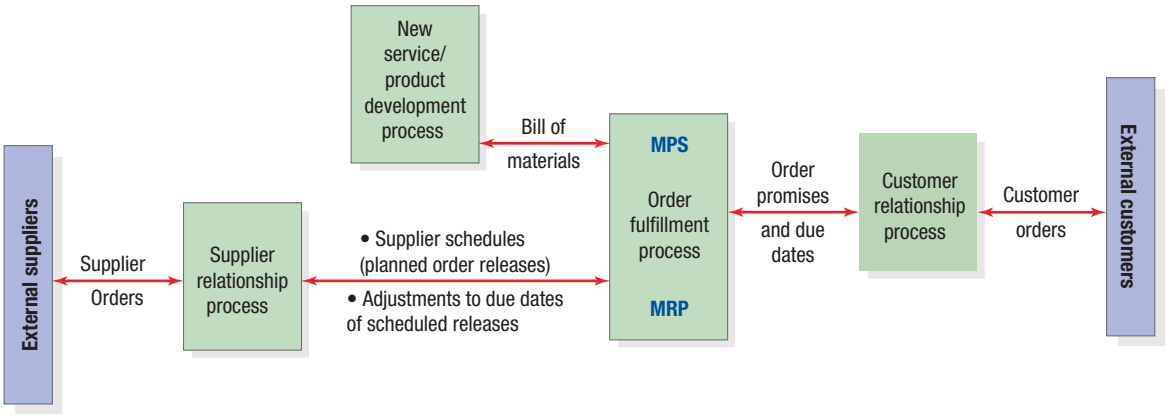
Workmen at the Winnebago factory in Forest City, Iowa, lift a completed side of a Winnebago, with interior and exterior surfaces, wiring, framework and foam insulation all in place, onto a cart so it can be hauled to the next step in the assembly process. Since 1995, Winnebago has spent approximately $5 million annually to automate some of its assembly-line processes, up from $2 million per year previously.

Source: "Road Rules: Creating and Adapting Homegrown Software is the Key to Winnebago's Drive for Business Self Sufficiency," October 15, 2006; **http://www.winnebagoind.com/** (May 28, 2011).

MRP, Core Processes, and Supply Chain Linkages

Among the four core processes of an organization that link activities within and across firms in a supply chain, the MRP system interacts with all of them either through its inputs or its outputs. It all begins with customer orders, which consist of orders for end items as well as replacement parts. MRP and resource planning typically reside inside the order fulfillment process. Master Production Scheduling is an integral part of MRP (see Figure 16.2). As shown schematically in Figure 16.20, the MPS drives the feedback between the order fulfillment process and the customer relationship process through confirmation of order receipts and promised due dates. MPS also provides guidance to the sales group within the customer relationship process with respect to when future orders can be promised, and whether the due dates for existing orders can be adjusted in the time frame requested. The new service and product development process provides an updated bill of materials to the MRP system, and makes sure that every component and assembly needed for manufacturing of end items is properly recognized.

FIGURE 16.20 ▶
MRP Related Information Flows in the Supply Chain



In a similar vein on the inbound side, orders to external suppliers are based on the planned order releases, which come directly from the output of MRP reports. The power of MRP, however, becomes evident when changes to an existing schedule are needed. These changes can be generated, for example, by changes to the MPS because a customer wants to change the timing or size of future orders, by some internal failure such as material shortages or unexpected machine downtime, or by supplier failure. In a supply chain, schedule changes have implications for customers as well as suppliers. Some firms, in partnership with their suppliers, have ERP/MRP systems that can actually "see" into their suppliers' inventory to determine if a particular item is in stock or, if not, when it can be expected. This is an advantage when contemplating a change to the original schedule of order releases. While systems such as this are powerful tools for making changes, care must be taken to avoid unnecessary fluctuations in the timing and size of PORs because of the choice of lot sizing policy. As we have seen, lot sizing rules such as POQ or L4L are susceptible to changes in requirements, and using them indiscriminately can cause instability in replenishment orders. In turn, this instability can be transmitted up the supply chain if the firm's MRP system is electronically linked to the production planning and control systems of its immediate suppliers.

Execution of MRP-based plans using the information flows between core processes as shown in Figure 16.20 properly link a firm with its upstream and downstream supply chain partners. Valid customer and supplier priorities would not be effectively recognized without such an MRP-based framework, which in many firms is actually implemented through an ERP system discussed earlier in the chapter.

MRP and the Environment

Consumer and governmental concern about the deterioration of the natural environment has driven manufacturers to reengineer their processes to become more environmentally friendly. The recycling of base materials is becoming more commonplace, and products are being designed in such a way that they can be remanufactured after their useful lives. Nonetheless, manufacturing processes often produce a number of waste materials that need to be properly disposed of. Wastes come in many forms:

- Effluents, such as carbon monoxide, sulfur dioxide, and hazardous chemicals associated with the processes used to manufacture the product
- Materials, such as metal shavings, oils, and chemicals associated with specific operations
- Packaging materials, such as unusable cardboard and plastics associated with certain products or purchased items
- Scrap associated with unusable products or component defects generated by the manufacturing process

Companies can modify their MRP systems to help them track these wastes and plan for their disposal. The type and amount of waste associated with each item can be entered into its BOM by treating the waste much like you would a component of the item. When the MPS is developed for a product, reports can be generated that project the amount of waste expected during the production process and when it will occur. Although this approach can require that a firm's BOM be modified substantially, the benefits are also substantial. Firms can identify their waste problems in advance to eliminate them in some cases (through process improvement efforts) or plan for their proper disposal in others. It also gives the firm a way to generate any formal documentation required by the government to verify that it has complied with environmental laws and policies.

Resource Planning for Service Providers

We have seen how manufacturing companies can disaggregate an MPS of finished products, which in turn must be translated into the needs for resources, such as staff, equipment, components, and financial assets. The driver for these resource requirements is a material requirements plan. Service providers, of course, must plan their resources just as manufacturers do. However, unlike finished goods, services cannot be inventoried. They must be provided on demand. In terms of resource planning then, service organizations must focus on maintaining the *capacity* to serve their customers. In this section, we will discuss how service providers use the concept of dependent demand and a bill of resources in managing capacity.

Dependent Demand for Services

When we discussed planning and control systems for manufacturers earlier in this chapter, we introduced the concept of *dependent demand*, which is demand for an item that is a function of the production plans for some other item the company produces. For service resource planning, it is useful to define the concept of dependent demand to include demands for resources that are driven by forecasts of customer requests for services or by plans for various activities in support of the services the company provides. Here are some other examples of dependent demands for service providers.

Restaurants Every time you order from the menu at a restaurant, you initiate the restaurant's need for certain types of goods (uncooked food items, plates, and napkins), staff (chef, servers, and dishwashers), and equipment (stoves, ovens, and cooking utensils). Using a forecast of the demand for each type of meal, the manager of the restaurant can estimate the need for these resources. Many restaurants, for example, feature "specials" on certain days, say, fish on Fridays or prime rib on Saturdays. Specials improve the accuracy of the forecasts managers need to make for different types of meals (and the food products that are required to make them) and typically signal the need for above-average staffing levels. How much of these resources will be needed, however, depends on the number of meals the restaurant ultimately expects to serve. As such, these items—food products and staff members—are dependent demands.

Airlines Whenever an airline schedules a flight, certain supporting goods are needed (beverages, snacks, and fuel), labor (pilots, flight attendants, and airport services), and equipment (a plane and airport gate). The number of flights and passengers the airline forecasts it will serve determines the amount of these resources needed. Just like a manufacturer, the airline can explode its master schedule of flights to make this determination.

Hospitals With the exception of the emergency room services, hospitals can use their admission appointments to create a master schedule. The master schedule can be exploded to determine the resources the hospital will need during a certain period. For example, when you schedule a surgical procedure, you generate a need for facilitating goods such as medicines, surgical gowns, linens, staff (a surgeon, nurses, and anesthesiologist), and equipment (an operating room, surgical tools, and recovery bed). As they build their master schedules, hospitals must ensure that certain equipment and personnel do not become overcommitted—that capacity is maintained, in other words. For example, an appointment for a key operation might have to be scheduled in advance at a time a surgeon is available to do it, even though the hospital's other resources—operating room, nurses, and so forth—might be currently be available.



Doctors and nurses in operation room during surgery

ochen Tack/PhotoLibrary

Hotels A traveler who makes a reservation at a hotel generates demand for facilitating goods (soap and towels), staff (front desk, housekeeping, and concierge), and equipment (fax, television, and exercise bicycle). To determine its dependent resource needs, a hotel adds the number of reservations already booked to the number of "walk-in" customers it forecasts it will have. This figure is used to create the hotel's master schedule. One resource a hotel cannot easily adjust, however, is the number of rooms it has. If the hotel is overbooked, for instance, it cannot simply add more rooms. If it has too few guests, it cannot "downsize" its number of rooms. Given the high capital costs needed

for this resource, hotels try to maintain as high a utilization rate as possible by offering group rates or special promotions at certain times of the year. In other words, they try to drive up dependent demand for this particular resource.

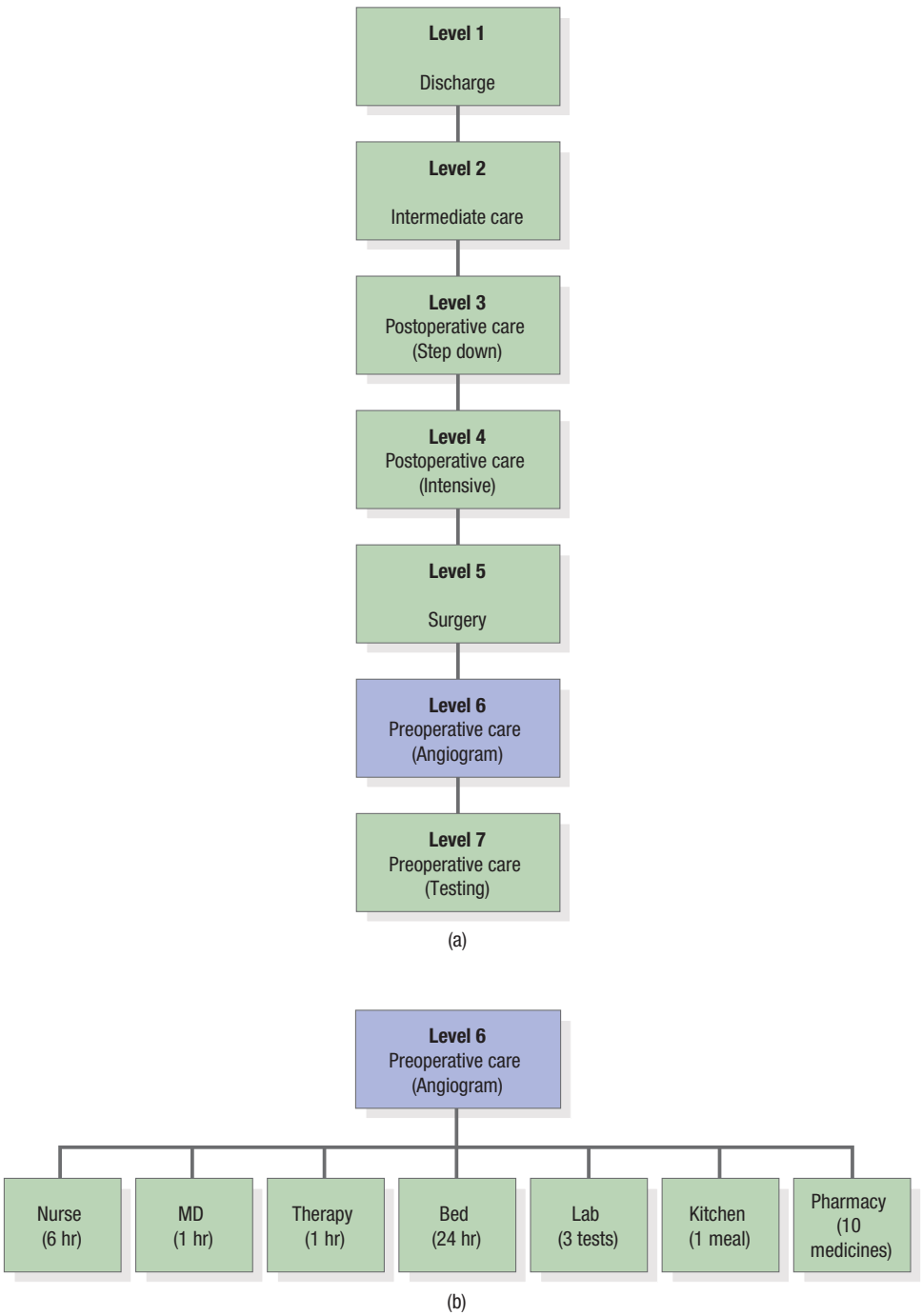
Bill of Resources

bill of resources (BOR)

A record of a service firm's parent-component relationships and all of the materials, equipment time, staff, and other resources associated with them, including usage quantities.

The service analogy to the bill of materials in a manufacturing company is the **bill of resources (BOR)**, which is a record of a service firm's parent–component relationships and all of the materials, equipment time, staff, and other resources associated with them, including usage quantities. Once the service firm has completed its master schedule, the BOR can be used to determine what resources the firm will need, how much of them it will need, and when. A BOR for a service provider can be as complex as a BOM for a manufacturer. Consider a hospital that just scheduled treatment of a patient with an aneurysm. As shown in Figure 16.21 (a), the BOR for treatment of an aneurysm has seven levels, starting at the top (end item): (1) discharge, (2) intermediate care, (3) postoperative care (step down), (4) postoperative care (intensive), (5) surgery, (6) preoperative

FIGURE 16.21 ▶
BOR for Treating an Aneurysm



care (angiogram), and (7) preoperative care (testing). Each level of the BOR has a set of material and resource requirements and an associated lead time. For example, at level 6, shown in Figure 16.21(b), the patient needs 6 hours of nurses' time, 1 hour of the primary MD's time, 1 hour of the respiratory therapist's time, 24 hours of bed time, 3 different lab tests, 1 meal, and 10 different medicines from the pharmacy. The lead time for this level is 1 day. The lead time for the entire stay for treatment of the aneurysm is 12.2 days. A master schedule of patient admissions and the BORs for each illness enable the hospital to manage its critical resources. Reports analogous to the MRP II reports we discussed earlier in the chapter can be generated for the people who manage the various functional areas of the hospital.

One resource every service provider needs, however, is cash. Service organizations have to forecast the number of customers they expect to serve so that they have enough cash on hand to purchase materials that support the services—labor and other products. Purchasing these items increases the firm's accounts payable. As services are actually completed for customers, the firm's accounts receivable increases. The firm's master schedule and its accounts receivable and payable help a company predict the amount and timing of its cash flows.

LEARNING GOALS IN REVIEW

1. **Explain how enterprise resource planning (ERP) systems can foster better resource planning**. Review the opening vignette on Dow Corning, and the section on "Enterprise Resource Planning," pp. 564–566. Pay attention to Figure 16.1 to understand how different application modules come together to create functionality and value in the ERP systems.
2. **Explain how the concept of dependent demand is fundamental to resource planning**. See the section on "Dependent Demand," p. 567, which shows how continuous independent demand can lead to lumpy requirements for dependent demand. Then, a separate system, called material requirements planning, is needed to manage dependent demand situations.
3. **Describe a master production schedule (MPS) and compute available-to-promise quantities.** The section "Master Production Scheduling," pp. 569–573, shows you how firms break down a production plan into more detailed schedules. Understand the key relationships between Figures 16.5, 16.6, and 16.8.
4. **Apply the logic of a material requirements planning (MRP) system to identify production and purchase orders needed for dependent demand items**. Using Figure 16.12, page 575, understand how an inventory record is created for a given lot size rule. The section on "Planning Factors," pp. 576–579, shows you how choice of different managerial policies affect material plans. Finally, focus on understanding the MRP explosion process as illustrated in Figure 16.18 on page 580 and Solved Problem 3 on pages 590–591.
5. **Apply MRP principles to the provision of services and distribution inventories.** The section, "Resource Planning for Service Providers," pp. 585–587, illustrates how the Bill of Resources can be used to plan dependent demand for services in settings such as the restaurants, airlines, hospitals, and hotels.

MyOMLab helps you develop analytical skills and assesses your progress with multiple problems on bills of material, parents, intermediate items, components, purchased items, lead times, master production schedules, MRP record, FOQ, L4L, POQ, scheduled receipts, and multi-level material requirements plans.

MyOMLab Resources	Titles	Link to the Book
Video	*Nantucket Nectars: ERP*	Enterprise Resource Planning
Active Model Exercise	16.1 Material Requirements Plan	Bill of Resources
OM Explorer Solvers	Master Production Scheduling Material Requirements Planning Single-Item MRP	Master Production Scheduling; Solved Problem 2 (pp. 589–590) Material Requirements Planning; Solved Problem 3 (pp. 590–591) Material Requirements Planning; Solved Problem 3 (pp. 590–591)
OM Explorer Tutors	16.1 Master Production Scheduling 16.2 FOQ, POQ, and L4L Rules	Master Production Scheduling; Solved Problem 2 (pp. 589–590) Planning Factors; Solved Problem 3 (pp. 590–591)
Tutor Exercises	16.1 Applying Different MRP Lot Sizing Rules	Planning Factors; Solved Problem 3 (pp. 590–591)
Virtual Tours	Vaughn Hockey Equipment Winslow Life Rafts	Bill of Materials Enterprise Resource Planning

MyOMLab Resources	Titles	Link to the Book
MyOMLab Supplements	H. Measuring Output Rates I. Learning Curve Analysis	MyOMLab Supplement H MyOMLab Supplement I
Internet Exercise	Schwinn Hewlett-Packard	Material Requirements Planning Material Requirements Planning
Key Equations		
Image Library		

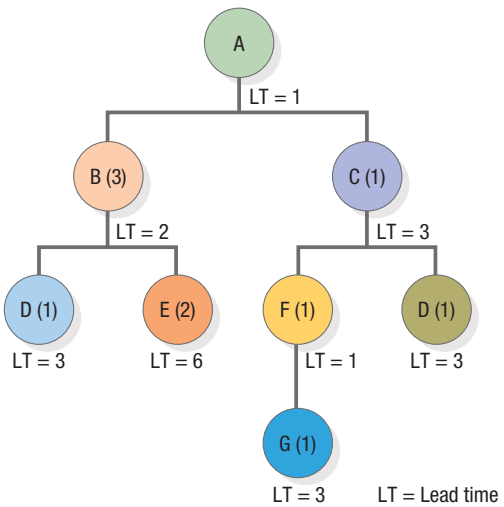
Key Terms

action notice 581
available-to-promise (ATP) inventory 572
bill of materials (BOM) 567
bill of resources (BOR) 586
capacity requirements planning (CRP) 581
component 567
dependent demand 567
end item 568
enterprise process 564
enterprise resource planning (ERP) systems 564
fixed order quantity (FOQ) 576
gross requirements 573
intermediate item 568
inventory record 573
lot-for-lot (L4L) rule 577
manufacturing resource planning (MRP II) 582
master production schedule (MPS) 569
material requirements planning (MRP) 566
MRP explosion 566
parent 567
part commonality 569
periodic order quantity (POQ) 576
planned order release 575
planned receipts 574
projected on-hand inventory 574
purchased item 569
resource planning 564
subassembly 568
usage quantity 568

Solved Problem 1

Refer to the bill of materials for product A shown in Figure 16.22.

FIGURE 16.22 ▶
BOM for Product A



If there is no existing inventory and no scheduled receipts, how many units of items G, E, and D must be purchased to produce 5 units of end item A?

SOLUTION

Five units of item G, 30 units of item E, and 20 units of item D must be purchased to make 5 units of A. The usage quantities shown in Figure 16.22 indicate that 2 units of E are needed to make 1 unit of B and that 3 units of B are needed to make 1 unit of A; therefore, 5 units of A require 30 units of E ($2 \times 3 \times 5 = 30$). One unit of D is consumed to make 1 unit of B, and 3 units of B per unit of A result in 15 units of D ($1 \times 3 \times 5 = 15$); 1 unit of D in each unit of C and 1 unit of C per unit of A result in another 5 units of D ($1 \times 1 \times 5 = 5$). The total requirements to make 5 units of A are 20 units of D ($15 + 5$). The calculation of requirements for G is simply $1 \times 1 \times 1 \times 5 = 5$ units.

Solved Problem 2

The order policy is to produce end item A in lots of 50 units. Using the data shown in Figure 16.23 and the FOQ lot-sizing rule, complete the projected on-hand inventory and MPS quantity rows. Then, complete the MPS start row by offsetting the MPS quantities for the final assembly lead time. Compute the available-to-promise inventory for item A. Finally, assess the following customer requests for new orders. Assume that these orders arrive consecutively and their affect on ATP is cumulative. Which of these orders can be satisfied without altering the MPS Start quantities?

a. Customer C requests 10 units in week 3.

b. Customer B requests 30 units in week 4.

c. Customer A requests 30 units in week 1.

d. Customer D requests 50 units in week 5.

Item: A — **Order Policy:** 50 units — **Lead Time:** 1 week

Quantity on Hand: 5	Week 1	2	3	4	5	6	7	8	9	10
Forecast	20	10	40	10	0	0	30	20	40	20
Customer orders (booked)	30	20	5	8	0	2	0	0	0	0
Projected on-hand inventory	25									
MPS quantity	50									
MPS start										
Available-to-promise (ATP) inventory										

▲ **FIGURE 16.23** MPS Record for End Item A

SOLUTION

The projected on-hand inventory for the second week is

$$\begin{pmatrix}\text{Projected on-hand inventory at end of week 2}\end{pmatrix} = \begin{pmatrix}\text{On-hand inventory in week 1}\end{pmatrix} + \begin{pmatrix}\text{MPS quantity due in week 2}\end{pmatrix} - \begin{pmatrix}\text{Requirements in week 2}\end{pmatrix}$$

$$= 25 + 0 - 20 = 5 \text{ units}$$

where requirements are the larger of the forecast or actual customer orders booked for shipment during this period. No MPS quantity is required.

Without an MPS quantity in the third period, a shortage of item A will occur: $5 + 0 - 40 = -35$. Therefore, an MPS quantity equal to the lot size of 50 must be scheduled for completion in the third period. Then, the projected on-hand inventory for the third week will be $5 + 50 - 40 = 15$.

Figure 16.24 shows the projected on-hand inventories and MPS quantities that would result from completing the MPS calculations. The MPS start row is completed by simply shifting a copy of the MPS quantity row to the left by one column to account for the 1-week final assembly lead time. Also shown are the available-to-promise quantities. In week 1, the ATP is

$$\begin{pmatrix}\text{Available-to-Promise in week 1}\end{pmatrix} = \begin{pmatrix}\text{On-hand quantity in week 1}\end{pmatrix} + \begin{pmatrix}\text{MPS quantity in week 1}\end{pmatrix} - \begin{pmatrix}\text{Orderes booked up to week 3 when the next MPS arrives}\end{pmatrix}$$

$$= 5 + 50 - (30 + 20) = 5 \text{ units}$$

The ATP for the MPS quantity in week 3 is

$$\begin{pmatrix}\text{Available-to-Promise in week 3}\end{pmatrix} = \begin{pmatrix}\text{MPS quantity in week 3}\end{pmatrix} - \begin{pmatrix}\text{Orderes booked up to week 7 when the next MPS arrives}\end{pmatrix}$$

$$= 50 - (5 + 8 + 0 + 2) = 35 \text{ units}$$

The other ATPs equal their respective MPS quantities because no orders are booked for those weeks. As for the new orders, Customer A's request for 30 units in week 1 cannot be accommodated; the earliest it can be shipped is week 3 because the ATP for week 1 is insufficient. Assuming that Customer A's order is rejected, Customer B's request may be satisfied. The ATP for week 1 will

Lot Size: 50 units
Lead Time: 1 week

Quantity on Hand: 5	Week 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Forecast	20	10	40	10			30	20	40	20					
Customer orders (booked)	30	20	5	8		2									
Projected on-hand inventory	25	5	15	5	5	3	23	3	13	43					
MPS quantity	50		50				50		50	50					
MPS start		50				50		50	50						
Available-to-promise (ATP) inventory	5		35				50		50	50					

FIGURE 16.24 ▲
Completed MPS Record for End Item A

stay at 5 units and the ATP for week 3 will be reduced to 5 units. This acceptance allows the firm the flexibility to immediately satisfy an order for 5 units or less, if one comes in. When the MPS is updated next, the customer orders booked for week 4 will be increased to 38 to reflect the new order's shipping date. Customer C's order for 10 units in week 3 is likewise accepted. The ATP for weeks 1 and 3 will be reduced to 0, and when the MPS is updated, the customer orders booked for week 3 will be increased to 15. Finally, Customer D's order for 50 units in week 5 cannot be satisfied without changing the MPS.

Solved Problem 3

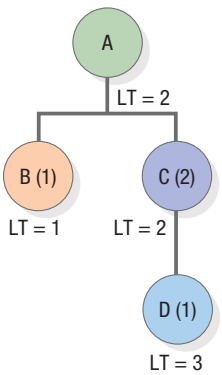


FIGURE 16.25 ▲
BOM for Product A

MyOMLab
Active Model 16.1 in MyOMLab provides additional insight on lotsizing decisions for this problem.

The MPS start quantities for product A calls for the assembly department to begin final assembly according to the following schedule: 100 units in week 2; 200 units in week 4; 120 units in week 6; 180 units in week 7; and 60 units in week 8. Develop a material requirements plan for the next 8 weeks for items B, C, and D. The BOM for A is shown in Figure 16.25, and data from the inventory records are shown in Table 16.1.

SOLUTION

We begin with items B and C and develop their inventory records, as shown in Figure 16.26. The MPS for product A must be multiplied by 2 to derive the gross requirements for item C because of the usage quantity. Once the planned order releases for item C are found, the gross requirements for item D can be calculated.

TABLE 16.1 | INVENTORY RECORD DATA

	ITEM		
Data Category	**B**	**C**	**D**
Lot-sizing rule	POQ ($P = 3$)	L4L	FOQ = 500 units
Lead time (LT)	1 week	2 weeks	3 weeks
Scheduled receipts	None	200 (week 1)	None
Beginning (on-hand) inventory	20	0	425

Item: B
Description:
Lot Size: POQ (P = 3)
Lead Time: 1 week

		Week									
		1	2	3	4	5	6	7	8	9	10
Gross requirements			100		200		120	180	60		
Scheduled receipts											
Projected on-hand inventory	20	20	200	200	0	0	240	60	0	0	0
Planned receipts			280				360				
Planned order releases		280				360					

Item: C
Description:
Lot Size: L4L
Lead Time: 2 weeks

		Week									
		1	2	3	4	5	6	7	8	9	10
Gross requirements			200		400		240	360	120		
Scheduled receipts		200									
Projected on-hand inventory	0	200	0	0	0	0	0	0	0	0	0
Planned receipts					400		240	360	120		
Planned order releases			400		240	360	120				

Item: D
Description:
Lot Size: FOQ = 500 units
Lead Time: 3 weeks

		Week									
		1	2	3	4	5	6	7	8	9	10
Gross requirements			400		240	360	120				
Scheduled receipts											
Projected on-hand inventory	425	425	25	25	285	425	305	305	305	305	305
Planned receipts					500	500					
Planned order releases		500	500								

FIGURE 16.26
Inventory Records for Items B, C, and D

Discussion Questions

1. Consider a service provider that is in the delivery business, such as UPS or FedEx. How can the principles of MRP be useful to such a company?
2. Form a group in which each member represents a different functional area of a firm. Provide a priority list of the information that could be generated from an MPS, from the most important to the least important, for each functional area. Rationalize the differences in the lists.
3. Consider the master flight schedule of a major airline, such as Air New Zealand. Discuss the ways in which it is analogous to a master production schedule for a manufacturer.
4. For an organization of your choice, such as where you previously worked, discuss how an ERP system could be used and whether it would increase effectiveness.

Problems

The OM Explorer and POM for Windows software is available to all students using the 10th edition of this textbook. Go to **www.pearsonglobaleditions.com/krajewski** website to download these computer packages. If you purchased MyOMLab, you also have access to Active Models software and significant help in doing the following problems. Check with your instructor on how best to use these resources. In many cases, the instructor wants you to understand how to do the calculations by hand. At the least, the software provides a check on your calculations. When calculations are particularly complex and the goal is interpreting the results in making decision, the software entirely replaces the manual calculations.

1. Consider the bill of materials (BOM) in Figure 16.27.
   - **a.** How many immediate parents (one level above) does item I have? How many immediate parents does item E have?
   - **b.** How many unique components does product A have at all levels?
   - **c.** Which of the components are purchased items?
   - **d.** How many intermediate items does product A have at all levels?
   - **e.** Given the lead times (LT) in weeks noted on Figure 16.27, how far in advance of shipment must a purchase commitment be for any of the purchased items identified in part (c)?
2. Product A is made from components B, C, and E. Item E is a subassembly that requires 2 units of D and 3 unit of C. Item B also is an intermediate item, made from F. All other usage quantities are 1. Draw the BOM for product A.
3. What is the lead time (in weeks) to respond to a customer order for product A, based on the BOM shown in Figure 16.28, assuming no existing inventories or scheduled receipts?
4. Product A is made from components B and C. Item B, in turn, is made from D and E. Item C also is an intermediate item, made from F and H. Finally, intermediate item E is made from H and G. Note that item H has two parents. The following are item lead times:

Item	A	B	C	D	E	F	G	H
Lead Time (weeks)	1	2	2	6	5	6	4	3

   - **a.** What lead time (in weeks) is needed to respond to a customer order for product A, assuming no existing inventories or scheduled receipts?
   - **b.** What is the customer response time if all purchased items (i.e., D, F, G, and H) are in inventory?
   - **c.** If you are allowed to keep just one purchased item in stock, which one would you choose?
5. Refer to Figure 16.22 and Solved Problem 1. If inventory consists of 2 units of B, 1 unit of F, and 3 units of G, how many units of G, E, and D must be purchased to produce 5 units of product A?

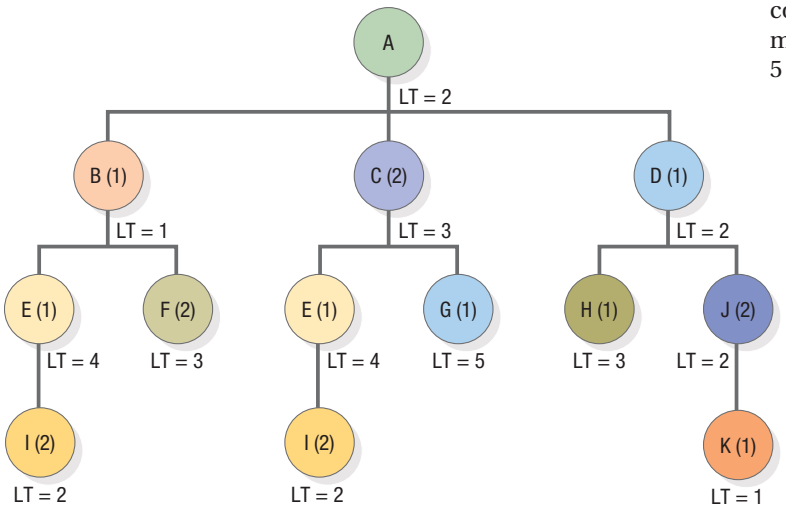


FIGURE 16.27 ▲
BOM for Product A

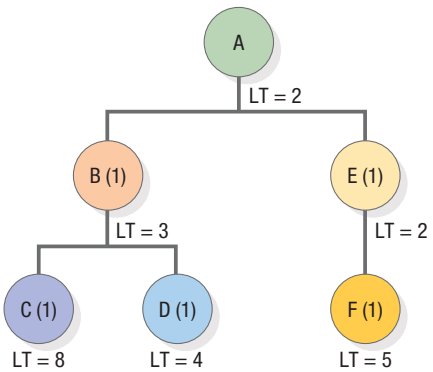


FIGURE 16.28 ▲
BOM for Product A

6. Complete the MPS record in Figure 16.29 for a single item.

FIGURE 16.29
MPS Record for Single Item

Item: A — **Order Policy:** 100 units — **Lead Time:** 1 week

Quantity on Hand: 30	**Week**							
	1	2	3	4	5	6	7	8
Forecast	20	18	31	28	26	30	31	35
Customer orders (booked)	15	17	9	17	11	0	11	0
Projected on-hand inventory								
MPS quantity								
MPS start								

7. Complete the MPS record shown in Figure 16.30 for a single item.

FIGURE 16.30
MPS Record for Single Item

Item: A — **Order Policy:** 100 units — **Lead Time:** 1 week

Quantity on Hand: 70	**January**				**February**			
	1	2	3	4	5	6	7	8
Forecast	70	70	70	50	55	50	50	55
Customer orders (booked)	40	15	90	0	35	65	0	0
Projected on-hand inventory								
MPS quantity								
MPS start								

8. An end item's demand forecasts for the next 10 weeks are 30, 20, 35, 50, 25, 25, 0, 40, 0, and 50 units. The current on-hand inventory is 80 units. The order policy is to produce in lots of 100. The booked customer orders for the item, starting with week 1, are 22, 30, 15, 9, 0, 0, 5, 3, 7, and 0 units. At present, no MPS quantities are on-hand for this item. The lead time is 2 weeks. Develop an MPS for this end item.

9. Figure 16.31 shows a partially completed MPS record for ball bearings.

a. Develop the MPS for ball bearings.

b. Four customer orders arrived in the following sequence:

Order	Quantity	Week Desired
1	500	4
2	400	5
3	300	1
4	300	7

Assume that you must commit to the orders in the sequence of arrival and cannot change the desired shipping dates or your MPS. Which orders should you accept?

10. The forecasted requirements for an electric hand drill for the next 6 weeks are 15, 40, 10, 20, 50, and 30 units. The marketing department has booked orders totaling 20, 25, 10, and 20 units for delivery in the first (current), second, third, and fourth weeks. Currently, 30 hand drills are in stock. The policy is to order in lots of 60 units. Lead time is one week.

a. Develop the MPS record for the hand drills.

b. A distributor of the hand drills places an order for 15 units. What is the appropriate shipping date for the entire order?

11. Figure 16.32 shows a partially completed MPS record for 2" pneumatic control valves. Suppose that you receive the following orders for the valves (shown in the order of their arrival). As they arrive, you must decide whether to accept or reject them. Which orders would you accept for shipment?

Order	Amount (Units)	Week Requested
1	15	2
2	30	5
3	25	3
4	75	7

12. Tabard Industries forecasted the following demand for one of its most profitable products for the next 8 weeks: 120, 120, 120, 100, 100, 100, 80, and 80 units. The booked customer orders for this product, starting in week 1 are: 100, 80, 60, 40, 10, 10, 0, and 0 units. The current on-hand inventory is 150 units, the order quantity is 200 units, and the lead time is one week.

a. Develop a MPS for this product.

b. The marketing department revised its forecast. Starting with week 1, the new forecasts are: 120, 120, 120, 150, 150, 150, 100, and 100 units. Assuming that the prospective MPS you developed in part (a) does not change, prepare a revised MPS record. Comment on the situation that Tabard now faces.

c. Returning to the original forecasted demand level and the MPS record you developed in part (a), assume that marketing accepted a new customer order for 200 units in week 2 and thereby booked orders in week 2 is now 280 units. Assuming that the prospective MPS you developed in part (a) does not change, prepare a revised MPS record. Comment on the situation that Tabard now faces.

13. A forecast of 240 units in January, 320 units in February, and 240 units in March has been approved for the

FIGURE 16.31 ▶
MPS Record for Ball Bearings

Item: Ball bearings — **Order Policy:** 500 units; **Lead Time:** 1 week

Quantity on Hand: 400	Week 1	2	3	4	5	6	7	8	9	10
Forecast	550	300	400	450	300	350	200	300	450	400
Customer orders (booked)	300	350	250	250	200	150	100	100	100	100
Projected on-hand inventory										
MPS quantity	500									
MPS start										
Available-to-promise (ATP) inventory										

Item: 2" Pneumatic control valve **Order Policy:** 75 units
Lead Time: 1 week

Quantity on Hand: 10	Week 1	2	3	4	5	6	7	8
Forecast	40	40	40	40	30	30	50	50
Customer orders (booked)	60	45	30	35	10	5	5	0
Projected on-hand inventory								
MPS quantity	75	75						
MPS start	75							
Available-to-promise (ATP) inventory								

◀ **FIGURE 16.32**
MPS Record for 2" Pneumatic Control Valve

seismic-sensory product family manufactured at the Rockport facility of Maryland Automated, Inc. Three products, A, B, and C, comprise this family. The product mix ratio for products A, B, and C for the past 2 years has been 35 percent, 40 percent, and 25 percent, respectively. Management believes that the monthly forecast requirements are evenly spread over the 4 weeks of each month. Currently, 5 units of product C are on hand. The company produces product C in lots of 40, and the lead time is 2 weeks. A production quantity of 40 units from the previous period is scheduled to arrive in week 1. The company has accepted orders of 20, 9, 8, 15, 2, and 6 units of product C in weeks 1 through 6, respectively. Prepare a prospective MPS for product C and calculate the available-to-promise inventory quantities.

14. The partially completed inventory record for the tabletop subassembly in Figure 16.33 shows gross requirements, scheduled receipts, lead time, and current on-hand inventory.

a. Complete the last three rows of the record for an FOQ of 120 units.

b. Complete the last three rows of the record by using the L4L lot-sizing rule.

c. Complete the last three rows of the record by using the POQ lot-sizing rule, with $P = 2$.

Item: M405—X **Lot Size:**
Description: Tabletop subassembly **Lead Time:** 2 weeks

Week	1	2	3	4	5	6	7	8	9	10
Gross requirements	95		80		80		40	95		
Scheduled receipts	120									
Projected on-hand inventory 30										
Planned receipts										
Planned order releases										

◀ **FIGURE 16.33**
Inventory Record for the Tabletop Subassembly

15. The partially completed inventory record for the driveshaft subassembly in Figure 16.34 shows gross requirements, scheduled receipts, lead time, and current on-hand inventory.

a. Complete the last three rows of the record for an FOQ of 60 units.

b. Complete the last three rows of the record by using the L4L lot-sizing rule.

c. Complete the last three rows of the record by using the POQ lot-sizing rule, with $P = 4$.

FIGURE 16.34 ▶
Inventory Record for the Driveshaft Subassembly

Item: Driveshaft subassembly — **Lot Size:** 60 units, **Lead Time:** 3 weeks

	Week							
	1	2	3	4	5	6	7	8
Gross requirements	30	30	20	15	35	35	55	55
Scheduled receipts	80							
Projected on-hand inventory 10								
Planned receipts								
Planned order releases								

16. The partially completed inventory record for the rotor subassembly in Figure 16.35 shows gross requirements, scheduled receipts, lead time, and current on-hand inventory.

a. Complete the last three rows of the record for an FOQ of 140 units.

b. Complete the last three rows of the record by using the L4L lot-sizing rule.

c. Complete the last three rows of the record by using the POQ lot-sizing rule, with $P = 2$.

17. Figure 16.36 shows a partially completed inventory record for the real wheel subassembly. Gross requirements, scheduled receipts, lead time, and current on-hand inventory are shown.

a. Complete the last three rows of the record for an FOQ of 200 units.

b. Complete the last three rows of the record by using an FOQ of 100 units.

c. Complete the last three rows of the record by using the L4L rule.

FIGURE 16.35 ▶
Inventory Record for the Rotor Subassembly

Item: Rotor subassembly — **Lot Size:** 140 units, **Lead Time:** 2 weeks

	Week							
	1	2	3	4	5	6	7	8
Gross requirements	55	20	45	40	70	70	70	70
Scheduled receipts	140							
Projected on-hand inventory 20								
Planned receipts								
Planned order releases								

Item: MQ–09 **Lot Size:**
Description: Rear wheel subassembly **Lead Time:** 1 week

	Week									
	1	2	3	4	5	6	7	8	9	10
Gross requirements	25	105	110	90		45	110	60		
Scheduled receipts										
Projected on-hand inventory 50										
Planned receipts										
Planned order releases										

◀ **FIGURE 16.36**
Inventory Record for the Rear Wheel Subassembly

18. A partially completed inventory record for the motor subassembly is shown in Figure 16.37.

a. If it costs the company $1 to hold a unit in inventory from one week to the next, and the cost to release an order is $50, which of the lot sizing rules used above will provide the lowest inventory holding + order release cost?

b. Complete the last three rows of the record by using the POQ rule with $P = 2$.

c. Complete the last three rows of the record by using the POQ rule with $P = 4$.

d. Complete the last three rows of the record by using the L4L rule.

Item: GF–4 **Lot Size:**
Description: Motor subassembly **Lead Time:** 2 weeks

	Week									
	1	2	3	4	5	6	7	8	9	10
Gross requirements		80	50	35	20	55	15	30	25	10
Scheduled receipts	60									
Projected on-hand inventory 20										
Planned receipts										
Planned order releases										

▲ **FIGURE 16.37**
Inventory Record for the Motor Subassembly

Advanced Problems

Depending on the time available to cover this chapter, your instructor may not assign these problems. Perhaps, your assignment for the multi-level problems allows the use of the *Material Requirements Planning* Solver of OM Explorer, with main requirement of looking for situations that need the inventory planner's attention (i.e., an action notice).

19. The BOM for product A is shown in Figure 16.38, and data from the inventory records are shown in Table 16.2. In the master production schedule for product A, the MPS start row has 100 units in week 3 and 200 in week 6. Develop the material requirements plan for the next 6 weeks for items B, C, and D.

a. Develop the material requirements plan for the next 6 weeks for items B, C, and D.

b. What specific managerial actions are required in week 1?

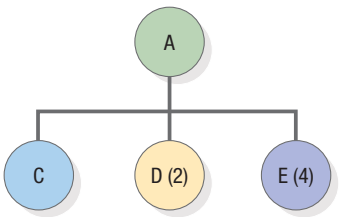


FIGURE 16.38 ▲
BOM for Product A

TABLE 16.2 | INVENTORY RECORD DATA

	ITEM		
Data Category	**C**	**D**	**E**
Lot-Sizing Rule	L4L	FOQ = 200	POQ (P = 3 weeks)
Lead time	2 weeks	1 week	1 week
Scheduled receipts	None	200 (in week 3)	200 (in week 3)
Beginning inventory	50	200	0

20. The BOMs for products A & B and data from the inventory records are shown in Figure 16.39. Data from the inventory records are shown in Table 16.3. In the master production schedule for product A, the MPS start row has 85 units in week 2 and 200 in week 4 and 50 in week 8. In the master production schedule for product B, the MPS start row has 65 units in week 3 and 50 in week 4 and 50 in week 5 and 75 in week 8.

a. Develop the material requirements plan for the next 8 weeks for items C, D, and E. Note any difficulties you observe in the inventory records.

b. Can the difficulties noted in part a be rectified by expediting any Scheduled Receipts?

FIGURE 16.39 ▶
BOMs for Product A and Product B

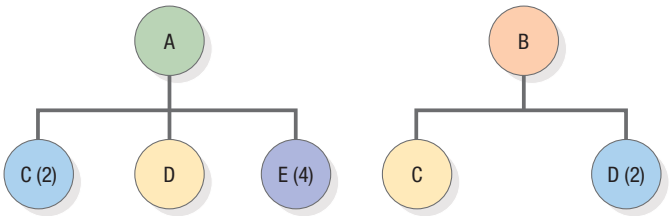
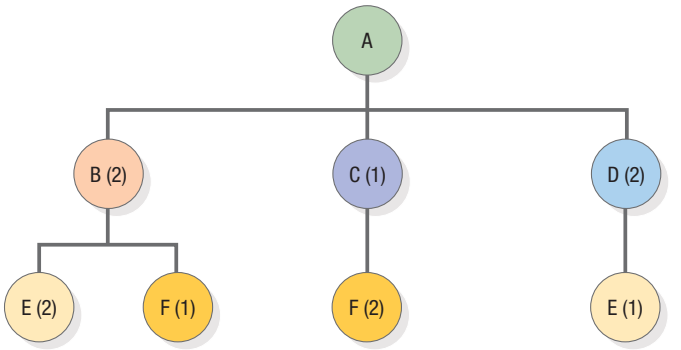


TABLE 16.3 | INVENTORY RECORD DATA

	ITEM		
Data Category	**C**	**D**	**E**
Lot-sizing rule	L4L	FOQ = 200	POQ (P = 2 weeks)
Lead time	2 weeks	1 week	1 week
Scheduled receipts	200 (in week 3)	0	0
Beginning inventory	0	0	200

21. Figure 16.40 illustrates the BOM for product A. The MPS start row in the master production schedule for product A calls for 50 units in week 2, 65 units in week 5, and 80 units in week 8. Item C is produced to make A and to meet the forecasted demand for replacement parts. Past replacement part demand has been 20 units per week (add 20 units to C's gross requirements). The lead times for items F and C are 1 week, and for the other items the lead time is 2 weeks. No safety stock is required for items B, C, D, E, and F. The L4L lot-sizing rule is used for items B and F; the POQ lot-sizing rule (P = 3) is used for C. Item E has an FOQ of 600 units, and D has an FOQ of 250 units. On-hand inventories are 50 units of B, 50 units of C, 120 units of D, 70 units of E, and 250 units of F. Item B has a scheduled receipt of 50 units in week 2. Develop a material requirements plan for the next 8 weeks for items B, C, D, E, and F.

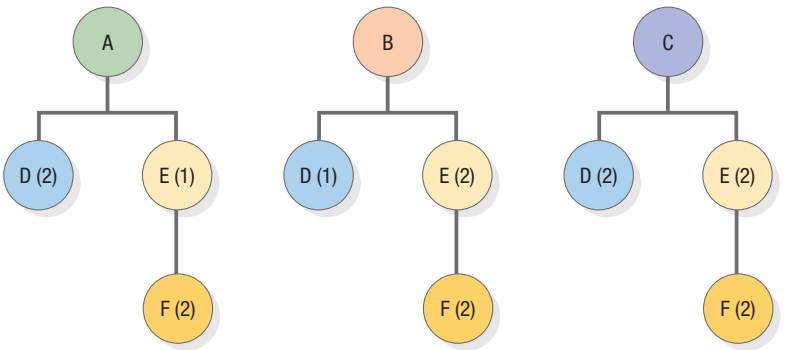


▲ **FIGURE 16.40**
BOM for Product A

22. The following information is available for three MPS items:

Product A	An 80-unit order is to be started in week 3. A 55-unit order is to be started in week 6.
Product B	A 125-unit order is to be started in week 5.
Product C	A 60-unit order is to be started in week 4.

Develop the material requirements plan for the next 6 weeks for items D, E, and F. The BOMs are shown in Figure 16.41, and data from the inventory records are shown in Table 16.4. (*Warning:* A safety stock requirement applies to item F. Be sure to plan a receipt for any week in which the projected on-hand inventory becomes less than the safety stock.)

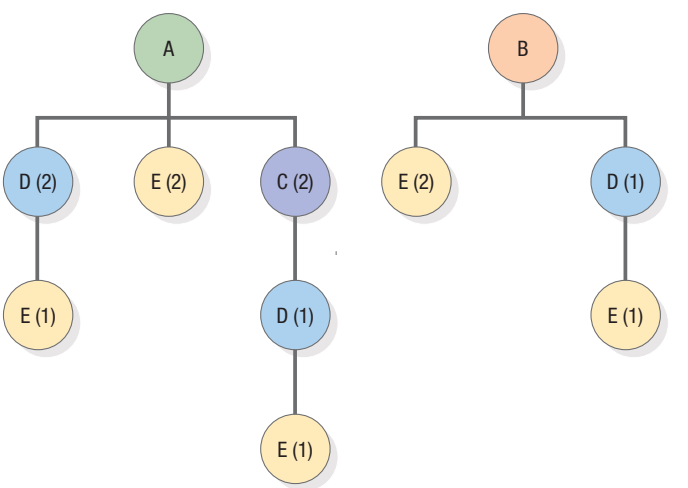


◀ **FIGURE 16.41**
BOMs for Products A, B, and C

TABLE 16.4 | **INVENTORY RECORD DATA**

	ITEM		
Data Category	**D**	**E**	**F**
Lot-sizing rule	FOQ = 150	L4L	POQ ($P = 2$)
Lead time	3 weeks	1 week	2 weeks
Safety stock	0	0	30
Scheduled receipts	150 (week 3)	120 (week 2)	None
Beginning inventory	150	0	100

▼ **FIGURE 16.42**
BOMs for Products A and B



23. Figure 16.42 shows the BOMs for two products, A and B. Table 16.5 shows the MPS quantity start date for each one. Table 16.6 contains data from inventory records for items C, D, and E. There are no safety stock requirements for any of the items.

a. Determine the material requirements plan for items C, D, and E for the next 8 weeks.

b. What specific managerial actions are required in week 1?

c. Suppose that a very important customer places an emergency order for a quantity of product A. In order to satisfy this order, a new MPS of 200 units of product A is now required in week 5. Determine the changes to the material requirements plan if this order is accepted and note any problems that you detect.

TABLE 16.5 | MPS QUANTITY START DATES

	DATE							
Product	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
A		125		95		150		130
B			80			70		

TABLE 16.6 | INVENTORY RECORD DATA

	ITEM		
Data Category	**C**	**D**	**E**
Lot-sizing rule	L4L	POQ ($P = 3$)	FOQ = 800
Lead time	3 weeks	2 weeks	1 week
Scheduled receipts	200 (week 2)	None	800 (week 1)
Beginning inventory	85	625	350

24. The BOM for product A is shown in Figure 16.43. The MPS for product A calls for 120 units to be started in weeks 2, 4, 5, and 8. Table 16.7 shows data from the inventory records.

a. Develop the material requirements plan for the next 8 weeks for each item.

b. What specific managerial actions are required in week 1? Make sure you address any specific difficulties you encounter in the inventory records.

FIGURE 16.43 ▶
BOM for Product A

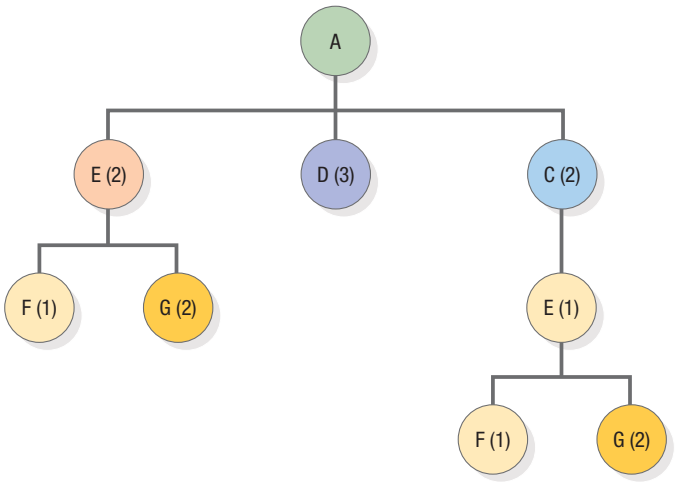


TABLE 16.7 | INVENTORY RECORD DATA

	ITEM				
Data Category	**C**	**D**	**E**	**F**	**G**
Lot-sizing rule	L4L	FOQ = 700	FOQ = 700	L4L	L4L
Lead time	3 weeks	3 weeks	4 weeks	2 weeks	1 week
Safety stock	0	0	0	50	0
Scheduled receipts	150 (week 2)	450 (week 2)	700 (week 1)	None	1,400 (week 1)
Beginning inventory	125	0	235	750	0

25. Refer to Solved Problem 1 (Figure 16.22) for the bill of materials and Table 16.8 for component inventory record information. Develop the material requirements plan for all components and intermediate items associated with product A for the next 10 weeks. The MPS for product A calls for 60 units to be started in weeks 2, 6, 8, and 9. (*Warning:* Safety stock requirements apply to items B and C.)

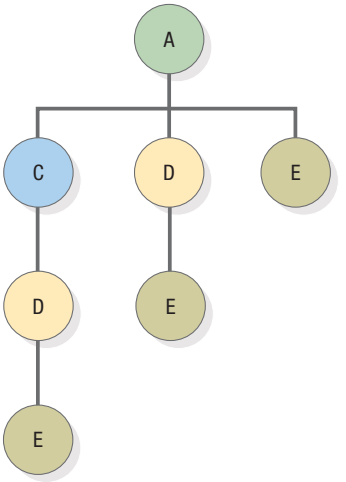
TABLE 16.8 | INVENTORY RECORD DATA

	ITEM					
Data Category	**B**	**C**	**D**	**E**	**F**	**G**
Lot-sizing rule	L4L	L4L	POQ ($P = 2$)	L4L	L4L	FOQ = 480
Lead time	2 weeks	3 weeks	3 weeks	6 weeks	1 week	3 weeks
Safety stock	20	15	0	0	0	0
Scheduled receipts	120 (week 2)	50 (week 2)	None	200 (week 6)	90 (week 3)	None
Beginning inventory	20	60	60	300	0	0

26. The bill of materials and the data from the inventory records for product A are shown in Figure 16.44. Assume that the MPS start quantities for A are 100 units in weeks 1, 2, 3, 4, 7, 8, 9, and 10.

Derive an MRP plan for the components going into product A using the data in Table 16.9.

What specific managerial actions are required in week 1? Make sure you address any specific difficulties you encounter in the inventory records.



▲ **FIGURE 16.44**
BOM for product A

TABLE 16.9 | INVENTORY RECORD DATA

	ITEM		
Data Category	**C**	**D**	**E**
Lot-sizing rule	L4L	L4L	FOQ = 300
Lead time	1 week	2 weeks	2 weeks
Scheduled receipts		100 (in week 2)	300 (in week 2)
Beginning Inventory	225	350	100

27. An end item's demand forecasts for the next 10 weeks are 35, 35, 35, 35, 20, 20, 30, 30, 30, and 30 units. The current on-hand inventory is 100 units. The order policy is to produce in lots of 75. The booked customer orders for the item, starting with week 1, are 15, 38, 7, 5, 0, 3, 10, 0, 0, and 0 units. The lead time is 2 weeks.

a. Develop an MPS for this end item.

b. The marketing department has received five orders for this item in the following sequence:

Order 1 is for 40 units to be delivered in period 1
Order 2 is for 75 units to be delivered in period 4
Order 3 is for 100 units to be delivered in period 6
Order 4 is for 125 units to be delivered in period 7
Order 5 is for 90 units to be delivered in period 10

Assuming that the prospective MPS you developed in part (a) does not change, which orders would you be able to accept based on the available to promise (ATP)?

28. An end item's demand forecasts for the next 6 weeks are 35 units, followed by forecasts of 30 units for weeks 7 though 10. The current on-hand inventory is 70 units. The order policy is to produce in lots of 120. The booked customer orders for the item, starting with week 1, are 17, 35, 15, 14, 0, 0, 12, 0, 0, and 0 units. The lead time is 2 weeks.

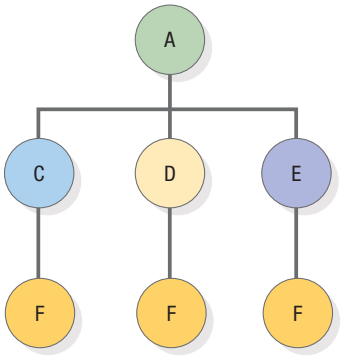
a. Develop an MPS for this end item.

b. The marketing department has received six orders for this item in the following sequence:

Order 1 is for 40 units to be delivered in period 3
Order 2 is for 60 units to be delivered in period 4
Order 3 is for 70 units to be delivered in period 6
Order 4 is for 40 units to be delivered in period 3
Order 5 is for 20 units to be delivered in period 5
Order 6 is for 115 units to be delivered in period 9

Assuming that the prospective MPS you developed in part (a) does not change, which orders would you be able to accept based on the available to promise (ATP)?

29. The bill of materials and the data from the inventory records for product A are shown in Figure 16.45. Assume that the MPS start quantities for A are 50 units in weeks 1, 2 and 3, and 150 units in weeks 6 and 8. Derive an MRP plan for the components going into product A using the data in Table 16.10.



▲ **FIGURE 16.45**
BOM for Product A

TABLE 16.10 | INVENTORY RECORD DATA

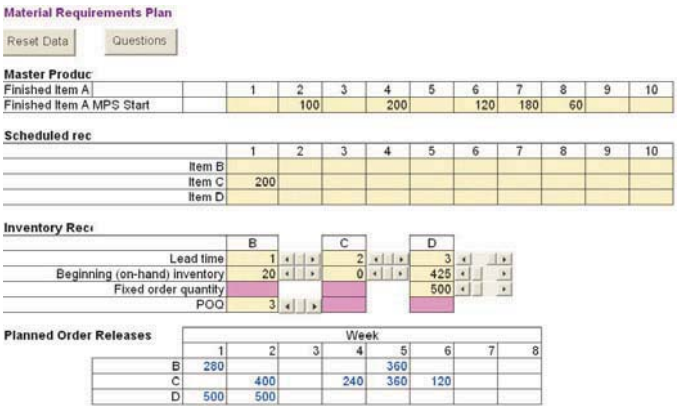
	ITEM			
Data Category	**C**	**D**	**E**	**F**
Lot-sizing rule	POQ ($P = 2$)	L4L	FOQ = 300	FOQ = 400
Lead time	1 week	1 week	2 weeks	4 weeks
Scheduled receipts		100 (week 2)		400 (week 1)
Beginning inventory	100	0	110	40

Active Model Exercise

This Active Model appears in MyOMLab. It allows you to evaluate the relationship between the inventory record data and the planned order releases.

QUESTIONS

1. As the lead time for item C changes, what happens to the order releases for items B, C, and D?
2. As the on-hand inventory for item C increases from 0 to 200, what happens to the order releases for items B, C, and D?
3. As the fixed order quantity (FOQ) for item D increases from 500 to 750, what happens to the order releases for items B, C, and D?
4. Suppose that the POQ for item B is changed from 3 weeks to 2 weeks. How does this change affect the order releases for items B, C, and D?



▲ **ACTIVE MODEL 16.1**
Material Requirements Planning Using Data from Solved Problem 3 and Table 16.1

CASE Flashy Flashers, Inc.

Flashy Flashers is a medium-sized firm employing 900 persons and 125 managerial and administrative personnel. The firm produces a line of automotive electrical components. It supplies about 95 auto parts stores and Moonbird Silverstreak car dealers in its region. Johnny Bennett, who serves as the president, founded the company by producing cable assemblies in his garage. By working hard, delivering consistent product quality, and by providing good customer service, he expanded his business to produce a variety of electrical components. Bennett's commitment to customer service is so strong that his company motto, "Love Thy Customers as Thyself," is etched on a big cast-iron plaque under his giant oil portrait in the building's front lobby.

The company's two most profitable products are the automotive front sidelamp and the headlamp. With the rising popularity of Eurosport sedans, such as the Moonbird Silverstreak, Flashy Flashers has enjoyed substantial demand for these two lamp items.

Last year, Kathryn Marley, the vice president of operations and supply chain management, approved the installation of a new MRP system. It is a first important step toward the eventual goal of a full-fledged ERP system. Marley worked closely with the task force that was created to bring MRP online. She frequently attended the training sessions for selected employees, emphasizing how MRP should help Flashy Flashers secure a better competitive edge.

A year later, the MRP system is working fairly well. However, Marley believes that there is always a better way and seeks to continually improve the company's processes. To get a better sense for potential improvements, she met with the production and inventory control manager, the shop supervisor, and the purchasing manager. Here are some of their observations.

Production and Inventory Control Manager

Inventory records and BOM files are accurate and well maintained. Inventory transactions are faithfully made when inventory is replenished or removed from the stockroom so that current on-hand balances are credible. There is a MRP explosion each week, which gives the company the new material

requirements plan. It provides information that helps identify when new orders need to be launched. Information can also be searched to help identify which scheduled receipts need to be expedited and which ones can be delayed by assigning them a later due date, thereby making room for more urgent jobs.

One planner suggested that the MRP outputs should be extended to provide priority and capacity reports, with pointers as to which items need their attention. The original plan was to get the order-launching capability implemented first. However, there is no formal system of priority planning, other than the initial due date assigned to each scheduled receipt when it is released, transforming it from a planned order release into a scheduled receipt. The due dates do not get updated later even when there are unexpected scrap losses, capacity shortages, short shipments, or last-minute changes in the MPS (responding to requests from favorite customers). Jobs are scheduled on the shop floor and by suppliers according to the EDD rule, based on their due dates. If due dates assigned to scheduled receipts were updated, it might help get open orders done when they are really needed. Furthermore, planned order releases in the action bucket are translated into scheduled receipts (using inventory transactions), after checking that its components are available. The current system does not consider possible capacity problems when releasing new orders.

Shop Supervisor

His primary complaint is that the shop workloads are anything but level. One week, they hardly have any work, and the supervisor overproduces (more than called for by the scheduled receipts) just to keep everyone busy. The next week can be just the opposite—so many new orders with short fuses that almost everyone needed to work overtime or else the scheduled receipt quantities are reduced to cover immediate needs. It is feast or famine, unless they make things work on the shop floor! They do make inventory transactions to report deviations from plan for the scheduled receipts, but these "overrides" make the scheduled receipt information in the MRP records more uncertain for the planners. A particular concern is to make sure that the bottleneck workstations are kept busy.

Purchasing

Buyers are putting out too many fires, leaving little time for creative buying. In such cases, their time is spent following up on orders that are required in the very near future or that are even late. Sometimes, the MRP plan shows planned order releases for purchased items that that are needed almost immediately, not allowing for the planned lead time. In checking the MRP records, the planned lead times are realistic and what the suppliers expect. Last week, things were fine for an item, and this week a rush order needs to be placed. What is the problem?

Marley tried to assimilate all this information. She decided to collect all the required information about the sidelamps and headlamps (shown in Table 16.11 through Table 16.14 and in Figure 16.46) to gain further insight into possible problems and identify areas for improvement.

Your Assignment

Put yourself in Marley's place and prepare a report on your findings. Specifically, you are required to do a manual MRP explosion for the sidelamps and headlamps for the next 6 weeks (beginning with the current week). Assume that it is now the start of week 1. Fill in the planned order releases form provided in Table 16.15. It should show the planned order releases for all items for the next 6 weeks. Include it in your report.

TABLE 16.11 | PART NUMBERS AND DESCRIPTIONS

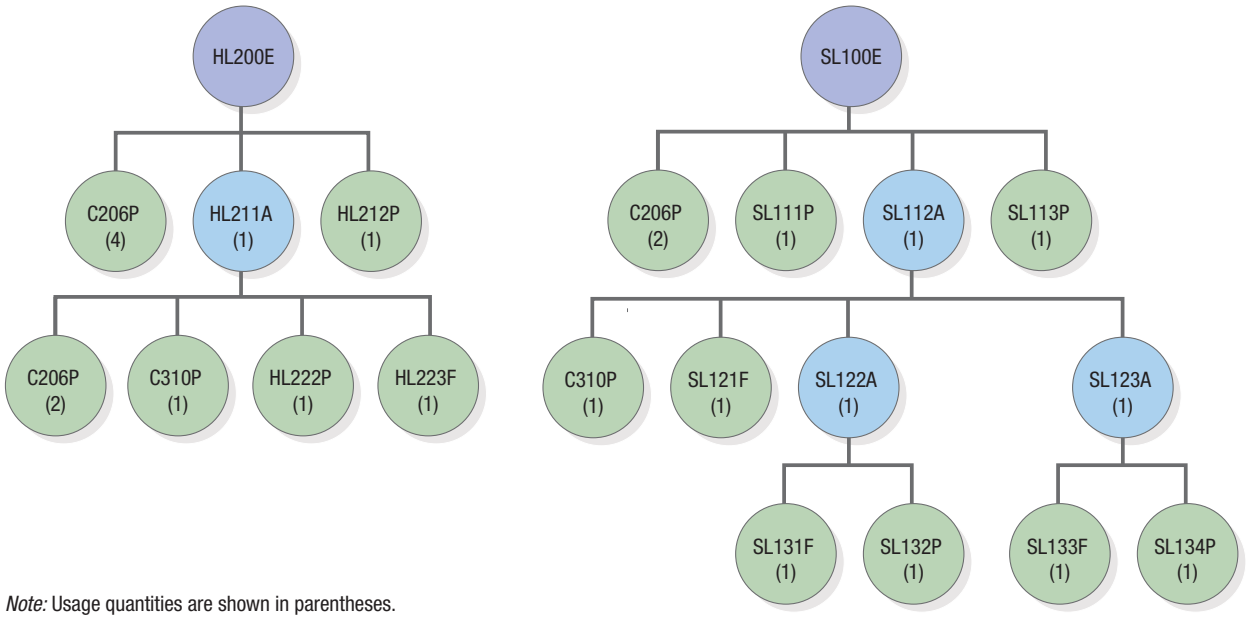
Part Number	Description
C206P	Screws
C310P	Back rubber gasket
HL200E	Headlamp
HL211A	Head frame subassembly
HL212P	Head lens
HL222P	Headlamp module
HL223F	Head frame
SL100E	Sidelamp
SL111P	Side lens
SL112A	Side frame subassembly
SL113P	Side lens rubber gasket
SL121F	Side frame
SL121F	Side frame
SL122A	Side bulb subassembly
SL123A	Flasher bulb subassembly
SL131F	Side cable grommet and receptacle
SL132P	Side bulb
SL133F	Flasher cable grommet and receptacle
SL134P	Flasher bulb

TABLE 16.12 | MASTER PRODUCTION SCHEDULE

Item Description and Part Number	Quantity	MPS Start Date
Headlamp (HL200E)	120	Week 4
	90	Week 5
	75	Week 6
Sidelamp (SL100E)	100	Week 3
	80	Week 5
	110	Week 6

TABLE 16.13 | REPLACEMENT PART DEMAND

Item Description and Part Number	Quantity	Date
Side lens (SL111P)	40	Week 3
	35	Week 6



Note: Usage quantities are shown in parentheses.

▲ **FIGURE 16.46**
BOMs for Headlamps and Sidelamps

Supplement your report with worksheets on the manual MRP explosion, and list the actions that planners should consider this week to (1) release new orders, (2) expedite scheduled receipts, and (3) delay a scheduled receipt's due date.

Finally, identify the good and bad points of MRP implementation at Flashy Flashers. Conclude by making suggestions on ways to improve its resource planning process.

TABLE 16.14 | SELECTED DATA FROM INVENTORY RECORDS

Part Number	Lead Time (Weeks)	Safety Stock (Units)	Lot-Sizing Rule	On-Hand (Units)	Scheduled Receipt (Units and Due Dates)
C206P	1	30	FOQ = 2,500	270	—
C310P	1	20	FOQ = 180	40	180 (week 1)
HL211A	3	0	L4L	0	
HL212P	2	15	FOQ = 350	15	—
HL222P	3	10	POQ (P = 4 weeks)	10	285 (week 1)
HL223F	1	0	POQ (P = 4 weeks)	0	120 (week 1)
SL111P	2	0	FOQ = 350	15	—
SL112A	3	0	L4L	20	80 (week 2)
SL113P	1	20	FOQ = 100	20	—
SL121F	2	0	L4L	0	80 (week 2)
SL122A	2	0	L4L	0	80 (week 2)
SL123A	2	0	FOQ = 200	0	—
SL131F	2	0	POQ (P = 2 weeks)	0	110 (week 1)
SL132P	1	25	FOQ = 100	35	100 (week 1)
SL133F	2	0	FOQ = 250		—
SL134P	1	0	FOQ = 400	100	—

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TABLE 16.15 | PLANNED ORDER RELEASE FORM

FILL IN THE PLANNED ORDER RELEASES FOR ALL COMPONENTS.

	Week					
Item Description and Part Number	**1**	**2**	**3**	**4**	**5**	**6**
Side lens (SL111P)						
Side lens rubber gasket (SL113P)						
Side frame subassembly (SL112A)						
Side frame (SL121F)						
Side bulb subassembly (SL122A)						
Flasher bulb subassembly (SL123A)						
Side cable grommet and receptacle (SL131F)						
Flasher cable grommet and receptacle (SL133F)						
Side bulb (SL132P)						
Flasher bulb (SL134P)						
Head frame subassembly (HL211A)						
Head lens (HL212P)						
Headlamp module (HL222P)						
Head frame (HL223F)						
Back rubber gasket (C310P)						
Screws (C206P)						

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